

Large Language Model Driven Reinforcement Learning for Portfolio Allocation

by

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Abstract

This thesis studies when large language models (LLMs) add incremental value to reinforcement learning (RL) for portfolio management beyond strong price-based baselines. Portfolio management matters because investors must allocate capital under transaction costs, drawdown constraints, and regime shifts. Classical models provide static snapshots of this trade-off, while deep-RL frameworks learn dynamic policies mostly from prices and technical indicators. In parallel, financial LLMs summarise earnings calls, regulatory filings, and news, yet it is unclear how such text signals affect an RL allocator in a portfolio setting.

We instantiate this question as a weekly allocation task over a 20-stock U.S. equity universe from mid-April 2019 to early September 2025, with proportional trading costs and a risk-aware reward. A multi-stage LLM analyst stack converts earnings calls, SEC filings, and news into structured signals and risk scores, which are fed to a Proximal Policy Optimisation (PPO) agent either as observations or as exogenous penalties. The empirical programme asks: (i) how LLM-derived observations compare with a tuned price-only PPO and equal-weight and minimum-variance baselines; (ii) whether LLM-based risk shaping improves downside protection; and (iii) whether curriculum-style feature introduction outperforms all-in-one multimodal training.

Experiments fix a reproducible environment and systematically vary observation sets, risk coefficients, and training schedules, with logging of turnover, concentration, and portfolio-weight heatmaps. First, a price-only PPO agent consistently matches or exceeds classical baselines in net asset value while respecting similar risk profiles. Second, LLM-enhanced agents change trading style—inducing sector tilts and higher turnover—but do not robustly outperform the price-only backbone, consistent with semi-strong market efficiency in this finite, non-stationary sample. Third, LLM-based risk shaping and curricula offer at best modest, fragile gains relative to a simpler all-in-one backbone.

The thesis contributes an end-to-end LLM-driven, risk-aware portfolio framework, an ablation-centred evaluation of multi-source integration, and design lessons for LLM–RL systems in similar equity-portfolio settings.

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by

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This is to certify that I have examined the above MPhil thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

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28 January 2026

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Portfolio allocation is a central problem in quantitative finance: given a set of tradable assets and a stream of market information, an investor must decide how to allocate capital over time while balancing expected return against risk. Classical models, beginning with Markowitz’s mean–variance theory, cast this as a single-period optimization problem: at a given rebalancing time, the investor chooses a fixed portfolio weight vector for the next horizon by trading off expected return against return variance under estimated moments of asset returns [1]. The Sharpe ratio and related performance measures extend this perspective by evaluating strategies over a backtest horizon, but still summarize each strategy primarily through its average excess return per unit volatility [2]. In practice, mean–variance style models are applied in a rolling fashion—re-estimating moments on moving windows and solving a fresh single-period problem at each date—but they remain myopic in two senses: they do not explicitly model how today’s decision affects the distribution of future states and actions, and they typically omit transaction costs, market frictions, and path-dependent risks such as drawdowns from the core optimization, handling these concerns instead via heuristic constraints or post-hoc evaluation.

Reinforcement learning (RL) offers a sequential alternative: given a stream of observations, an agent repeatedly selects actions and receives rewards, and a policy-gradient algorithm such as Proximal Policy Optimization (PPO) [3] can directly optimize a long-horizon objective. In finance, open-source frameworks such as FinRL and FinRL-Meta [4, 5] demonstrate that deep RL can learn non-trivial trading and allocation policies from price- and volume-based signals in realistic market environments. However, most practical RL setups remain predominantly price-driven, with limited use of long-horizon fundamental disclosures or rich textual context.

At the same time, large language models (LLMs) have substantially improved the state of financial natural language processing. Domain-adapted models such as FinBERT [6] and large financial LLMs such as BloombergGPT and FinGPT [7, 8], as surveyed by [9], can summarise earnings calls, parse regulatory filings, and extract sentiment or risk signals from news at scale. A growing line of work on financial LLM agents and multi-agent trading systems shows that such

models can be embedded into more complex decision pipelines, often with promising backtest performance.

Despite this progress, it remains unclear how much incremental value LLM-extracted information adds on top of strong price-based baselines in a realistic portfolio allocation task. Weekly rebalancing is a particularly interesting regime: it is slow enough that fundamental and disclosure-driven signals might matter, but frequent enough that transaction costs, turnover, and drawdowns cannot be ignored. In this regime, classical equal-weight and minimum-variance portfolios remain robust benchmarks, and prior evidence of LLM-based outperformance is limited and often evaluated in different settings (e.g., single-asset or daily trading).

This thesis takes an empirical, systems-oriented perspective on this question, by constructing a weekly, U.S.-equity portfolio environment spanning April 2019 to September 2025, design a multi-source LLM pipeline that produces structured signals from earnings calls, SEC filings, news, and technical commentary, and couple these signals to a PPO-based portfolio agent via both observation features and risk-aware reward shaping. The goal is not to propose a new RL algorithm, but to stress test a plausible LLM→RL integration under realistic data and compute constraints, and to quantify when such a system behaves sensibly and how its performance compares to strong non-LLM baselines.

Empirically, the extensive experiments in Chapter 4 show that the proposed LLM–RL agents achieve risk-adjusted performance that is broadly comparable to price-only RL and classical benchmarks; clear and consistent outperformance is rare. The main contributions are therefore methodological and diagnostic: we demonstrate how to build and instrument such a system, and we analyse where multi-genre, multi-horizon textual information appears to help, where it does not, and what this implies for future research on LLM-driven portfolio allocation.

1.2 Problem Statement

We study a weekly portfolio allocation problem over a fixed universe of liquid U.S. equities. At each decision week t , the agent observes a feature vector constructed from (i) portfolio state (current weights and cash), (ii) optionally technical indicators derived from daily OHLCV data, (iii) optionally structured LLM signals generated from earnings calls, SEC filings, news, and technical commentary, and (iv) global risk-related features. The agent then chooses an action—a target allocation over the risky assets plus cash—which is executed by a transaction-cost-aware environment. Between t and $t + 1$, asset prices evolve according to the historical data; at $t + 1$ the environment computes the portfolio value, a scalar reward that combines log return and risk

penalties, and the next observation.

Formally, we model this as an episodic, partially observable decision problem with a fixed sequence of weekly decision dates. The environment dynamics are derived from historical prices and engineered features; the reward function combines return with endogenous (behavioural) and exogenous (information-based) risk components, as detailed in Chapter 3. We do not assume that LLM signals are inherently superior to price-based indicators; rather, we treat them as an additional, potentially noisy information channel.

Within this setting, the practical questions are:

- How does a PPO-based agent, operating on a structured observation space that includes LLM-derived signals, perform in terms of risk-adjusted returns when compared against strong non-LLM baselines such as equal-weight and minimum-variance portfolios?
- When multi-source textual information is added—via leakage-safe retrieval windows and structured analyst schemas—does it materially alter the learned trading behaviour, risk profile, or robustness, regardless of whether headline performance improvements are large?
- How do alternative integration mechanisms for LLM-derived information (purely as additional observations versus as risk-sensitive penalties in the reward) interact with the finite data horizon, transaction costs, and PPO hyperparameters in terms of training stability and out-of-sample behaviour?
- How does the training mechanism itself—in particular, curriculum-style staged introduction of modalities versus all-in-one multimodal training— affect optimisation dynamics and the eventual trade-off between return, risk, and turnover?

These research questions are studied under realistic constraints. The historical sample consists of a single finite weekly window (2019–2025) with uneven and often sparse textual coverage across tickers; many decision dates have no associated earnings calls or filings in their look-back windows, and the raw disclosures themselves can be noisy or only loosely related to short-horizon returns. The underlying market is also markedly non-stationary over this period, spanning pre-COVID conditions, the 2020 crash, subsequent monetary tightening, and the recent AI-driven rally; any learned policy, whether price-only or LLM-enhanced, must implicitly average over shifting regimes rather than a stable environment. On the modeling side, the LLM pipeline is implemented using cost-constrained access to GPT-4o-mini, which limits both the number of prompts that can be issued and the breadth of hyperparameter or model-variant sweeps that are feasible. Moreover, because GPT-4o-mini is a frontier model trained on large-scale web and financial corpora, it may internally encode information about parts of the 2019–2025 win-

dow that are “in the future” relative to a given decision date. The system mitigates explicit look-ahead by only retrieving documents whose timestamps fall within leakage-safe windows, and by prompting the model to summarise supplied documents rather than to recall external knowledge, but some residual entanglement between pre-training knowledge and the backtest horizon cannot be ruled out. Within these constraints and caveats, the goal of this thesis is not to claim a universally superior trading strategy, but to provide a carefully instrumented case study of how multi-genre, multi-horizon LLM-derived signals interact with a risk-aware RL allocator, with an emphasis on controlled ablations and honest analysis of both positive and negative findings.

1.3 Research Questions and Objectives

Guided by the above problem formulation, the thesis is organised around three research questions. Each is grounded in the literature reviewed in Chapter 2 and instantiated in the experimental programme of Chapter 4.

RQ1: LLM signals as observation features. How does multi-genre, multi-timeframe textual information, when distilled into structured LLM-derived signals, affect an RL portfolio allocator relative to price-only inputs? We start from a price-only backbone (portfolio features, technical indicators, and global factors) and incrementally add signals from individual Tier 1 analysts (news, earnings calls, SEC filings, and technical commentary) and from a Tier 2 “senior” analyst that aggregates them. The objective is to measure the marginal impact of each information source and aggregation level on test performance (NAV multiple, Sharpe ratio, drawdown) and on qualitative behaviour (turnover, concentration), while keeping the environment and training protocol fixed.

RQ2: LLM-derived risk scores for risk-aware RL. Building on risk-aware and constrained RL methods that incorporate coherent risk measures or explicit cost criteria [10–13], we ask whether LLM-extracted risk assessments can usefully shape the reward. In our design, the reward combines weekly log return with (i) an endogenous risk term that summarizes recent downside behaviour, and (ii) exogenous risk penalties derived from LLM risk scores and market turbulence indicators. We systematically vary the strength and source of these penalties to examine whether they produce more downside-averse policies, how they trade off return versus risk, and whether they improve or degrade out-of-sample robustness.

RQ3: Training mechanisms and curriculum-style feature introduction. Finally, we investigate training mechanisms that control how and when high-dimensional LLM features are introduced to the agent. Besides a standard “all-in-one” setup where all modalities are available from the start, we consider curriculum-style schedules in which the policy first learns on a simpler observation space (e.g., prices and technical indicators only) before gradually incorporating longer-horizon textual signals. The goal is to understand whether such curricula help PPO find more stable or interpretable solutions, and how sensitive the resulting agents are to training budgets and hyperparameters, given our limited data horizon.

Across these three research questions, the thesis pursues the following objectives:

- to design and document a reproducible weekly portfolio environment and PPO-based training pipeline that can accommodate multi-source LLM signals;
- to provide a careful empirical assessment of when LLM-derived information offers incremental value over strong price-based and classical baselines, and when it does not;
- to extract methodological lessons—about feature design, reward shaping, training schedules, and evaluation practice—that can inform future work on LLM-driven decision systems in finance and related domains.

1.4 Contributions and Thesis Organisation

This thesis makes contributions along three main axes: an instantiated environment and pipeline, an empirical assessment of LLM integration against strong baselines, and a set of design lessons and diagnoses that connect the results to broader financial theory and LLM–RL practice.

C1: A weekly, risk-aware RL portfolio environment with multi-source LLM signals. The first contribution is a reproducible weekly portfolio environment and training pipeline that combines: (i) a 20-asset U.S. equity universe over 2019–04–20 to 2025–09–14; (ii) a structured observation space including portfolio state, technical indicators, global risk features, and multi-genre LLM-derived signals from earnings calls, SEC filings, news, and technical commentary; and (iii) a risk-aware reward that penalises endogenous downside risk and exogenous LLM-based risk exposures under realistic transaction costs. The environment is implemented with a PPO backbone and carefully instrumented to log trading behaviour (turnover, concentration, cash usage) as well as standard risk–return metrics, enabling controlled ablations and robustness checks.

C2: An empirical case study of LLM integration versus strong baselines. The second contribution is an extensive empirical study of how LLM-derived signals affect a weekly RL allocator relative to both classical baselines (equal-weight and minimum-variance) and a tuned price-only PPO agent. Across RQ1–RQ3, the experiments show that: (i) the price-only PPO agent consistently matches or modestly outperforms the classical baselines on test Sharpe and terminal NAV, with comparable drawdowns; (ii) adding LLM-derived observation features yields mixed trading performance: some configurations remain competitive with equal-weight, but none reliably exceed the price-only PPO once transaction costs and non-stationary regimes are accounted for; and (iii) LLM-based risk shaping slightly improves tail behaviour and encourages more defensive allocations, but again does not deliver large, robust gains over the price-only backbone. These findings are interpreted in light of semi-strong-form market efficiency in deep U.S. large-cap markets and the non-stationarity of the 2019–2025 period: public textual information is largely reflected in prices, and any residual edge from LLM-compressed signals is small, fragile, and easily eroded by regime shifts and trading frictions.

C3: Design lessons for LLM–RL integration and behavioural analysis. The third contribution is a set of design takeaways for LLM-driven portfolio agents and, more generally, LLM–RL systems: (i) a *baseline-first, guarantee-seeking* stance in which price-only PPO and classical portfolios serve as anchors for evaluating any LLM augmentation; (ii) an architectural pattern that treats LLMs as structured information providers—via carefully designed schemas and leakage-safe retrieval—while leaving control and constraint enforcement to a separate RL layer; and (iii) an evaluation methodology that looks beyond aggregate performance to behavioural diagnostics such as turnover, Herfindahl–Hirschman concentration, and regime-specific allocation patterns. In particular, the thesis documents distinct qualitative behaviours across integration mechanisms: for example, SEC-augmented agents that tilt more aggressively into AI-related names after mid-2025, and all-in-one multi-modal agents that become highly active and concentrated when trained for longer horizons, consistent with overfitting in a finite, non-stationary backtest. These observations provide concrete examples of how LLM integration can change trading style without delivering commensurate risk-adjusted outperformance.

Thesis organisation. The remainder of the thesis is organised as follows. Chapter 2 surveys prior work on RL for trading and portfolio allocation, risk-aware and constrained RL, financial LLMs, and LLM-based trading agents, and positions this thesis with respect to those strands. Chapter 3 describes the weekly portfolio environment, the risk-aware reward formulation, the

PPO policy architecture, and the multi-source LLM analyst pipeline, including leakage-safe document retrieval and structured schemas. Chapter 4 presents the experimental programme and results for RQ1–RQ3, covering price-only baselines, LLM-derived observation features, risk shaping with LLM-based exposures, and curriculum versus all-in-one multimodal training, along with detailed trading-behaviour analysis. Chapter 5 synthesises the empirical findings, discusses limitations and threats to validity (including regime non-stationarity and the use of pre-trained LLMs that may encode future information), outlines directions for future work, and includes the overall concluding remarks of the thesis.

CHAPTER 2

RELATED WORK

This chapter reviews four strands of literature most relevant to our problem: (i) reinforcement learning (RL) for trading and portfolio allocation, (ii) risk-aware and constrained RL, (iii) large language models (LLMs) in financial analysis, and (iv) LLM-augmented decision-making and trading agents. We conclude with a gap analysis that motivates the design choices in this thesis: a weekly, risk-aware RL portfolio allocator driven by multi-source, multi-horizon LLM-derived signals.

2.1 Reinforcement Learning for Trading and Portfolio Allocation

2.1.1 From classical portfolio theory to sequential decisions

Modern portfolio theory originates from Markowitz’s mean-variance formulation, which casts portfolio selection (given return beliefs) as a single-period trade-off between expected return and return variance, characterized by the efficient frontier. A key implication is that effective diversification depends on cross-asset covariances: because asset returns are intercorrelated, portfolio risk is governed by the correlation structure[1]. Risk-adjusted performance measures such as the Sharpe ratio operationalize the trade-off between excess return and volatility over a backtest horizon [2]. Subsequent factor models, notably the three-factor model of Fama and French, attribute cross-sectional variation in average returns to market, size (SMB), and value (HML) factors beyond the market portfolio [14].

Alongside these portfolio and factor models, the efficient market hypothesis (EMH) provides a theory of how quickly information is incorporated into prices. In its semi-strong form, EMH posits that all publicly available information—including regulatory filings, earnings announcements, and news—is rapidly reflected in security prices, especially in deep, liquid large-cap markets [15]. Under semi-strong EMH, systematic excess returns from trading purely on public textual information should be small, unstable, and easily eroded by transaction costs and regime shifts. This perspective underpins the interpretation in Chapters 4–5, where LLM-extracted sig-

nals rarely deliver large, robust improvements over a strong price-only PPO backbone.

These formulations are mathematically tractable and widely adopted, but largely static. They either treat each rebalancing date as a separate mean–variance optimization, or rely on heuristic rolling-window procedures. Transaction costs, path-dependent risks (e.g., drawdowns), and non-stationary dynamics are typically handled outside the core optimization. In contrast, RL provides a sequential decision-making framework that directly optimizes long-horizon objectives under uncertainty: an agent repeatedly observes state, chooses actions, and receives rewards, learning a policy that maps states to actions.

Policy-gradient methods such as Proximal Policy Optimization (PPO) are commonly used for continuous-control tasks such as portfolio weight selection due to their stability and implementation simplicity [3].

2.1.2 Deep RL for trading and portfolio allocation

A substantial body of work applies deep RL to trading and portfolio management. Many studies frame single-asset or small-basket trading as an RL problem where states comprise price and technical indicators, actions specify position or trade directions, and rewards reflect profit-and-loss (PnL) or risk-adjusted returns. Open-source frameworks have helped standardize data pipelines, environments, and agents. For example, FinRL provides a modular deep-RL library for automated stock trading and portfolio allocation, including data preprocessing, environment wrappers, and a suite of DRL algorithms [4]. FinRL-Meta extends this ecosystem with more realistic market environments and benchmark tasks, with an emphasis on reproducibility and bias control [5].

For portfolio allocation specifically, Jiang et al. propose an end-to-end RL architecture that maps recent price and volume tensors into portfolio weights (EIIE), explicitly modeling transaction costs and demonstrating improved returns over heuristic baselines [16]. Ye et al. argue that portfolio management requires heterogeneous signals and introduce state-augmented RL (SARL), where RL states include outputs of auxiliary asset movement prediction models alongside price-based features [17]. These works illustrate two trends: (i) learning portfolio weights directly, and (ii) enriching RL state with auxiliary predictors.

Despite rapid progress, the dominant pattern in RL-for-portfolio research remains:

- Observation spaces are primarily numerical, built from prices, volumes, and technical indicators; textual information is often absent or reduced to coarse sentiment scores.
- Decision frequencies are typically daily or intraday; weekly rebalancing over multi-year

horizons is less explored.

- Evaluation focuses on aggregate return metrics (e.g., mean return, Sharpe, maximum drawdown), with limited analysis of policy behaviour, risk posture, and robustness to non-stationarity.

The environment developed in this thesis differs along all three axes: it operates on weekly rebalancing over a multi-year U.S. equity history, integrates LLM-derived signals from multiple textual sources, and provides rich behavioural logging to study learned policies beyond scalar performance metrics.

2.2 Risk-Aware and Constrained Reinforcement Learning

Classical RL is typically *risk-neutral*: a policy π_θ is chosen to maximise the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right], \quad (2.1)$$

where r_t is a scalar reward and $\gamma \in (0, 1]$ is a discount factor. This objective only depends on the *mean* of the return distribution, and does not distinguish policies that have the same expected return but very different variability or tail behaviour. In many safety-critical or financially-constrained applications, this is insufficient: practitioners care about the probability and severity of bad outcomes (large negative returns, catastrophic failures, or long drawdowns), not only the average.

Risk-aware and constrained RL extend the basic formulation by introducing explicit *risk functionals* and/or *constraints* on random returns or costs. This subsection briefly reviews the concepts that motivate our reward design, without claiming to implement the full algorithms in this thesis.

2.2.1 Risk measures for sequential decision making

Let $G(\pi)$ denote a random performance measure under policy π ; for example, the (discounted) cumulative *loss* along a trajectory. A *risk measure* is a functional $\rho(G)$ that maps the distribution of G to a real number representing risk. A particularly important class are *coherent risk measures*, which satisfy monotonicity, subadditivity, positive homogeneity, and translation invariance. Intuitively, these axioms encode that (i) strictly worse outcomes should not be rated as less risky, (ii) diversification should not increase risk, (iii) scaling the position scales risk, and

(iv) adding a sure loss increases risk additively.

For a loss random variable L , Conditional Value-at-Risk (CVaR) at level $\alpha \in (0, 1)$ is a canonical coherent risk measure [10]. $\text{CVaR}_\alpha(L)$ can be interpreted as the *expected loss conditional on being in the worst α fraction of outcomes*. Unlike variance, which penalises upside and downside symmetrically, CVaR focuses entirely on the tail. This makes it attractive for RL problems where rare but severe failures are more important than small fluctuations. Tamar et al. show how policy-gradient methods can be extended to optimise coherent risk measures such as CVaR directly in Markov decision processes [12].

In this language, risk-sensitive RL can be viewed as replacing the risk-neutral objective $J(\theta)$ with a risk functional, for example

$$J_{\text{risk}}(\theta) = -\text{CVaR}_\alpha(L(\pi_\theta)) \quad \text{or} \quad J_{\text{mean-risk}}(\theta) = \mathbb{E}[G(\pi_\theta)] - \lambda \rho(G(\pi_\theta)), \quad (2.2)$$

where $\lambda > 0$ controls the trade-off between mean performance and risk.

2.2.2 Risk-constrained RL and safe policy optimisation

A complementary approach is to keep the reward and risk signals *separate* and impose constraints on the latter. This leads to *risk-constrained MDPs*, where we introduce an auxiliary cost signal $c_t \geq 0$ (for example, a per-step risk proxy) and define

$$J_r(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right], \quad (2.3)$$

$$J_c(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T-1} \gamma^t c_t \right]. \quad (2.4)$$

The design goal is then

$$\max_{\theta} J_r(\theta) \quad \text{subject to} \quad J_c(\theta) \leq d, \quad (2.5)$$

where d is a risk budget. Chow et al. formulate percentile and CVaR-based constraints in this framework and derive policy-gradient and actor–critic methods for such constrained MDPs [11]. Achiam et al. propose Constrained Policy Optimisation (CPO), which modifies the trust-region update used in TRPO to enforce a first-order approximation of the risk constraint at each policy update [13]. At a high level, CPO maintains separate value functions for rewards and costs, estimates the gradients $\nabla_{\theta} J_r$ and $\nabla_{\theta} J_c$, and solves a small quadratic program at each iteration to choose a policy update direction that improves reward while respecting the cost constraint up

to approximation error.

A simpler and widely used alternative is *risk-aware reward shaping*, in which risk is incorporated directly into the scalar reward. Given a per-step risk signal c_t and trade-off parameter $\lambda > 0$, one defines a shaped reward

$$\tilde{r}_t = r_t - \lambda c_t, \quad (2.6)$$

and applies standard policy-gradient or actor–critic methods to $\tilde{r}_t$. In this case there is no explicit constraint; instead, risk aversion is encoded in the reward itself. Tamar et al. discuss such mean–risk trade-offs in terms of coherent risk measures [12], and many applied RL systems (including financial ones) use variants of this pattern (e.g., return minus volatility, or return minus downside penalties) because it is straightforward to implement with off-the-shelf algorithms such as PPO.

2.2.3 Implications for portfolio RL

For portfolio allocation, these ideas translate naturally:

- The *reward* r_t can be a function of log return or PnL.
- The *risk signal* c_t can be based on tail-sensitive quantities such as CVaR-like estimates, downside deviation, loss frequency, or exposure to external risk factors.
- One can either (i) integrate c_t into the reward via shaping, or (ii) treat it as a separate cost and aim to satisfy a constraint $J_c(\theta) \leq d$.

This thesis follows the first route. We adopt a *risk-aware reward shaping* design in which the per-step reward combines log return with penalties derived from endogenous risk statistics and exogenous LLM-based risk signals. We draw on the risk-sensitive RL literature to justify using tail-focused metrics and explicit penalty terms, but we deliberately keep the optimisation algorithm itself close to standard PPO.

2.3 Large Language Models in Financial Analysis

2.3.1 Classical textual analysis in finance

Textual finance research established that news and regulatory disclosures contain predictive information about returns, volatility, and trading volume. Tetlock shows that media pessimism predicts short-horizon stock market movements and reversion, demonstrating that aggregated

news tone is informative for returns [18]. For regulatory filings, Loughran and McDonald argue that generic sentiment dictionaries misclassify many financially neutral or positive terms as negative, and propose finance-specific lexicons tailored to 10-K and 10-Q filings [19]. These works justify the premise that financial text is informative, but also warn that naive NLP pipelines can mismeasure sentiment and risk.

2.3.2 Domain language models and financial LLMs

Transformer-based pretraining led to domain-adapted language models for financial text. FinBERT adapts BERT to financial communications and achieves improved sentiment classification performance on earnings and news corpora [6]. BloombergGPT extends this idea to a 50B-parameter model trained on a mixture of proprietary financial and general-domain corpora, showing strong performance across a range of financial NLP benchmarks [7]. In parallel, the AI4Finance community introduced FinGPT, an open-source family of financial LLMs built via a data-centric pipeline that curates Internet-scale financial text and employs lightweight fine-tuning [8]. FinGPT aims to democratize access to financial LLMs and demonstrates applications such as robo-advising and sentiment-based trading.

Survey work by Li et al. provides an overview of LLMs in finance, including applications to information extraction and summarization for earnings releases and filings, sentiment and event classification for news and social media, question answering, and code generation for quantitative workflows [9]. They emphasize the trade-off between general frontier models accessed via prompting and domain-specific models obtained via fine-tuning, and highlight the need for domain-grounded evaluation protocols.

2.3.3 Limitations for sequential portfolio decisions

Most FinLLM studies treat models as stand-alone predictors or explainers. LLM outputs are typically converted into numerical features (e.g., sentiment scores, estimated earnings changes) and fed into traditional predictive models or simple trading rules. The coupling between LLM-derived signals and sequential portfolio decisions is either indirect or absent. Furthermore, temporal alignment and causality are non-trivial: disclosures arrive at irregular intervals, and their effects can be delayed or persistent over weeks or quarters. Mapping documents to decision dates without look-ahead bias requires careful design, particularly in a weekly RL setting where decisions are spaced further apart than news arrivals. These challenges motivate a structured, leakage-safe interface between LLM outputs and RL states and rewards.

2.4 LLM-Augmented Decision-Making and Trading Agents

2.4.1 General agentic patterns

LLM-based agents extend static prompting by allowing models to reason, act, and interact with tools and other agents. ReAct formalizes an interaction pattern that interleaves natural-language “thoughts” and actions, enabling LLMs to plan while querying external tools or environments [20]. AutoGen generalizes this idea into an engineering framework for multi-agent conversation and coordination, in which LLM-based agents with different roles collaborate via message passing and tool use [21]. These patterns—tool-augmented reasoning and role-specialized multi-agent systems—have influenced many applied designs beyond finance.

2.4.2 Finance-oriented LLM agents

In finance, LLM agents are often designed to emulate research and trading workflows. FinMem proposes a trading agent with layered memory and character/profiling components to improve decision consistency over time [22]. FinAgent introduces a multimodal foundation agent for financial trading that integrates textual and numerical inputs via tools and specialized modules [23]. TradingAgents presents a multi-agent framework with specialized analyst roles (e.g., fundamental, sentiment, technical) and risk management components that synthesize agent debates into trading decisions [24]. FinCon further emphasizes hierarchical manager–analyst communication and a verbal reinforcement mechanism to update beliefs across episodes [25].

These systems demonstrate that LLMs can be embedded inside agentic architectures that resemble real-world investment processes: analysts generate views and rationales, higher-level agents aggregate and arbitrate among them, and an execution layer translates decisions into trades. Empirically, they show encouraging backtest performance and provide qualitative interpretability via natural-language rationales.

2.4.3 Limitations for weekly portfolio allocation

Despite their promise, current finance-oriented LLM agents leave several gaps relative to the weekly RL portfolio problem studied here:

- Many systems are evaluated on single-asset or small-basket trading, or at daily/intraday frequencies. Weekly portfolio rebalancing over a broad equity universe is less explored.

- The mechanism by which textual information is integrated varies widely (direct action suggestions, intermediate signals, or post-hoc portfolio optimization), making it hard to attribute performance gains to specific information sources or timeframes.
- Risk control is often heuristic; few systems combine LLM-derived risk assessments with principled risk-aware RL objectives or constraints.

This thesis adopts a simpler but more controlled design: LLMs are used solely as signal-extraction modules that produce structured numeric features and risk assessments, which are then fed into a conventional RL agent (PPO) operating in a fixed weekly environment. This separation enables clearer ablations over sources, horizons, and integration mechanisms.

2.5 Benchmarks, Competitions, and Evaluation Pitfalls

Because financial RL is vulnerable to data leakage, non-stationarity, and evaluation artefacts, benchmark environments and standardized protocols are increasingly important. FinRL and FinRL-Meta move toward this goal by providing configurable market environments, unified data pipelines, and reproducible DRL baselines for both trading and portfolio management tasks [4, 5]. These frameworks emphasize realistic frictions (e.g., transaction costs, slippage) and highlight issues such as survivorship bias and overfitting in naive backtests.

Beyond static libraries, the FinRL community has also organized a series of contests that showcase current practice in financial RL and LLM-driven signal engineering. Wang et al. describe GPU-optimized parallel market environments used in recent FinRL contests and provide standardized pipelines for large-scale sampling and backtesting [26]. Contest solutions illustrate emerging design patterns: Grover’s FinRLlama solution focuses on LLM-engineered signals and prompt-based feature construction for the LLM-signal challenge [27], while FLAG-Trader fuses LLM agents with gradient-based RL for financial trading and demonstrates how LLM-derived signals can be tightly coupled with policy optimization [28]. These developments reinforce the need for reproducible environments, clear separation between signal engineering and policy learning, and careful ablations when evaluating LLM-enhanced RL strategies.

2.6 Gap Analysis and Positioning of This Thesis

2.6.1 Gaps across RL-for-finance, FinLLM, and agentic systems

The literature reviewed above suggests that RL for trading, financial language models, and LLM-based agents have all progressed rapidly, but in different directions.

First, most portfolio-RL studies focus on price- and volume-based features and relatively short decision horizons. Deep RL architectures such as EIIE/PVM learn portfolio weights directly from technical indicators [16], and subsequent work augments price-based state with auxiliary predictions or sentiment, often at daily or even higher frequency [17]. Libraries such as FinRL and FinRL-Meta standardise environments and baselines for both trading and portfolio allocation [4, 5], but the default benchmarks remain dominated by technical features and daily trading calendars. There is relatively little work that systematically studies *weekly* portfolio rebalancing driven by slower-moving, disclosure-oriented information.

Second, textual finance and financial LLMs establish that news, earnings calls, and regulatory filings contain rich predictive signals, but they are usually evaluated on static NLP tasks (classification, regression, QA) rather than sequential allocation. Classical studies show that media tone and 10-K language correlate with returns and volatility [18, 19], and FinBERT exemplifies domain adaptation of transformer models for financial sentiment [6]. More recent FinLLMs such as BloombergGPT and FinGPT scale this idea to large models and curated financial corpora [7–9]. However, these systems typically output predictions or labels that are either fed into separate portfolio optimisers or evaluated in isolation; they rarely appear inside a unified RL environment where textual information is consumed at multiple horizons and directly shapes a trading policy.

Third, LLM-based trading agents demonstrate promising results but under evaluation settings that differ substantially from a weekly portfolio allocator. FinMem, FinAgent, TradingAgents, and FinCon all employ multi-role or hierarchical LLM agent architectures for trading workflows, combining fundamental, sentiment, technical, and risk perspectives in single-stock or small-basket settings [22–25]. FLAG-Trader further integrates LLM agents with gradient-based RL for financial trading [28], and FinRLlama showcases LLM-engineered signals in the FinRL contest series [27]. While these works are important demonstrations, they typically (i) operate at daily frequency, (ii) mix information sources without clean ablations by source and timeframe, and (iii) evaluate under heterogeneous environments and metrics. This makes it hard to isolate how specific textual sources and horizons contribute to an RL allocator under a fixed

decision calendar.

Fourth, risk-aware RL provides principled formulations for tail-sensitive objectives and constraints, but most financial RL applications still treat risk primarily as an ex-post evaluation criterion (Sharpe ratio, drawdown) instead of as a first-class signal inside the RL objective. Coherent risk measures such as CVaR [10] and risk-constrained MDP formulations [11–13] show how policies can be optimised under explicit risk objectives or constraints, yet there is limited work that couples learned, information-derived risk signals (for example, from textual analysis) with these risk-aware RL principles in a multi-asset portfolio setting.

Finally, evaluation pitfalls remain a central concern. The Probability of Backtest Overfitting (PBO) and Deflated Sharpe Ratio (DSR) quantify how multiple testing and non-normal returns can distort performance estimates [29, 30], and practitioner-oriented treatments emphasise cross-validation and robust experimental design [31]. Parallel market environments and contest infrastructure such as those described by Wang et al. [26] highlight best practices for large-scale evaluation, but many LLM-for-finance demonstrations do not clearly separate signal engineering, policy learning, and hyperparameter search. This amplifies the need for controlled, ablation-oriented studies rather than purely end-to-end claims.

2.6.2 Positioning and scope of this thesis

Against this backdrop, the present thesis is deliberately scoped as a controlled study of how multi-genre, multi-timeframe LLM-extracted information interacts with an RL portfolio allocator under a weekly decision calendar, rather than as a new universal trading system or a new RL algorithm. The positioning can be summarised along four design axes:

1. **Weekly, multi-asset portfolio environment.** We construct a weekly rebalancing environment over a diversified equity universe with transaction costs and realistic market frictions, following the DRL-for-trading tradition [4, 5, 16] but aligning the decision cadence with slower-moving fundamental and disclosure signals. This provides a fixed testbed in which all subsequent ablations are run.
2. **Structured LLM signals as observations and risk proxies.** Building on textual finance and FinLLM work [6–9, 18, 19], we design a multi-stage LLM analyst pipeline that converts heterogeneous text (news, earnings calls, SEC filings, and technical commentary) into structured per-ticker signals and risk assessments. These are integrated into the RL observation space and into a risk-aware reward as endogenous and exogenous risk components. The focus is not on pre-training a new FinLLM, but on how to interface existing

models with an RL allocator in a systematic way.

3. **Risk-aware reward shaping, not full constrained optimisation.** Inspired by risk-sensitive RL [10–13], we adopt a risk-aware reward shaping approach: the per-step reward combines log return with penalties derived from endogenous downside statistics and LLM-based risk exposures. We intentionally keep the optimisation algorithm close to standard PPO [3], leaving fully constrained policy optimisation (e.g., CPO-style updates with explicit risk budgets) as future work. This design keeps the algorithmic layer simple and makes it easier to attribute differences in behaviour to information and reward design rather than to changes in the RL algorithm itself.
4. **Ablation-first evaluation under a unified protocol.** Informed by contest-style benchmarking [5, 26], we fix a single environment and training protocol and organise experiments into targeted families: (i) adding/removing information sources and timeframes, (ii) toggling LLM-based risk shaping, and (iii) comparing curriculum-style versus all-in-one feature introduction. The goal is to produce coherent evidence on *when* and *how* textual information is useful for weekly portfolio allocation, even if the overall performance level is modest and far from production-grade.

Within this scope, the thesis should be read as an empirical, RL-for-finance contribution that bridges existing strands of work rather than as a definitive solution to LLM-driven trading. It aims to clarify design trade-offs and failure modes that are likely to be relevant for future, more sophisticated LLM–RL systems in multi-asset portfolio settings.

CHAPTER 3

PROBLEM FORMULATION AND METHODOLOGY

This chapter formalises the weekly portfolio allocation problem studied in this thesis and describes the full methodology used to couple reinforcement learning (RL) with large language model (LLM) signals. Section 3.1 introduces the problem setup and notation. Section 3.2 describes the market environment and data construction. Section 3.3 specifies the Markov decision process (MDP), action parameterisation, and observation design. Section 3.4 presents the two-tier LLM architecture used to extract structured signals from heterogeneous financial information. Section 3.5 defines the risk-aware reward, including endogenous and LLM-derived risk components. Finally, Section 3.6 summarises the training protocol and explains how the experimental program in Chapter 4 instantiates the three research questions.

3.1 Problem Definition

The overarching question in this thesis is:

Can large language models provide informative, structured signals and risk scores from multiple textual sources (e.g., news, earnings-call transcripts, SEC filings) that help a reinforcement learning (RL) agent make better portfolio allocation decisions than (i) classical allocation rules such as equal-weight and minimum-variance, and (ii) a price-only RL agent?

This question is broken down into three research questions that organise the rest of the thesis:

1. **RQ1: LLM signals as information.** When leakage-safe, schema-constrained LLM signals from SEC filings, earnings calls, and news are added on top of prices and technical indicators, how does a PPO-based portfolio agent perform relative to strong non-RL baselines such as equal-weight and minimum-variance, and relative to a price-only PPO baseline? Do the text-derived features materially change test-time NAV, Sharpe ratio, or drawdowns?
2. **RQ2: LLM-informed risk shaping.** Given multi-source textual information, does using LLM-derived risk assessments as explicit penalties in the reward, rather than solely

as additional observations, materially alter the learned trading behaviour, risk profile, or robustness of the RL agent? In particular, can LLM-based risk scores help an RL policy avoid portfolios that accumulate downside risk, especially in turbulent market regimes?

3. **RQ3: Curriculum vs. all-in-one integration.** How do alternative integration mechanisms for LLM-derived information — exposing the full multimodal state from the start versus introducing modalities sequentially via curricula (e.g., SEC/EC first, then news/technical, or the reverse) — interact with the finite weekly data horizon, transaction costs, and PPO hyperparameters in terms of training stability and out-of-sample behaviour?

To study these questions, the thesis instantiates a concrete weekly portfolio allocation problem. The agent allocates wealth across a fixed universe of $N = 20$ large-cap U.S. equities and a cash position over the October 2023–early September 2025 backtest window. At each weekly decision date, the agent observes a structured state summarising portfolio holdings, prices, technical indicators, and text-derived signals aligned on a canonical (ticker, week) panel, chooses new portfolio weights, pays proportional transaction costs, and holds the resulting portfolio for one week. The decision rule is implemented as a PPO-based RL policy, and strong non-LLM baselines (equal-weight, minimum-variance) are used as external comparators.

A key modelling choice is that the large language model is used purely as a frozen generic summariser. Heterogeneous textual sources — SEC filings, earnings calls, and news — are aligned to decision weeks and passed through a multi-agent LLM pipeline (based on a model such as gpt-4o-mini) that produces bounded numeric signals (direction, risk, confidence) and short summaries. These LLM outputs are pre-computed for the whole sample and then treated as fixed features: the RL agent is trained only on realised returns and never fine-tunes or adapts the LLM itself.

The remainder of this chapter makes this setup precise. Section 3.2 details the market environment, asset universe, and data construction. Section 3.3 formalises the RL environment, including the observation design, action parameterisation, and transition dynamics. Section 3.4 describes the multi-agent LLM pipeline that produces leakage-safe, structured signals, and Section 3.5 defines the risk-aware reward that combines returns, endogenous risk, and LLM-based risk exposures. Section 3.6 then explains the training protocol and how the experimental program in Chapter 4 instantiates RQ1–RQ3 on top of this common framework.

3.2 Market Environment and Data Sources

3.2.1 Asset universe and time horizon

The environment trades a universe of $N = 20$ liquid U.S. equities selected from the S&P 500 based on liquidity and sector representativeness, together with a cash asset. To support technical indicator computation and rolling normalisation, at least one additional year of historical market data prior to the first trading week is included. Weekly decision dates are defined as the last trading day of each week on a canonical U.S. equity-market calendar (NYSE/Nasdaq close).

In the main experimental program, the backtest window for RL training and evaluation spans decision dates from October 2023 to early September 2025, with the terminal portfolio value marked one week after the final decision date. Each RL episode corresponds to traversing a contiguous sequence of weekly decision dates within this backtest window: at each date the agent observes the current state, chooses a new portfolio allocation, pays proportional transaction costs, and then holds the resulting portfolio for one week until the next decision date. The exact chronological split between training, validation, and test within this backtest window is described in Section 3.6.1 and revisited in Section 4.1.1.

3.2.2 Underlying data modalities

The environment is driven by a weekly panel aligned to decision dates, built from the following modalities:

- **Trading data (OHLCV).** Daily open, high, low, close, and volume are used to derive weekly close prices and a set of technical indicators. The indicators are computed on the daily OHLCV series over rolling windows and then sampled at the weekly decision dates, so that each portfolio decision uses only information available strictly before that week’s close.
- **News.** Unstructured news items are aggregated into ticker-specific bundles for each decision week. Coverage is uneven across tickers and time; the pipeline logs article counts and coverage statistics to support robustness analysis. The raw feed consists mainly of headlines plus short, sometimes truncated descriptions rather than full articles, which limits textual depth and introduces additional noise and sparsity in the information available to the LLM analysts.
- **Earnings calls (EC).** Earnings-call transcripts are aligned to the weekly calendar and

treated as medium-horizon fundamental information around quarterly results and management guidance.

- **SEC filings.** 10-K and 10-Q filings are parsed and aligned to the weekly calendar to provide slow-moving structural information about the business, balance sheet, and long-lived risk factors. In practice, the pipeline focuses on the Management’s Discussion and Analysis (MD&A) and Risk Factors sections of each filing, discarding boilerplate and purely legal material to concentrate the LLM analysis on economically meaningful content.

All non-price modalities are pre-processed into per-ticker, per-week bundles before being passed to the LLM pipeline described in Section 3.4.

3.2.3 Technical indicators

From OHLCV data the environment computes a standard set of technical indicators to capture momentum, trend, and volatility dynamics. Typical examples include moving averages and their slopes, MACD, RSI, stochastic oscillators, realised volatility, and turbulence-like measures at the individual ticker level. These indicators are used in two roles:

1. as numeric features directly observed by the RL agent; and
2. as the input to an LLM-based technical analyst that produces compact, semantically labelled technical signals.

All indicators are computed over rolling windows ranging from several weeks to several months and are normalised using trailing-history statistics. Taken together, these features provide a compact characterisation of recent price and volume regimes that is most informative for the technical analyst’s intended 1–8 week horizon described in Section 3.4.2.

3.2.4 Weekly canonical panel and ticker alignment

All information is assembled into a canonical weekly panel defined on the decision dates introduced in Section 3.2.1. Each decision date corresponds to the last trading day of the week on a standard U.S. equity-market calendar and is stored with a UTC timestamp at second-level resolution. All raw events — prices, trades, earnings calls, SEC filings, and news items — carry at least second-level timestamps and are first mapped into the containing decision week before aggregation. The resulting panel has exactly N rows per decision date, one per ticker. Rows are sorted by (decision date, ticker), yielding a stable ticker order that is used consistently across all per-ticker components:

- price and technical-indicator features,
- LLM-derived features (signals, risk scores, confidences, timing metadata).

The same canonical ticker order is also used whenever the RL environment and policy represent per-ticker quantities as vectors or matrices (for example, action vectors and logged holdings or trades), so that the j -th element of any such vector always corresponds to the same ticker across time and across experiments. This is crucial for reproducibility and for interpreting allocation heatmaps and concentration measures. The canonical panel itself is materialised as columnar files (e.g., Parquet) that feed both the LLM extraction pipeline and the RL environment.

3.3 Reinforcement Learning Framework

3.3.1 MDP formulation

We formulate weekly portfolio allocation as an episodic Markov decision process (MDP) over a fixed sequence of decision weeks. For each data split (train, validation, test) we obtain an ordered list of weekly decision dates $\{t_0, t_1, \dots, t_H\}$ from the canonical calendar described in Section 3.2.1. Conceptually, an episode corresponds to traversing this sequence once: at each week t , the agent observes a feature vector, chooses a new portfolio allocation, and receives a scalar reward based on the resulting portfolio performance and risk penalties. During training we run many such episodes in parallel over the training window with a stochastic policy. During evaluation on validation and test windows we replay the same historical paths but apply the learned policy in deterministic mode (no exploration noise), so that the trading trajectory and performance metrics are reproducible given the observation sequence.

Formally, we write the MDP as $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where $\mathcal{S}$ is the set of observations, $\mathcal{A}$ is the set of portfolio allocations, P encodes the portfolio and price dynamics, r is the risk-adjusted reward, and γ is the discount factor. At each decision week t , the environment exposes an observation $x_t \in \mathbb{R}^d$ that summarises the information relevant for rebalancing: portfolio features (current weights and cash), technical indicators derived from daily OHLCV data, and LLM-based signals extracted from earnings calls, SEC filings, and news (see Section 3.3.4). The agent chooses an action $a_t = w_t^{\text{target}}$, a target weight vector over the N risky assets plus cash, constrained to be non-negative and to sum to one. Given x_t and w_t^{target} , the environment executes trades at the close price of the current decision date under proportional transaction costs, updates the portfolio to a new realised weight vector w_t , and advances to the next decision week. In practice, transaction costs and cash-feasibility scaling mean that w_t may differ slightly from

w_t^{target} ; the realised weights w_t are what enter the reward and risk computations in Sections 3.5 and 3.6.

The action space explicitly includes a cash asset alongside the N equities. Allocating more weight to cash reduces exposure to equity risk and allows the agent to learn conservative behaviour in turbulent periods. Apart from this cash channel, the task remains a standard multi-asset allocation problem over a fixed 20-stock universe.

The reward r_t at week t combines the portfolio’s log return over $(t, t+1]$ with endogenous and exogenous risk penalty terms, as detailed in Section 3.5. The learning objective is to find a policy $\pi_\theta(a_t | x_t)$ that maximises the expected sum of weekly rewards over an episode.

For clarity, the main quantities in this formulation are:

- $t \in \{t_0, \dots, t_H\}$: weekly decision dates in a given episode;
- $x_t \in \mathbb{R}^d$: observation vector at week t (portfolio features, technical indicators, LLM-derived signals);
- $a_t = w_t^{\text{target}}$: target portfolio weights over the N assets plus cash, constrained to be non-negative and to sum to one;
- w_t : realised portfolio weights immediately after rebalancing at week t under transaction costs and cash constraints;
- r_t : scalar reward at week t , combining log return and risk penalties;
- $\pi_\theta(a_t | x_t)$: stochastic policy parameterised by θ .

The underlying market and firm state contains many unobserved factors (future earnings, order flow, macro shocks), and the agent only observes the constructed feature vector x_t , which is a deterministic function of recent prices, volumes, portfolio holdings, and LLM summaries. Following common practice in financial reinforcement learning, we treat x_t as an approximate Markov state: the feature design (portfolio features, rolling technical indicators, and windowed LLM signals) is chosen so that x_t captures the recent history most relevant for near-term weekly decisions. The policy $\pi_\theta(a_t | x_t)$ is then learned on this approximate state space using PPO, which optimises the expected cumulative reward under the environment dynamics induced by historical data and the transaction cost model.

3.3.2 Action space: weight-based allocation

Rather than predicting discrete trade quantities, the policy outputs target weights. Let $\mathbf{z}_t \in \mathbb{R}^{N+1}$ denote the raw logits produced by the policy network at time t . These logits are mapped to target

weights via a softmax transformation:

$$\mathbf{w}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau}\right), \quad \sum_{j=0}^N w_{t,j} = 1, \quad w_{t,j} \geq 0, \quad (3.1)$$

where index $j = 0$ corresponds to cash and $\tau > 0$ is a temperature hyperparameter. Optionally, a minimum cash constraint $w_{t,0} \geq \underline{w}_{\text{cash}}$ can be enforced by projection onto the feasible simplex.

This softmax parameterisation ensures that allocations are non-negative and sum to one by construction, while keeping PPO’s optimisation in the unconstrained logit space and avoiding any manual projection or constrained optimisation step. As a result, the policy output admits a straightforward interpretation as percentage portfolio weights over cash and equities.

3.3.3 Trading execution and transaction costs

Given pre-trade holdings and the close prices on the weekly decision date, the environment rebalances the portfolio towards the target weights $\mathbf{w}_t^{\text{target}}$ using a sell-first-then-buy routine with proportional transaction costs. Let $h_{t,i}$ denote the number of shares held in ticker i at decision time t (after executing the previous action a_{t-1} but before applying a_t), and let $p_{t,i}$ denote its closing price on the decision date. Let cash_t denote the cash position. The portfolio net asset value (NAV) at time t is

$$V_t = \text{cash}_t + \sum_{i=1}^N h_{t,i} p_{t,i}. \quad (3.2)$$

At each decision time t the environment proceeds as follows:

1. It computes the dollar target for each asset and for cash from $\mathbf{w}_t^{\text{target}}$ and V_t .
2. Assets that are overweight relative to their dollar targets are sold first, incurring a proportional sell cost c_s per dollar traded.
3. The freed cash is then used to buy assets that are underweight relative to their dollar targets, incurring a proportional buy cost c_b .
4. If necessary, buy orders are scaled down to respect cash feasibility; no borrowing or shorting is allowed.

Fractional shares are permitted, which simplifies the mapping between weights and holdings on the canonical weekly panel. The post-trade holdings define the realised weights w_t used in the reward and risk calculations. Prices then evolve from t to $t + 1$, and the environment computes the realised one-week portfolio return and the corresponding reward r_t based on Section 3.5.

This execution model is deliberately simplified relative to real markets: it assumes instantaneous execution at the weekly close with proportional costs but no explicit liquidity or impact constraints. It is nonetheless sufficient to capture the first-order interaction between turnover, transaction costs, and portfolio performance for the purposes of this thesis.

3.3.4 Observation space and feature organisation

The environment exposes a structured state s_t implemented as a dictionary with four components:

- **Portfolio state** s_t^{port} : current cash weight, current asset weights, and optionally additional portfolio-level diagnostics such as realised volatility or turnover.
- **Valuation state** s_t^{val} : a per-ticker feature matrix that concatenates price-based indicators with LLM-derived signals and optional fundamental variables, all aligned on the canonical weekly panel.
- **Trading state** s_t^{trade} : a rolling history of normalised prices (or log returns) over the last H weeks for each ticker. This state underlies the *trading_mode* configurations in the feature extractor.
- **Global state** s_t^{glob} : optional macro or market-wide features shared across tickers (e.g. index returns, volatility indices).

This factorisation allows the same environment to support multiple ablations by toggling subsets of features, without altering the core transition dynamics. In all experiments, the portfolio state and trading state are always present, while the valuation and global states are selectively enabled to study the incremental impact of technical indicators and LLM-derived signals.

On top of this structured observation, a custom feature extractor maps s_t into a fixed-dimensional fused representation $\mathbf{z}_t$ for use by the PPO policy. The extractor operates in two stages.

Per-ticker temporal backbone (trading_mode). For each ticker i and decision week t , the trading state provides a short history of recent daily normalised prices over a fixed window of length H (the trading-state history length). This one-dimensional sequence is the input to a per-ticker temporal encoder, and the *trading_mode* flag specifies how it is mapped into a fixed-dimensional representation $\mathbf{h}_{i,t}^{\text{trade}}$:

- **mlp**: flatten the last H normalised prices into a vector and pass it through a small multi-layer perceptron, yielding a compact embedding of the recent price path;

- **conv1d**: treat the sequence as a 1D time series and apply a shallow temporal convolutional network (Conv1D) followed by non-linearities and pooling, yielding a translation-insensitive summary of short-horizon price dynamics.

The valuation-state features (technical indicators, LLM-derived signals, and fundamentals) are encoded in parallel into a second embedding $\mathbf{h}_{i,t}^{\text{val}}$. If ticker ID embeddings are enabled, a learned vector $\mathbf{e}_i$ is also included. Concatenating these components yields a unified per-ticker representation

$$\mathbf{z}_{i,t} = [\mathbf{h}_{i,t}^{\text{trade}}, \mathbf{h}_{i,t}^{\text{val}}, \mathbf{e}_i].$$

Cross-ticker aggregation (per_ticker_mode). The per_ticker_mode flag specifies how the set of per-ticker representations $\{\mathbf{z}_{i,t}\}_{i=1}^N$ is aggregated into a single cross-ticker representation $\mathbf{z}_t$:

- **flatten**: concatenate all per-ticker embeddings $(\mathbf{z}_{1,t}, \dots, \mathbf{z}_{N,t})$ in the canonical ticker order into a long vector and project it back to a fixed-size representation before fusing with portfolio and global features;
- **pool**: apply simple mean or max pooling across tickers to obtain a cross-sectional summary of $\{\mathbf{z}_{i,t}\}_{i=1}^N$, which is then fused with portfolio and global features via a small fully-connected layer;
- **attention**: apply a lightweight self-attention block over the per-ticker embeddings to learn ticker-specific interaction weights, then flatten and project the attended tokens to a fixed-size global representation.

In all cases, the result is a single fused feature vector $\mathbf{z}_t \in \mathbb{R}^{d_{\text{fuse}}}$ per decision week, which is passed to the actor–critic heads of the PPO policy (Section 3.3.6). Different choices of trading_mode and per_ticker_mode trade off architectural simplicity against representational capacity; Chapter 4 empirically compares these configurations.

3.3.5 Normalisation and warm-up

Market and LLM-derived features are heterogeneous in scale and exhibit non-stationarity. To stabilise training we apply:

- per-ticker rolling z-score normalisation to dynamic features (prices and technical indicators) computed over a trailing window;
- cross-sectional rank scaling for selected features where relative ranking is more meaningful than absolute magnitude;

- clipping of all numeric features to bounded ranges to avoid extreme outliers.

An initial warm-up prefix is reserved at the beginning of each episode. During this period the environment updates rolling buffers and computes valid normalisation statistics, but no trades are executed and rewards are not used for learning. Trading begins only once all required histories are available.

3.3.6 Policy learning algorithm: Proximal Policy Optimisation

The policy and value functions are trained with Proximal Policy Optimisation (PPO), chosen for its robustness in continuous-control settings and its mature implementation in Stable-Baselines3. PPO is well suited to this task because it directly supports continuous actions (portfolio weights over assets) and uses a clipped objective to stabilise updates in a data-scarce regime.

Let $\pi_\theta(a_t | x_t)$ denote the stochastic policy parameterised by θ , and let $\pi_{\theta_{\text{old}}}$ be the behaviour policy used to generate a batch of trajectories. PPO maximises the clipped surrogate objective

$$L_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right], \quad (3.3)$$

where $r_t(\theta) = \pi_\theta(a_t | x_t) / \pi_{\theta_{\text{old}}}(a_t | x_t)$ is the policy ratio, A_t is the advantage estimate computed from discounted sums of per-step rewards, and $\epsilon > 0$ is a small clipping parameter. The clipping term prevents large, destabilising updates by limiting how far the new policy can move from the old one in a single optimisation step.

The fused feature vector $\mathbf{z}_t$ produced by the feature extractor (Section 3.3.4) is shared by an actor–critic architecture:

- **Actor head.** A small fully-connected network maps $\mathbf{z}_t$ to an unconstrained $(N+1)$ -dimensional vector of logits $\mathcal{L}_t$, with one logit for each risky asset and for cash. A temperature-scaled softmax maps these logits to target portfolio weights $\mathbf{w}_t^{\text{target}}$ on the simplex.
- **Critic head.** A second fully-connected network maps $\mathbf{z}_t$ to a scalar value estimate $V_\phi(x_t)$, used for advantage estimation and as a baseline in the PPO objective.

In this way, the architectural novelty of the thesis is concentrated in the observation design, cross-ticker feature extractor, and risk-aware reward (Sections 3.2–3.5), while the policy learning algorithm itself remains deliberately close to the vanilla PPO implementation provided by Stable-Baselines3.

3.4 LLM-Based Signal Extraction

The reinforcement learning environment in Section 3.3 assumes that, at each weekly decision date, the agent receives a compact feature representation of heterogeneous information: prices, technical indicators, and textual disclosures aligned on the canonical (ticker, week) panel. This section describes how long and noisy textual inputs are converted into stable, bounded signals suitable for RL. The design follows a two-tier analyst architecture with explicitly defined roles, time horizons, and numeric schemas.

3.4.1 Multi-agent LLM architecture

Recent work on agentic LLM systems and multi-agent trading frameworks suggests that decomposing complex decisions into specialised roles can improve robustness and interpretability. Motivated by this, the thesis adopts a lightweight two-tier analyst architecture for weekly equity signals.

We implement a lightweight two-tier analyst architecture:

Tier 1: source-specialised analysts. For each ticker and decision week, a set of Tier 1 analyst agents operates on heterogeneous information channels: earnings-call transcripts, SEC filings, textual news, and a compact bundle of technical indicators. Each analyst is prompted with a source-specific role and is instructed to produce a schema-constrained record rather than free-form prose. Conceptually, each record contains three components:

1. a short structured summary of the analyst’s view;
2. a small set of numeric fields that quantify direction, risk, and confidence;
3. a brief free-text rationale or caveat.

Long documents such as earnings calls and SEC filings are processed via a simple map–reduce pattern. The raw text is first divided into overlapping segments; the model produces chunk-level extractions (e.g., candidate risks and opportunities) for each segment; and a subsequent reduce step merges these into a single document-level summary with basic deduplication and consistency checks. This pattern allows the same prompts to scale from short news items to very long filings while keeping the interface to the RL environment unchanged.

Tier 2: senior aggregation and risk governance. On top of the Tier 1 analysts, a second layer of senior agents operates at the ticker–week level. These Tier 2 agents read the full collection

of Tier 1 records and timing metadata (e.g., days since the last filing or earnings call), and then produce integrated signals and risk assessments. The intended roles mirror an investment committee:

- an integrating analyst that synthesises the various Tier 1 views into a single directional tilt;
- a risk analyst that highlights concentrated or systemic downside risks;
- a skeptic that stress-tests optimistic narratives and points out missing or conflicting evidence;
- a judge that reconciles these perspectives into a consolidated numeric signal and concise commentary.

Only the numeric outputs of Tier 1 and Tier 2 enter the RL system: they appear as part of the valuation state in Section 3.3.4 and, for Research Question 2, as inputs to the LLM-based risk term in the reward (Section 3.5). Textual summaries and rationales are retained only for interpretability and for qualitative analysis in later chapters.

3.4.2 Analyst roles and time horizons

Each Tier 1 analyst is given a clearly scoped information role and time horizon, so that their outputs form a multi-timescale feature stream rather than overlapping heuristics.

SEC analyst (structural, 1–3 years). The SEC analyst reads the Management Discussion & Analysis (MD&A) and Risk Factors sections of 10-K/10-Q filings for a given ticker. The prompt emphasises structural and long-lived aspects of the business: balance-sheet strength, refinancing risk, regulatory and legal overhangs, business-model vulnerabilities, and persistent concentration risks (e.g., customers, suppliers, geographies). The analyst is asked to form a view on the firm’s risk-adjusted structural risks, opportunities and exposures over a 1–3 year horizon.

Earnings-call analyst (fundamental, 1–4 quarters). The earnings-call analyst processes quarterly earnings-call transcripts. Its focus is on recent earnings performance and guidance over the next one to four quarters: revenue and margin trends, segment dynamics, demand signals in management commentary, and credible forward-looking statements from the Q&A. The task is explicitly framed in terms of fundamentals at the quarterly horizon, rather than day-to-day price moves. The analyst condenses the call into a medium-horizon directional view and a correspond-

ing risk assessment that reflects plausible downside mechanisms such as execution slippage or demand shocks.

News analyst (material, weeks to quarters). The news analyst aggregates short news items and headlines associated with a given ticker and decision week. The instructions ask it to identify developments that are material and persistent enough to matter beyond the next few trading days, such as regulatory actions, major customer wins or losses, rating changes, large transactions, or repeated negative coverage. At the same time, the prompt discourages overreacting to isolated, purely short-term events (e.g., a single rumour) unless they fit into a broader pattern. The news analyst reports its view using the same numeric schema as the fundamental analysts, so that news-derived signals can be compared and combined consistently.

Technical analyst (price-volume only, 1–8 weeks). The technical analyst does not see any raw text. Instead, it receives a compact set of technical indicators derived from daily prices and volumes, sampled at the weekly decision dates as described in Section 3.3.4. It is asked to interpret these indicators in terms of short- to medium-horizon price-action regimes (trending, mean-reverting, range-bound, volatile, or unclear) and to output a directional signal and risk assessment that are consistent with the inferred regime. This provides a purely price-based reference channel that can be compared against and combined with the text-based analysts.

In combination, these roles produce naturally multi-timescale information: slow-moving structural SEC signals, medium-horizon earnings-call and news signals, and faster technical signals.

Tier 2 senior agents are then prompted to read all Tier 1 records for a ticker and week, reconcile potentially conflicting views, and separate directional signal from risk in their integrated outputs.

3.4.3 Numeric interface and schemas

To make LLM outputs usable by the RL environment, we design light-weight schemas that enforce a structured numeric interface while preserving traceability back to the underlying text. Each Tier 1 analyst produces, for every (ticker, week, source) tuple, a single JSON record with three main components:

- a small set of numeric fields, typically a signed directional signal $s_{i,t}$ (near-term return view), a non-negative risk score $r_{i,t}$, and a discrete confidence level $c_{i,t}$;

- a short, structured textual summary that captures the main arguments behind the view in a few sentences;
- an optional free-text rationale or caveat where the analyst can record missing information, scenario assumptions, or reasons for low confidence.

The schemas are intentionally minimal: the goal is not to reproduce all nuances of the source documents, but to expose a stable, machine-readable signal that can be aggregated across tickers and over time. Pydantic validation enforces basic type and range constraints (e.g., clipping $s_{i,t}$ to $[-10, 10]$, requiring non-negative $r_{i,t}$, and ensuring that required fields are present), and automatically rejects malformed responses. In the data pipeline we keep the full JSON record, including IDs and timestamps, and log the raw text alongside the structured fields.

On the RL side, only a subset of the numeric outputs of Tier 1 and Tier 2 enter the RL system: in the experiments in Chapter 4, directional signals are used as observation features (valuation state in Section 3.3.4), while selected risk-score fields are used only via the LLM-based risk term in the reward (Section 3.5). Textual summaries and rationales are retained only for interpretability and for qualitative analysis in later chapters.

3.4.4 Prompt design, rubrics, and calibration

To make the LLM outputs both interpretable and suitable for RL, the thesis adopts explicit qualitative rubrics for each analyst role, documented in Appendix A. These rubrics are used in two ways: (i) as prompt-side instructions that steer the models toward the desired behaviour, and (ii) as an ex-post evaluation tool for auditing sample outputs.

For Tier 1 analysts operating on SEC filings, earnings calls, news and technical indicators, the rubrics score each ticker-week output (signal, risk_score, confidence, rationale) along dimensions such as horizon alignment, business anchoring, risk specificity, calibration of signal and risk_score, and rationale quality. For example, the EC rubric checks whether the rationale is clearly framed at a 1–4 quarter horizon and grounded in concrete call content, whether downside mechanisms underlying risk_score are specific, and whether the numeric scores use the $[-10, 10]$ scale in a calibrated way rather than collapsing near zero or at extremes.¹ Similar dimensions are defined for SEC (structural 1–3 year horizon) and technical outputs (regime-consistent mapping from indicators to scores, confidence calibration, and appropriate use of discard behaviour when inputs are unreliable).

¹See Appendix A.1 for the EC rubric and Appendix A.2 for the SEC rubric.

For Tier 2 “investment-committee” agents, separate rubrics evaluate the Business Analyst, Risk Analyst, Skeptic, and Judge roles. The Business Analyst rubric focuses on horizon discipline, business anchoring and evidence use, signal_proposal calibration, thesis quality, and the concreteness of key_drivers. The Risk Analyst rubric emphasises near-term risk focus (1–4 weeks), mechanism-based risk factors, identification of genuine tail events, and calibration of risk_score_proposal. Skeptic and Judge rubrics capture the quality of challenge, consistency checks across sources, and the coherence of the final integrated verdict.

In practice, these rubrics are used in two ways. First, rubric-style instructions are woven into the prompts to steer the models toward the desired behaviour. Second, an LLM-as-judge process is applied on a development subset to audit sample outputs against the rubrics. Prompt wording, guardrails (for example, discouraging next-week price prediction), and numeric scaling instructions are iteratively refined until sample outputs achieve consistently acceptable to strong ratings across the relevant dimensions. This rubric-driven process encourages that the structured LLM signals entering the RL environment are horizon-consistent, business-grounded, and numerically stable, rather than arbitrary scores produced by loosely specified prompts.

3.4.5 Model choice and temporal considerations

The LLM backbone used throughout this thesis is a single, general-purpose commercial model (gpt-4o-mini) accessed via an API. This choice is deliberate: the focus of the work is on how a risk-aware RL portfolio manager can make use of structured LLM-derived signals, not on training or benchmarking financial language models per se. A production-grade generic model offers (i) strong summarisation and information-distillation ability across heterogeneous inputs (news, earnings calls, SEC filings), (ii) stable JSON-style output under schema constraints, and (iii) predictable latency and cost suitable for large-scale offline extraction. No task-specific fine-tuning or in-domain pretraining is performed; the same model and prompts are used for all tickers, weeks, and experiments.

Textual signals are generated in an offline, frozen fashion. For each (ticker, week) point on the canonical panel, the data pipeline selects only documents whose timestamps fall strictly before that week’s decision date, builds modality-specific bundles (news, EC, SEC), and then calls the LLM to produce schema-constrained analyst records. The resulting numeric fields (e.g., directional signal, risk score, confidence) are stored as fixed features and subsequently treated as ordinary inputs by the RL environment. During RL training and evaluation, the LLM is never called, and its parameters remain unchanged; only the policy and value-function parameters are

updated on historical data.

Under this setup, the experiments are best interpreted as evaluating what a risk-aware RL portfolio manager could do if a powerful generic text summariser such as gpt-4o-mini, in its current form, had been available throughout the October 2023–early September 2025 backtest period, rather than as evidence from an actual live deployment of that model over that period. This design cannot remove any temporal bias already present in the pre-trained LLM (for example, knowledge of events that occur after a given decision date), but it does minimise additional leakage: the LLM is not fine-tuned on realised returns, is not iteratively adapted based on backtest performance, and does not see future outcomes as part of the RL loop. The counterfactual interpretation is therefore: given fixed, leakage-safe document cut-offs and a frozen generic summariser, how much incremental value can RL extract from LLM-derived signals when trained only on information available before each portfolio decision?

3.5 Risk-aware reward design

The environment provides a single scalar reward at each weekly decision date that combines (i) scale-free portfolio log returns with (ii) penalties based on both endogenous and exogenous risk. The goal is to encourage policies that achieve high long-run performance while avoiding portfolios that accumulate downside risk, especially in turbulent regimes or when LLM analysts flag elevated firm-level risk.

Base return term. Let V_t denote the portfolio value at decision week t just before the action is applied, and V_{t+1} the value at the next decision week after prices have evolved and trades at week t have settled. The base reward is the weekly log return

$$r_t^{\text{return}} = \log\left(\frac{V_{t+1}}{V_t}\right), \quad (3.4)$$

which is additive across time and invariant to choice of currency scale.

Endogenous risk from recent performance. To measure the portfolio’s own recent downside behaviour, the environment maintains a rolling window of the most recent W weekly log returns $\{r_{t-W+1}^{\text{return}}, \dots, r_t^{\text{return}}\}$ and derives three non-negative risk statistics:

- *Downside deviation* σ_t^- , defined as the standard deviation of negative returns within the window;

- *Tail-loss magnitude* CVaR_t , defined as the average magnitude of the worst $\alpha\%$ negative returns in the window (a Conditional Value-at-Risk over losses);
- *Loss frequency* f_t , defined as the proportion of weeks in the window with $r_\tau^{\text{return}} < 0$.

Each component is normalised by a fixed scale parameter so that values of order one correspond to “typical” downside, and values greater than one represent elevated risk. Writing

$$\tilde{\sigma}_t^- = \frac{\sigma_t^-}{s_{\text{down}}}, \quad \widetilde{\text{CVaR}}_t = \frac{\text{CVaR}_t}{s_{\text{cvar}}}, \quad \tilde{f}_t = \frac{f_t}{s_{\text{freq}}},$$

the endogenous risk index at week t is

$$E_t = w_{\text{vol}} \tilde{\sigma}_t^- + w_{\text{mag}} \widetilde{\text{CVaR}}_t + w_{\text{freq}} \tilde{f}_t, \quad (3.5)$$

with non-negative weights $w_{\text{vol}}, w_{\text{mag}}, w_{\text{freq}}$ controlling the relative importance of volatility, tail losses, and loss frequency. By construction $E_t \geq 0$, and it increases when the portfolio experiences more frequent, more volatile, or more severe losses in the recent window.

Exogenous market and LLM-based risk exposures. In addition to endogenous risk, the environment includes two exogenous signals:

- a *market turbulence* series summarising system-wide stress, and
- an *LLM-based firm-level risk exposure* derived from the analyst pipeline described in Section 3.4.2.

At each week t , let $w_t \in \mathbb{R}^N$ denote the portfolio weights on the risky assets (excluding cash) after rebalancing at that decision. The LLM-based exposure is defined as a portfolio-weighted average of per-ticker risk scores:

$$r_t^{\text{llm}} = \sum_{i=1}^N w_{t,i} \text{llm_risk_exposure}_{t,i}, \quad (3.6)$$

where $\text{llm_risk_exposure}_{t,i}$ is the integrated LLM risk score for ticker i at week t . This raw exposure is normalised by a constant $\text{max_llm_risk_score}$, estimated from the training data, and clipped to the unit interval:

$$z_t^{\text{llm}} = \text{clip}\left(\frac{r_t^{\text{llm}}}{\max(\text{max_llm_risk_score}, \epsilon)}, 0, 1\right), \quad (3.7)$$

with a small $\epsilon > 0$ for numerical stability. Similarly, the market turbulence observable at week t is normalised by its training-split maximum `max_turbulence` to obtain a bounded market stress indicator

$$z_t^{\text{mkt}} = \text{clip}\left(\frac{\text{turbulence}_t}{\max(\text{max_turbulence}, \epsilon)}, 0, 1\right). \quad (3.8)$$

Both z_t^{llm} and z_t^{mkt} therefore lie in $[0, 1]$ and can be interpreted as normalised risk intensities.

Final reward. The final weekly reward subtracts risk penalties from the base log return, scaled by a global factor $\lambda > 0$:

$$r_t = \lambda \left(r_t^{\text{return}} - \beta z_t^{\text{mkt}} E_t - \gamma z_t^{\text{llm}} \right), \quad (3.9)$$

where $\beta \geq 0$ controls the strength of the interaction between market turbulence and endogenous risk, and $\gamma \geq 0$ controls the direct penalty from LLM-flagged firm-level risk. When $\beta = \gamma = 0$, the reward reduces to pure log return; when the coefficients are positive, the agent is incentivised to (i) reduce its own accumulation of downside risk, especially during turbulent markets (through the $\beta z_t^{\text{mkt}} E_t$ term), and (ii) moderate exposure to firms that LLM analysts judge to be especially risky (through the γz_t^{llm} term).

In the experiments for Research Questions 1 and 3 (Chapter 4), the reward is a pure log-return objective: we set $\beta = 0$ and $\gamma = 0$ so that the agent is trained only on r_t^{return} , and comparisons isolate the effect of changing observation features or training curricula under otherwise identical optimisation. Research Question 2 explicitly studies LLM-informed risk shaping: the construction of E_t , z_t^{mkt} and z_t^{llm} is held fixed, while we vary the choice of LLM risk source (news, technical, EC, SEC, or senior) and the coefficient γ , and include ablations in which one or both risk terms ($\beta z_t^{\text{mkt}} E_t$ and γz_t^{llm}) are switched off. This design keeps the reward function simple enough for standard PPO training while explicitly encoding the thesis goal of risk-aware portfolio management informed by both market conditions and LLM-derived signals.

3.6 Training procedure

3.6.1 Train/validation/test protocol

The full weekly sample from April 2019 to early September 2025 is first split chronologically into a training window and a test window using an approximately 70%/30% ratio. An initial warm-up prefix at the start of the training window is used to initialise technical indicators, rolling statistics,

and normalisation constants; no trading decisions in this prefix contribute to the learning signal or evaluation.

Within the training window we further perform a single, fixed *train/validation* split. Concretely, the early part of the training window is used for gradient-based learning, while a short suffix of the same window is reserved as a validation segment on which the current policy is periodically evaluated in a purely deterministic manner (no exploration noise). This validation segment is never used for parameter updates; it is only used to select hyperparameters (e.g., learning rate, total training steps) and to detect obviously unstable configurations.

In principle one could perform multi-fold forward-rolling validation, training on a prefix and validating on the next slice, then rolling the window forward. In our weekly setting, however, the total number of decision points is modest, and repeatedly carving out validation slices would undercut the amount of data available for actual RL training. We therefore adopt a pragmatic single-split protocol: hyperparameters are chosen based on the held-out validation segment, and the final configuration is then used consistently across all experiments in Chapter 4. The held-out *test* window is only used once per configuration, for deterministic rollouts that produce the reported performance metrics.

3.6.2 Logging and evaluation readiness

To support the multi-faceted evaluation in Chapter 4, the environment and training loop record a set of per-step and per-episode statistics:

- **Per-step logs:** executed portfolio weights, portfolio net asset value (NAV), realised log returns, the decomposed reward (return term, endogenous risk index, market and LLM risk penalties), and trading quantities sufficient to compute turnover and concentration (e.g., Herfindahl–Hirschman index).
- **Per-episode logs:** full NAV trajectory, final test NAV multiple, Sharpe ratio, maximum drawdown, total and average transaction costs, and episode-level summaries of risk components.
- **Training diagnostics:** PPO statistics such as approximate KL divergence, policy entropy, value loss, and explained variance, used to monitor optimisation stability and detect divergence or collapse.

All logs are written in a consistent, analysis-friendly format (JSONL for per-step records, CSV/Parquet for episode summaries and monitor files), and the same logging schema is used across all experiments. This ensures that differences observed in Chapter 4 can be attributed to

changes in features, rewards, or training curricula rather than to changes in data collection or evaluation code.

3.6.3 Experimental program mapping

Chapter 4 instantiates three families of experiments on top of the common environment and training protocol described above:

- **RQ1: LLM signals as observations.** Starting from a *price_only* backbone (historical prices only, without any technical indicators or LLM-derived signals), we add LLM-derived *directional* signals as additional observation features. Each RQ1 configuration augments the backbone with *exactly one* LLM signal source at a time (for example, a single Tier 1 analyst such as SEC, EC, news, or technical, or a single Tier 2 senior signal); combinations of multiple LLM sources are *not* used within RQ1. In this family the reward disables all explicit risk penalties, with $\beta = 0$ and $\gamma = 0$ in Section 3.5, so LLM information affects behaviour only via the observation space.
- **RQ2: LLM-informed risk shaping.** Holding the observation space fixed, we vary the LLM risk coefficient γ and the choice of risk source (news, technical, EC, SEC, or senior) in the risk-aware reward of Section 3.5. In this family both the endogenous risk term and the exogenous penalties are active, with $\beta > 0$ and $\gamma > 0$ for the configurations listed in Chapter 4. This isolates the impact of LLM-based risk shaping on test NAV, Sharpe ratio, and drawdown relative to purely return-driven training.
- **RQ3: Curriculum vs. all-in-one training.** Using the same PPO backbone and the pure log-return reward as in RQ1 (again with $\beta = 0$ and $\gamma = 0$), we compare all-in-one training on the full multimodal observation state (prices, technical indicators, and multiple LLM signals) against curricula that stage the introduction of modalities over training. Concretely, we study short→long curricula (faster sources such as technical indicators first, then news, then slower EC/SEC signals) and long→short curricula (EC/SEC first, then news/technical). Each curriculum keeps the environment dynamics and reward fixed and differs only in the schedule with which features are exposed to the policy during training.

By fixing the environment implementation, logging schema, and train/validation/test protocol across all runs, and varying only (i) how LLM information enters the system (as observations versus as risk penalties) and (ii) when different modalities are made available (all-in-one versus staged curricula), the experimental results in Chapter 4 can be interpreted directly in terms of

the questions posed in RQ1–RQ3: the value of LLM signals as observations, the role of LLM-based risk shaping, and the effect of curriculum-style feature introduction on optimisation and out-of-sample behaviour.

CHAPTER 4

EXPERIMENTAL SETUP AND RESULTS

4.1 Datasets and Preprocessing

4.1.1 Asset universe and horizon

The trading universe consists of 20 liquid US equities drawn from a large-cap, S&P 500–style universe.¹ After cleaning for missing prices and ensuring full cross-sectional coverage at each weekly decision date, this yields a panel of approximately 334 weekly decision dates from 18 April 2019 to 5 September 2025. As described in Chapter 3, all technical indicators and LLM-derived features are aligned to this canonical weekly calendar. The last trading decision date in the sample is 29 August 2025; the final evaluation date for portfolio value is 5 September 2025.

This canonical panel underlies both the descriptive analysis of LLM-derived signals in this chapter and the reinforcement learning experiments in Chapters 3 and 4. The exact train/validation/test segmentation and warm-up conventions used by the RL environment follow the protocol in Section 3.6.1.

4.1.2 Feature construction and weekly alignment

All experiments are based on a canonical weekly panel with one row per (ticker, decision week), aligned to the calendar described in Section 4.1.1. This panel aggregates the data modalities introduced in Chapter 3 into a single, analysis-ready table that serves as the interface between the data pipeline, the LLM signal-extraction system, and the RL environment.

For each ticker-week, the panel contains five broad groups of variables:

- **Market variables.** Weekly price and volume information derived from daily OHLCV data, together with simple return measures over recent horizons (for example, the one-week log return) and basic weekly aggregates, such as the number of trading days observed. These quantities provide the raw market context for both the technical indicators

¹The full ticker list is reported in Appendix A.

and the RL agent.

- **Technical indicators.** A set of standard technical indicators computed from the daily OHLCV history and then sampled at the weekly decision dates. These include moving averages and their slopes, trend and momentum measures, volatility proxies such as realised volatility and average true range, and simple price/volume normalisations. The lookback windows are chosen to be compatible with the weekly rebalancing frequency and with the short- to medium-horizon regime interpretation used by the technical analyst in Section 3.4.2.
- **Global risk factors.** Market-wide summary variables that capture the current risk regime, most notably a turbulence index constructed from recent cross-sectional equity returns following the “market turbulence” measure used in the FinRL framework and related literature, together with simple higher-moment statistics such as short-horizon kurtosis. These features appear in the global component of the, RL state and are also used to modulate the risk-aware reward in Section 3.5, allowing the agent to condition its behaviour on stressed versus normal market regimes.
- **LLM-derived signals.** For each of the Tier-1 analysts (SEC filings, earnings calls, news, and technical narratives) the panel stores a directional signal, a downside risk score, and a confidence level on the common numeric scales described in Section 3.4.3. In addition, the panel records the corresponding Tier-2 senior verdict for each ticker-week: an integrated senior signal, senior risk score, and senior confidence produced by the investment-committee layer that reconciles the source-specific views. Minimal metadata and recency information (such as the time since the most recent filing or earnings call) are kept alongside these scalar values to track the freshness of each information source and to support later diagnostic analysis. In the RL experiments of this thesis, only the directional signals from the chosen LLM sources enter the observation space; risk scores are used solely in the risk-aware reward for RQ2, and confidence values are retained as diagnostics and for potential future architectures.
- **Coverage and agreement indicators.** Finally, a small set of derived indicators summarises, for each ticker-week, which information sources produced valid signals and whether their directional views are broadly aligned or in conflict. These variables are used solely for descriptive statistics and quality checks; they are not exposed to the RL agent and do not enter the observation space.

All numeric fields in this panel are aligned to the weekly decision dates so that, at each step, the RL environment observes a self-consistent snapshot of prices, indicators, and LLM

signals that would have been available strictly before that week’s close. Future information is excluded by construction: daily series are truncated at the decision date before any aggregation, and textual sources are filtered by publication time relative to the decision week. Within the environment, additional rolling histories (for example, short windows of recent daily prices) are constructed on top of this weekly panel as needed, but the underlying alignment ensures that all experiments in Chapter 4 share a common, leakage-safe feature construction. Together with the modality-specific roles and horizons in Section 3.4.2, this panel therefore realises a genuinely multi-channel, multi-timescale information set: slow structural SEC signals, medium-horizon earnings-call and news signals, and faster technical signals, all fed into a single RL framework under leakage-safe cut-offs.

4.1.3 Descriptive statistics of LLM-derived signals

The LLM-based analysts are run on the full weekly sample from 12 April 2019 to 29 August 2025, producing one set of signals for each ticker–week. Before training the reinforcement learning agent on these outputs, it is important to understand how the resulting scores behave in aggregate. The descriptive analysis in Tables 4.1–4.4 serves two purposes. First, it verifies that the prompt-engineered scales behave in a numerically stable way across modalities, with no gross violations of the intended bounds (for example, signals outside $[-10, 10]$ or risk scores outside $[0, 10]$) and no degenerate confidence distributions. Second, it checks whether the empirical distributions are qualitatively consistent with the economic interpretations of each source: SEC as a slow, risk-focused structural view; EC as a medium-horizon fundamental update; news and technical analysis as higher-frequency, more volatile channels; and the senior layer as a cautious aggregate of these perspectives. Strong deviations from these expectations would cast doubt on the prompt design and on the usefulness of the resulting features for RL.

Each Tier-1 analyst (SEC, EC, news, technical) outputs a directional signal $s \in [-10, 10]$, a risk score $r \in [0, 10]$, and a confidence $c \in [0, 1]$ according to the schema in Section 3.4.3. The Tier-2 senior layer then aggregates these source-specific views into an integrated signal, risk score and confidence on the same scale for each ticker–week.

Directional signals across modalities. Table 4.1 shows that the four Tier-1 modalities occupy distinct but interpretable regions of the common $[-10, 10]$ signal scale. SEC-based signals are mildly negative on average (mean ≈ -1.10 , IQR $[-3, 1]$), reflecting the fact that mandatory risk disclosures tend to emphasise downside scenarios and rarely justify strongly bullish views on their own. EC signals are positive and tightly clustered (mean ≈ 3.36 , IQR $[3, 4]$), in line with

the observation that earnings calls are typically framed around growth narratives, with strongly negative calls appearing only occasionally. News signals display a much wider spread (std. $\approx$ 3.76, range $[-9, 8]$), which is consistent with their role in capturing short-horizon sentiment and idiosyncratic events. Technical signals have the largest dispersion (std. $\approx$ 5.91, IQR $[-7, 7]$), as they respond mechanically to strong price trends and momentum reversals in both directions.

The senior signal lies between these modalities. It has a modestly positive mean (mean $\approx$ 0.34) and a dispersion (std. $\approx$ 2.71, IQR $[-2, 2]$, range $[-8, 7]$) that is narrower than the raw news and technical signals but still makes substantial use of the scale. This matches its intended role as a cautious “investment-committee” summary: it tempers extreme views from any single source, yet preserves clear directional tilts when multiple modalities align. The fact that SEC signals are on average more negative, EC signals are more positive, news and technical signals are more volatile, and the senior signal sits in between is qualitatively consistent with practitioners’ understanding of these information sources, and supports the face validity of the prompt design.

Risk scores across modalities. Table 4.2 reports the corresponding risk scores. SEC filings exhibit the highest average risk ($\bar{r}_{\text{sec}} \approx 6.34$) with most mass in the 6–8 range, which is natural for structural risk disclosures that emphasise potential downside over a multi-year horizon. EC risk scores are lower and more concentrated ($\bar{r}_{\text{ec}} \approx 4.01$, IQR $[4, 4]$): management tends to highlight relevant risks over the next few quarters but rarely signals severe structural fragility. News and technical risks lie between these extremes, capturing a combination of event risk and elevated volatility in turbulent periods. The senior risk score sits below the SEC average but above the EC/news/technical means ($\bar{r}_{\text{senior}} \approx 5.48$), indicating that the senior layer inherits the conservative stance of the structural view while smoothing over the noisier fluctuations in shorter-horizon channels. Again, the ordering of risk levels across modalities accords with the intended semantics of each source.

Because all risk scores are explicitly bounded in $[0, 10]$ and the observed values respect these bounds, additional global rescaling is not necessary for the baseline experiments. Instead, we treat the prompt-induced scale, and the resulting differences in level and dispersion across modalities, as part of the modelling choice. The RL agent is allowed to learn how sensitive its policy should be to each risk channel.

Confidence scores. Table 4.3 shows that confidence values are generally high but concentrated in a relatively narrow band: most mass lies between 0.6 and 0.8 for the Tier-1 analysts.

Senior confidence is slightly lower on average (mean ≈ 0.59) and even more tightly clustered (std. ≈ 0.06), reflecting a cautious stance when sources disagree, or coverage is thin. Importantly, there is no collapse towards very low confidence across the board, which would call into question the reliability of the underlying signals. Given this limited dynamic range, we do not treat confidence as a primary input for the RL agent or as an explicit weight in reward shaping. Instead, it is retained as a diagnostic quantity for inspecting individual cases and for potential use in more elaborate architectures (for example, confidence-based gating or curriculum scheduling) in future work.

Recency and coverage. Table 4.4 summarises how recent the fundamental information is at each weekly decision date. For earnings calls, we measure the number of calendar days between the release date of the most recent call for a given ticker and the corresponding decision date, an analogous measure is constructed for the most recent SEC filing. These recency lags are also provided as context to the EC and SEC analysts when they form their views, so that the LLMs can distinguish between genuinely fresh disclosures and older information that may already have been superseded. For both earnings calls and filings, the median lag is around 43–44 days and the upper quartile around 66 days, indicating that the agent typically operates with relatively up-to-date fundamental inputs. A small fraction of ticker-weeks have much larger lags, reflecting periods with no new filings; in such cases, the pipeline either carries forward older structural assessments or defaults to neutral scores, so that the feature matrix remains dense while still encoding information age.

Implications for RL and interpretation. Taken together, these descriptive statistics provide a sanity check on the validity of the LLM-based analysts. The distributions of signals, risk scores, and confidence values respect the designed ranges, differ across modalities in ways that align with the real-world role of each information source, and avoid obvious numerical pathologies. On this basis, we adopt the following usage in the RL experiments:

- In RQ1 and RQ3, the agent’s observation space is built from the price-only backbone plus the directional LLM signals selected for that experiment.
- In RQ2, the LLM risk scores serve as the exogenous risk channel in the reward shaping term of Section 3.5, while the observation space is kept fixed.
- Confidence values are not used directly by the policy or the reward; they are retained as diagnostic quantities and as potential gating signals for future architectures.

Table 4.1: Descriptive statistics of LLM directional signals $s \in [-10, 10]$ across all ticker-weeks ($N = 6,680$).

Statistic	SEC	EC	News	Technical	Senior
Count	6680	6680	6680	6680	6680
Mean	-1.10	3.36	0.98	-0.06	0.34
Std. deviation	2.48	1.37	3.76	5.91	2.71
Min	-6.00	-5.00	-9.00	-8.00	-8.00
25th percentile	-3.00	3.00	-2.00	-7.00	-2.00
50th percentile	-2.00	3.50	2.00	0.00	1.00
75th percentile	1.00	4.00	4.00	7.00	2.00
Max	5.00	5.00	8.00	8.00	7.00

Table 4.2: Descriptive statistics of LLM risk scores $r \in [0, 10]$ across all ticker-weeks ($N = 6,680$).

Statistic	SEC	EC	News	Technical	Senior
Count	6680	6680	6680	6680	6680
Mean	6.34	4.01	3.84	4.20	5.48
Std. deviation	2.03	0.80	1.99	1.15	0.68
Min	0.00	0.00	0.00	2.00	2.00
25th percentile	6.00	4.00	3.00	3.00	5.00
50th percentile	7.00	4.00	4.00	4.00	5.00
75th percentile	7.00	4.00	5.00	5.00	6.00
Max	8.00	6.00	9.00	7.00	8.00

The differing centres and spreads across SEC, EC, news, technical, and senior channels, together with the recency patterns in Table 4.4, are therefore treated as informative structure rather than noise: they encode the intended multi-horizon, multi-source nature of the problem. For example, SEC signals tend to be more conservative due to their regulatory, risk-disclosure focus, EC signals skew more optimistic as they are framed by company management, news and technical signals exhibit higher volatility as they react to short-horizon events and price action, and the senior signal sits in between as a cautious aggregate. For EC and SEC in particular, the recency statistics in Table 4.4 show median lags of roughly 45 days but with a long tail of cases where disclosures are several weeks to months old; these lags are explicitly surfaced to the analysts (e.g. via `days_since_ec` and `days_since_sec`) and the prompts instruct the models to treat stale information more cautiously when forming their views. The rubric-based prompt design in Section 3.4.4 is precisely aimed at inducing such horizon-consistent, numerically calibrated behaviour, which the RL agent is then tasked with exploiting under the leakage-safe temporal cut-offs described in Chapters 3–4.

Table 4.3: Descriptive statistics of LLM confidence scores $c \in [0, 1]$ across all ticker-weeks ($N = 6,680$).

Statistic	SEC	EC	News	Technical	Senior
Count	6680	6680	6680	6680	6680
Mean	0.70	0.76	0.64	0.69	0.59
Std. deviation	0.21	0.14	0.27	0.10	0.06
Min	0.00	0.00	0.00	0.50	0.30
25th percentile	0.70	0.80	0.70	0.60	0.60
50th percentile	0.75	0.80	0.70	0.70	0.60
75th percentile	0.80	0.80	0.80	0.80	0.60
Max	0.85	0.80	0.90	0.85	0.80

Table 4.4: Descriptive statistics of recency features

Statistic	days_since_ec	days_since_sec
Count	6607	6484
Mean	45.37	44.74
Std. deviation	29.27	28.01
Min	0.00	0.00
25th percentile	22.00	21.00
50th percentile	44.00	43.00
75th percentile	66.00	66.00
Max	198.00	198.00

4.2 Evaluation Metrics

Our empirical analysis examines each experimental configuration through four complementary lenses: (i) out-of-sample trading performance on the held-out test window, (ii) the numerical stability of PPO training, (iii) the trading behaviour implied by the learned policy, and (iv) generalisation from the train/validation prefix to the test segment. All quantities are computed from the unified per-step and per-episode logging schema described in Chapter 3, so that price-only baselines and LLM-augmented agents are evaluated in a strictly comparable way.

4.2.1 Trading performance metrics

The primary evaluation of each strategy is based on its realised trading performance on the 100-week chronological test window defined in Section 4.1.1. Let $p_{t,i}$ denote the close price of asset i at decision date t , and let $w_{t,i}$ be the portfolio weight held *after* rebalancing at that date, including the cash position. Throughout, we compute returns at a weekly frequency.

Weekly portfolio returns and NAV. Given the vector of post-rebalance weights w_t and next-week gross asset returns $R_{t+1,i} = p_{t+1,i}/p_{t,i}$, the one-week gross portfolio return is

$$R_{t+1}^{\text{port}} = \sum_i w_{t,i} R_{t+1,i},$$

where transaction costs are deducted at each rebalance as a proportional cost on traded notional. We work with weekly log-returns

$$r_{t+1} = \log R_{t+1}^{\text{port}},$$

and define the portfolio net-asset value (NAV) process recursively via

$$\text{NAV}_0 = 1, \quad \text{NAV}_{t+1} = \text{NAV}_t \exp(r_{t+1}).$$

For each strategy we report the *NAV multiple* over the full test window,

$$\text{NAV}_{\text{test}} = \frac{\text{NAV}_T}{\text{NAV}_0},$$

where T is the number of test weeks (here $T = 100$).

Annualised return. To facilitate comparison with standard buy-and-hold benchmarks, we convert the NAV multiple over the T -week test horizon into an annualised gross return assuming 52 trading weeks per year:

$$r_{\text{ann}} = \text{NAV}_{\text{test}}^{52/T} - 1.$$

In all tables, NAV refers to the 100-week NAV multiple, while “Annualised return” denotes r_{ann} computed via the above formula.

Weekly Sharpe ratio. The Sharpe ratio measures the mean excess return per unit volatility. Since our evaluation is already carried out at a weekly frequency, and the test horizon is fixed across all methods, we report the *weekly* Sharpe ratio rather than an annualised version. Let μ_r and σ_r denote the sample mean and standard deviation of the weekly log-returns $\{r_t\}_{t=1}^T$; using the 3-month U.S. Treasury yield as a proxy for the risk-free rate would only introduce a negligible shift at the horizon considered, so we treat returns as excess returns. The weekly Sharpe is then

$$\text{Sharpe}_{\text{weekly}} = \frac{\mu_r}{\sigma_r}.$$

If desired, an annualised Sharpe can be obtained as $\text{Sharpe}_{\text{ann}} = \sqrt{52} \text{Sharpe}_{\text{weekly}}$, but we keep the weekly version in all main tables for clarity and consistency.

Maximum drawdown. To quantify downside risk, we report the realised maximum drawdown (MDD) over the test window. Let $M_t = \max_{0 \leq s \leq t} \text{NAV}_s$ denote the running peak of the NAV process. The drawdown at time t is

$$\text{DD}_t = \frac{\text{NAV}_t - M_t}{M_t},$$

and the maximum drawdown on the test horizon is

$$\text{MDD} = \min_{0 \leq t \leq T} \text{DD}_t,$$

a negative quantity indicating the largest peak-to-trough loss in percentage terms. All MDD values reported in Chapter 4 are computed directly from the realised test NAV path under the specified transaction-cost and cash-floor assumptions.

4.2.2 Training stability

To ensure that observed performance differences are not artefacts of unstable optimisation, we monitor several standard diagnostics during PPO training:

- **Smoothed episode reward.** Mean return per episode over a sliding window, which should increase and then plateau for well-behaved runs. Erratic swings or sustained deterioration can indicate optimisation difficulties or severe non-stationarity.
- **Approximate policy KL divergence.** The Kullback–Leibler divergence between successive policy iterates, which should remain within a small range for PPO-style clipped updates. Persistent large spikes typically signal overly aggressive updates or an unsuitable learning rate.
- **Explained variance of the value function.** The proportion of variance in returns explained by the critic’s value predictions. Values approaching 1 indicate that the critic provides a useful baseline; sharp collapses towards 0 mark episodes where value estimation fails.

In particular, we treat large, sustained increases in policy KL or abrupt drops in explained variance as indicators of pathological updates and verify that configurations incorporating LLM-derived features do not introduce additional instability relative to the pure-price baselines.

4.2.3 Trading behaviour

Scalar performance metrics do not reveal *how* a policy trades. We therefore analyse the trading patterns implied by each learned policy using step-level diagnostics:

- **Turnover.** The per-step ℓ_1 distance between consecutive portfolio weight vectors (scaled by 0.5), summarised by its mean, median, and upper quantiles over the test window. This quantity measures how aggressively the agent adjusts its positions from week to week.
- **Concentration (HHI).** The Herfindahl–Hirschman index of portfolio weights, computed as the sum of squared weights and averaged over time. Higher values correspond to more concentrated portfolios (capital placed in fewer names), while lower values indicate more diversified allocations.
- **Risk penalties and exposures.** For experiments that include explicit risk penalties in the reward (in practice, the RQ2 runs), we track the contributions of the market-risk and LLM-derived risk terms to the total reward and their correlations with realised step returns. This helps assess whether the agent internalises these signals in a meaningful way or effectively treats them as noise rather than information. Curriculum experiments in RQ3 reuse the pure log-return reward from RQ1, so they are evaluated only through the return- and behaviour-based metrics above.

For a small number of representative configurations (typically the best-performing model per research question), we additionally visualise the full test-time trajectories of NAV, turnover, concentration, and reward components, and relate salient episodes to major market moves.

4.2.4 Generalisation

Finally, we assess how well each configuration generalises from the train/validation prefix to the held-out test window. We adopt two simple but informative diagnostics:

- **Performance gaps.** For each configuration, we compute NAV, Sharpe, and MDD on both the train and test segments and summarise the differences as $\Delta\text{NAV} = \text{NAV}_{\text{test}} - \text{NAV}_{\text{train}}$, and analogously for Sharpe and MDD. Large negative gaps indicate over-fitting to the training episodes; small or slightly positive gaps suggest more robust policies.
- **Distributional shifts.** We compare quantiles of episode-level NAV and log returns between the train and test windows, which helps detect whether a policy performs well only on a small subset of favourable regimes or maintains its behaviour across a broader range of market conditions.

These generalisation metrics are reported for the main baselines and for the selected best-of-grid configurations in RQ1–RQ3. Taken together, the four evaluation lenses—trading performance, training stability, trading behaviour, and generalisation—provide a structured framework for interpreting how LLM-derived information affects the learned portfolio policies, even when headline performance improvements are modest.

In preliminary experiments, we also applied these metrics within a forward-rolling validation scheme based on three folds and multiple training budgets (20k, 60k, and 80k PPO steps) for a price-only baseline. The overall data set contains 334 weekly decision dates; as in Section 4.1.1 we first fix a 70% prefix comprising the earliest 234 weeks. Within this prefix we construct three chronological folds by always training from the common start date and rolling a 30-week validation window forward in time: in Fold 0 the model is trained on weeks 1–144 and validated on weeks 145–174; in Fold 1 it is trained on weeks 1–174 and validated on weeks 175–204; and in Fold 2 it is trained on weeks 1–204 and validated on weeks 205–234. For each (fold, budget) combination we train three independent PPO runs with different random seeds and average the resulting validation metrics. As summarised in Appendix C, the first two folds, which have relatively short effective training prefixes, yield systematically negative test NAV multiples (all < 1) and negative test Sharpe ratios across all budgets, together with large negative performance gaps. Only the third fold, whose training prefix is longest, produces clearly positive test Sharpe ratios and NAV multiples above 1. These results are consistent with underfitting in the early folds due to limited data and motivate the simpler single-split protocol adopted in Section 4.1.1, where the agent is trained on the full 234-week prefix and evaluated once on a clean chronological hold-out window.

Each of these metric families is tied to one or more of the research questions posed in Chapter 3. For RQ1 (LLM signals as observations), the primary focus is on test-window trading performance and on trading behaviour (turnover and concentration) relative to price-only and classical allocation baselines, with training stability used to check that any differences are not artefacts of optimisation failure. For RQ2 (LLM-informed risk shaping), the interaction between performance metrics and downside measures (MDD, drawdown paths, and the decomposed risk penalties) is central: the question is whether the risk-aware reward meaningfully alters the realised risk profile without collapsing overall returns. For RQ3 (curriculum vs. all-in-one training), the emphasis shifts towards training stability and generalisation diagnostics: we compare how different feature-introduction schedules affect PPO optimisation behaviour and out-of-sample performance on the same price and LLM feature sets. In this way, the four evaluation lenses introduced above are not generic statistics, but are chosen to align directly with the

thesis research questions.

4.3 Price-only PPO benchmark

Before introducing LLM-derived signals, we first establish how far a carefully tuned, price-only PPO agent can go on this weekly allocation task. This section (i) summarises the performance of the classical equal-weight and minimum-variance baselines; (ii) describes the hyperparameter selection procedure; and (iii) presents the resulting test-set performance of the price-only agent relative to these baselines under a realistic transaction-cost regime.

Throughout this section, trading performance is evaluated on the 100-week test window introduced in Section 4.2.1. We report four primary statistics: (i) the realised wealth multiple (“NAV multiple”) over the 100-week horizon; (ii) the corresponding annualised NAV, obtained by compounding the realised multiple over a 52-week year; (iii) the weekly Sharpe ratio; and (iv) the maximum drawdown (MDD) computed from the sequence of weekly portfolio returns.

4.3.1 Classical portfolio baselines

We consider two classical, non-LLM baselines: (i) an equal-weight (EW) portfolio that rebalances all tickers to the same weight at each decision date; and (ii) a minimum-variance (MV) portfolio computed from a 26-week rolling covariance matrix under standard long-only constraints. Both baselines are rebalanced weekly using the same 20-asset universe and trading calendar as the RL agent, and they incur the same proportional transaction costs of 10 basis points (bp) for both buys and sells.

Table 4.5 summarises the realised performance of these baselines on the 100-week test window. The EW strategy provides a strong benchmark with an annualised NAV growth of roughly 30% and a weekly Sharpe ratio above 1.6, while MV achieves lower volatility at the cost of materially weaker growth and a deeper maximum drawdown. In other words, even simple, fully mechanical allocation rules achieve substantial compounding over this period: despite several constituent names (for example, INTC, UNH, AMT) experiencing pronounced drawdowns or prolonged sideways regimes, the EW and MV portfolios still grow total wealth by about 65% and 45%, respectively. Any RL-based strategy must therefore improve on already strong baselines rather than correcting obviously poor benchmark choices.

Table 4.5: Performance of classical baselines on the 100-week test window with 10 bp transaction costs. “NAV multiple” denotes the realised wealth multiple over the test horizon ; “Annualised NAV” compounds this multiple to a 52-week year.

Strategy	NAV multiple	Annualised NAV	Weekly Sharpe	MDD
Equal-weight (EW)	1.6539	1.299	1.64	−0.16
Minimum variance (MV)	1.4573	1.216	1.27	−0.20

4.3.2 Hyperparameter selection for price-only PPO

The price-only PPO agent observes only price-related information (daily close prices projected to weekly decisions, without technical indicators or LLM-derived signals) and acts in the same weekly environment with a 10 bp transaction cost and no explicit cash floor (i.e. the agent is free to allocate the entire portfolio across the 20 assets plus cash, but is not forced to hold a minimum cash weight).

As discussed in Section 4.1.1 and Appendix C, we initially explored a fixed-start forward-rolling validation scheme on the 70% prefix of the data (234 weeks), with three folds that always trained from week 1 and moved a 30-week validation window forward. In the earlier folds, where the effective training prefixes were short, the price-only PPO agent underfit: validation NAV multiples and Sharpe ratios were weak, and the test NAV multiples were often below 1. Only the longest prefix (204 training weeks and a 30-week validation suffix) produced clearly positive Sharpe and NAV metrics. These preliminary experiments motivated the single-split protocol formalised in Section 3.6.1: hyperparameters are selected on the 204/30 train/validation split within the 234-week prefix, and the final agent is retrained on the full 234-week prefix before being evaluated once on the held-out 100-week test window.

Concretely, for the price-only agent we perform a compact grid search on this 204/30 split around the configuration used in prior financial RL work: two learning rates (1×10^{-4} and 3×10^{-4}), two rollout lengths (1,024 and 2,048 environment steps per update), two total training budgets (20k and 60k PPO steps), and two minimum cash-allocation floors (0 and 2% of portfolio value). All runs use the same feature-extraction backbone (“flatten” per-ticker aggregation plus a 1D convolutional trading head, as described in Section 3.3.6) and are averaged over three random seeds.

Table 4.6 reports a representative subset of the validation results on the 204/30 split. Configuration A uses a conservative learning rate, a 20k-step training budget, and no explicit cash floor, i.e., the policy is free to allocate almost all wealth to risky assets subject only to transaction costs. It attains the highest validation NAV multiple and competitive weekly Sharpe and

Table 4.6: Representative validation results for the price-only PPO hyperparameter grid on the 204/30 train/validation split. All configurations use the same feature extractor (per-ticker “flatten” and a 1D-convolution trading head). Metrics are averaged over three seeds and evaluated on the 30-week validation window.

Config	LR	Rollout steps	Train steps	Cash floor	Val. NAV $\times$	Val. Sharpe	Val. MDD
A	1×10^{-4}	1,024	20k	0.0	1.20	2.36	−0.06
B	3×10^{-4}	1,024	20k	0.2	1.17	2.45	−0.05
C	3×10^{-4}	1,024	60k	0.2	1.17	2.39	−0.05

drawdown. Configuration B keeps the same optimisation hyperparameters but imposes a 2% minimum cash allocation. Its validation performance is very similar to A, with slightly higher Sharpe and a marginally lower maximum drawdown. Configuration C extends the training horizon to 60k steps; although its validation NAV is comparable, the corresponding Sharpe and drawdown metrics vary more across seeds and offer no clear improvement over A or B. Given (i) the lack of a systematic validation advantage for larger training budgets, and (ii) the desire to maintain a like-for-like comparison with the fully invested equal-weight and minimum-variance baselines, we adopt Configuration A (20k steps, no minimum cash floor) as the default price-only PPO setup for all subsequent experiments. This ensures that any differences observed in later sections are attributable to the addition of LLM signals or curriculum schedules rather than to changes in core RL hyperparameters.

4.3.3 Test-set performance (price-only PPO) versus baselines

Using the selected hyperparameters from Table 4.6 (configuration A), we retrain the price-only PPO agent on the full 234-week prefix and evaluate it once on the 100-week test window. Table 4.7 compares the resulting price-only PPO agent against the EW and MV baselines on the 100-week test window. Each PPO configuration is trained on the full 70% prefix (234 weeks) with the hyperparameters selected above and evaluated on the held-out 30% suffix (100 weeks). We report the three individual seeds and their cross-seed average.

The price-only PPO agent consistently matches or slightly improves on the EW benchmark across seeds, both in terms of NAV growth and weekly Sharpe ratio, while maintaining a comparable maximum drawdown. The cross-seed mean achieves an annualised NAV of approximately 1.32 versus 1.30 for EW, and a NAV multiple of about 1.71 $\times$ versus 1.65 $\times$ for EW over the same 100-week horizon. In cumulative-return terms, the best seed grows wealth by roughly 73% compared with about 65% for EW, an improvement of around 8 percentage points. Maximum drawdowns remain in a tight band around −16% to −17% for both strategies, indicating

Table 4.7: Test-set performance of the price-only PPO agent compared with the classical EW and MV baselines on the 100-week test window (10 bp transaction costs, zero cash floor). “NAV multiple” and “Annualised NAV” are defined as in Table 4.5.

Model / seed	NAV multiple	Annualised NAV	Weekly Sharpe	MDD
Equal-weight (EW)	1.6539	1.299	1.64	−0.16
Minimum variance (MV)	1.4573	1.216	1.27	−0.20
PPO (seed 42)	1.7267	1.328	1.67	−0.16
PPO (seed 137)	1.7191	1.325	1.65	−0.17
PPO (seed 256)	1.6902	1.314	1.61	−0.16
PPO (mean over seeds)	1.7120	1.323	1.64	−0.17

that the price-only agent achieves its modest outperformance without materially increasing tail risk.

Qualitatively, this indicates that, given a sufficiently expressive feature extractor and modest hyperparameter tuning, PPO can learn a sensible weekly rebalancing strategy from prices alone. It exploits cross-sectional differences between names such as META and NFLX (which experienced strong upward trends) versus AMT, INTC, and UNH (which suffered drawdowns or prolonged weakness), and converts these into a small but consistent improvement over a strong equal-weight benchmark, without changing the overall risk profile on this test window.

These results serve two purposes for the remainder of the chapter:

1. They demonstrate that, under a realistic 10 bp transaction-cost regime and weekly rebalancing, a well-tuned price-only PPO agent is competitive with, and slightly stronger than, a robust equal-weight baseline on a diversified 20-asset universe.
2. They provide a clean reference point for the LLM-augmented experiments. In the remainder of Chapter 4, we treat this price-only configuration as the default RL backbone and gradually augment it with LLM-derived directional signals (RQ1), LLM-derived risk-shaping terms (RQ2), and curriculum-style feature-introduction schemes (RQ3) to investigate under what conditions additional information sources help or hurt policy learning.

4.4 Trading performance with LLM integration

4.4.1 RQ1: LLM directional signals as observation features

In this subsection we fix the PPO hyperparameters and network architecture from Section 4.3.3, and vary only which LLM-derived directional signals are appended to the price-based obser-

Table 4.8: RQ1 trading performance with different LLM directional signals (cash floor = 0, 10 bp trading cost). All agents share the same PPO hyperparameters and feature extractor (flatten+conv1d); results are averaged over three seeds on the 100-week test window. “NAV mult.” is the realised wealth multiple over 100 weeks; “NAV (ann.)” annualises this multiple as $M^{52/100}$; Sharpe is the weekly Sharpe ratio; MDD is maximum drawdown.

Strategy	NAV mult.	NAV (ann.)	Weekly Sharpe	MDD
PPO (price-only)	1.7120	1.3228	1.6446	-0.1651
PPO (+ SEC signal)	1.6919	1.3130	1.6035	-0.1637
PPO (+ EC signal)	1.6807	1.3080	1.5792	-0.1633
PPO (+ senior signal)	1.6747	1.3052	1.5573	-0.1626
PPO (+ news signal)	1.6665	1.3016	1.5484	-0.1623
PPO (+ technical-text signal)	1.6515	1.2951	1.5203	-0.1662

variations. Throughout, the reward reduces to the pure log-return objective (i.e. $\beta = \gamma = 0$ in Equation (3.8)), the environment uses a zero cash floor, and the transaction cost is 10 bp per trade. Each configuration is trained with three random seeds; we report the mean test performance over the 100-week evaluation window.

Table 4.8 compares the price-only PPO agent against variants that additionally consume one of the five single-source LLM signals: earnings-call (EC) signal, SEC-filing signal, news signal, technical-text signal, and the “senior” aggregated signal. For reference, Section 4.3.3 has already shown that the price-only PPO matches the equal-weight baseline in weekly Sharpe while improving NAV and maximum drawdown relative to both equal-weight and minimum-variance portfolios (Table 4.7).

A first observation is that adding any single LLM directional signal does not improve upon the tuned price-only PPO. The SEC signal variant is the strongest among the LLM configurations, but it still loses roughly 0.04 Sharpe points and about 0.01 in annualised NAV relative to the price-only agent. EC and senior signals form a second tier, followed by news and technical-text signals. All LLM variants retain broadly similar maximum drawdowns, suggesting that the deterioration is mainly in risk-adjusted return rather than in tail risk.

Comparing these numbers with Table 4.7, we see that all RQ1 agents still outperform the EW baseline in terms of NAV multiple (all NAV multiples remain above 1.65) and have weekly Sharpe ratios well above 1.5. In other words, starting from a strong price-only backbone, feeding a single LLM directional channel does not destroy performance; but at this horizon it does not create a consistent edge beyond what prices already deliver. The additional directional information extracted by the LLM analysts is at best weakly exploitable by PPO, and in several cases appears to act as additional noise that washes out under transaction costs.

4.4.2 RQ2: LLM risk scores and endogenous risk as reward shaping

In all experiments except RQ2, the PPO agent is trained on a purely return-based objective, namely the weekly log return of the portfolio (cf. Equation (3.4)). RQ2 activates the risk-aware reward of Equation (3.8) and asks whether combining an endogenous price-based risk term with exogenous LLM risk scores can improve the risk–return profile:

$$r_t = \lambda \left(r_t^{\text{return}} - \beta z_t^{\text{mkt}} E_t - \gamma z_t^{\text{llm}} \right), \quad (4.1)$$

where r_t^{return} is the log return of the portfolio over week t , E_t is the endogenous tail-risk statistic (constructed from downside deviation, CVaR-like losses, and loss frequency), $z_t^{\text{mkt}} \in [0, 1]$ is the normalised turbulence level, and $z_t^{\text{llm}} \in [0, 1]$ is the portfolio-level LLM risk exposure defined in Equations (3.5)–(3.7). In our implementation $\lambda = 1$. The price-only baseline corresponds to $\beta = 0$ and $\gamma = 0$, i.e., no explicit risk penalty. In RQ2 we *jointly* sweep the endogenous and exogenous risk coefficients β and γ while keeping all other aspects of the PPO setup (policy architecture, observations, training horizon) fixed.

For each of the five LLM risk sources—earnings calls (EC), SEC filings, news, technical narratives, and the senior (Tier-2) analyst—we construct a portfolio-level exposure z_t^{llm} by plugging the corresponding per-ticker risk scores into the same normalisation and clipping pipeline as in Chapter 3. For each source we sweep a small grid over the risk-shaping coefficients, $\gamma \in \{0.005, 0.010\}$ and $\beta \in \{0.01, 0.02, 0.03\}$, and train three seeds per (source, γ , β) configuration on the 234-week train/validation prefix. Each trained policy is then evaluated on the 100-week test window; we record the NAV multiple, annualised NAV via 52-week compounding, weekly Sharpe, and maximum drawdown.

Table 4.9 summarises, for each risk source, the best-performing configuration on the 100-week test window when ranked by weekly Sharpe ratio, and compares it against the price-only PPO agent.² We again report the realised NAV multiple over the 100-week horizon, the corresponding annualised NAV via 52-week compounding, the weekly Sharpe ratio, and maximum drawdown.

A few patterns emerge from Table 4.9:

- **Joint risk shaping is mildly detrimental on average.** For all five sources, the best (γ, β) combination delivers slightly lower NAV and Sharpe than the price-only baseline. The

²The full γ, β grid is omitted for brevity; it does not contain any configuration that strictly dominates the price-only baseline on both NAV and Sharpe.

Table 4.9: Best Sharpe configurations for RQ2: risk-aware reward with endogenous and LLM-based penalties. “NAV multiple” denotes the realised wealth multiple over the 100-week test window; “NAV (ann.)” is the corresponding annualised value obtained by compounding over a 52-week year.

Strategy	γ	β	NAV mult.	NAV (ann.)	Weekly Sharpe	MDD
PPO (price-only)	–	–	1.7120	1.3228	1.6446	-0.1651
PPO (+ EC risk penalty)	0.005	0.03	1.6893	1.3134	1.6153	-0.1613
PPO (+ News risk penalty)	0.005	0.03	1.6868	1.3124	1.6044	-0.1611
PPO (+ SEC risk penalty)	0.005	0.02	1.6840	1.3113	1.6028	-0.1605
PPO (+ Senior risk penalty)	0.010	0.01	1.6711	1.3061	1.5862	-0.1639
PPO (+ Technical risk penalty)	0.010	0.02	1.7013	1.3183	1.6290	-0.1627

strongest variant (technical risk, $\gamma = 0.010$, $\beta = 0.02$) attains a weekly Sharpe of 1.6290 and annualised NAV 1.3183, just below the baseline 1.6446 and 1.3228.

- **Drawdown reductions are modest.** All risk-shaped runs reduce maximum drawdown by a small margin (from -0.1651 to the range $[-0.1639, -0.1605]$), indicating that the endogenous and LLM penalties do curb tail risk. However, these gains are not large enough to compensate for the loss in average return.
- **No clearly superior combination of endogenous and LLM risk.** Across sources and (γ, β) pairs, we do not observe a configuration that improves upon the price-only agent on both NAV and Sharpe. Technical and EC risk shaping come closest, and several RQ2 configurations still outperform EW on both NAV and Sharpe, but none yields a robust edge over the price-only PPO backbone.

In summary, with the current linear reward-shaping design and the considered (γ, β) grid, augmenting the PPO objective with both endogenous and LLM-derived risk penalties is performance-neutral to mildly harmful: it slightly smooths drawdowns but does not unlock higher risk-adjusted returns beyond the return-only objective. This suggests that, at least for our weekly allocation task and this horizon, the primary value of the LLM risk scores is diagnostic rather than as a direct additive penalty; Chapter 5 returns to this point and discusses alternative integration strategies.

4.4.3 RQ3: Curriculum vs. all-in-one integration

Table 4.10 compares the two curriculum schedules—*Long*→*Short* (SEC + EC → +News → +Tech) and *Short*→*Long* (Tech → +News → +SEC + EC)—against an all-in-one integration of all modalities. All configurations use the same PPO hyperparameters and feature-extractor design as in RQ1–RQ2, with cash floor 0 and transaction cost 10 bp. For each regime and stage

Table 4.10: RQ3: curriculum vs. all-in-one integration (cash floor 0, cost 10 bp). Results are averaged over three seeds; best value in each column is highlighted in bold.

Regime / Stage	NAV mult.	NAV (ann.)	Weekly Sharpe	MDD
Long→Short, Stage 1	1.6770	1.308	1.5622	-0.1658
Long→Short, Stage 2	1.6836	1.311	1.5695	-0.1669
Long→Short, Stage 3	1.6528	1.299	1.4905	-0.1748
Short→Long, Stage 1	1.6550	1.299	1.5206	-0.1662
Short→Long, Stage 2	1.6048	1.279	1.3965	-0.1733
Short→Long, Stage 3	1.5462	1.254	1.2870	-0.1782
All-in-one, 20k steps	1.6595	1.301	1.5381	-0.1627
All-in-one, 60k steps	1.6854	1.312	1.5630	-0.1584

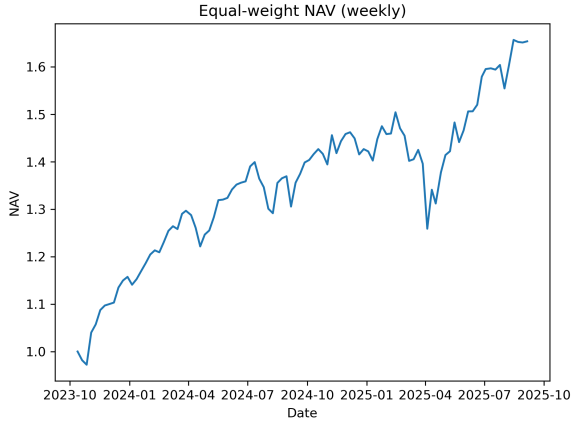
we report the mean over three random seeds (42, 137, 256); annualised NAV is computed from the test NAV multiple over a 100-week test horizon. As in RQ1, the reward reduces to pure log return ($\beta = \gamma = 0$), so RQ3 isolates the effect of introducing modalities via the *observation space* and training schedule rather than via reward shaping.

Overall, the performance differences across curriculum stages and against the all-in-one baseline are modest. Annualised NAV lies in a narrow band between 1.25 and 1.31 for all curriculum stages, with weekly Sharpe ratios between roughly 1.29 and 1.57. The *Long→Short* curriculum shows a slight improvement from Stage 1 to Stage 2 (NAV(ann.) 1.308 $\rightarrow$ 1.311, Sharpe 1.5622 $\rightarrow$ 1.5695), followed by a deterioration in Stage 3. The *Short→Long* curriculum is even less favourable: both NAV(ann.) and Sharpe degrade almost monotonically as more modalities are added, and Stage 3 is clearly the weakest regime.

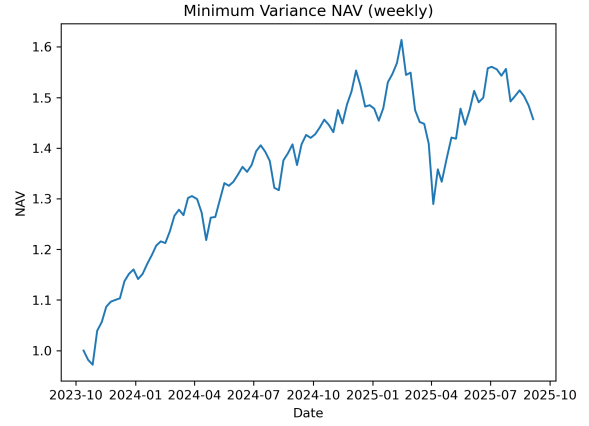
Compared to curricula, the all-in-one configuration at 60k updates achieves the highest NAV multiple and annualised NAV, with Sharpe comparable to the best curriculum stage and a slightly smaller drawdown. The 20k all-in-one run sits between the first stages of the two curricula. Taken together, these results suggest that, in this finite-horizon weekly setting, neither curriculum direction delivers a robust advantage over simply training a single all-in-one model longer: any gains are within noise and not systematically associated with staged introduction of modalities. Several RQ3 configurations remain competitive with EW (Table 4.16), but none surpasses the price-only PPO baseline in a consistent way.

4.4.4 Net asset value trajectories over the test period

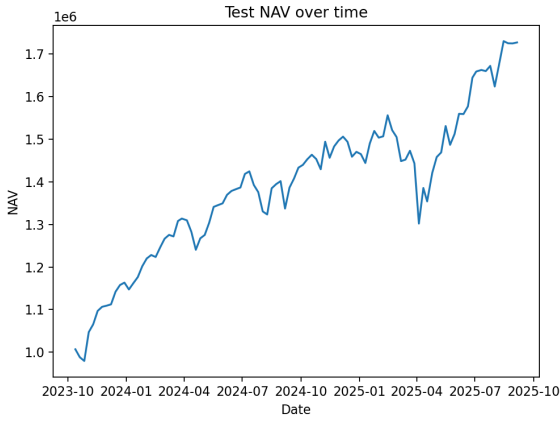
To complement the aggregate statistics in Section 4.3 and Section 4.4, we visualise the cumulative net asset value (NAV) paths over the 2023-10 to 2025-09 test window for four representative



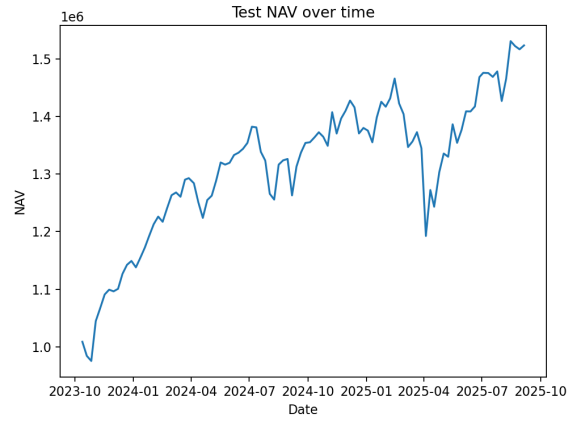
(a) Equal-weight benchmark



(b) Minimum-variance benchmark



(c) Price-only PPO (best feature extractor)



(d) RQ3: Short→long curriculum, Stage 3 (seed 137)

Figure 4.1: Weekly test-period NAV trajectories for two classical baselines, the price-only PPO agent, and the RQ3 curriculum agent at Stage 3.

configurations: equal-weight, minimum-variance, price-only PPO, and the RQ3 short→long curriculum agent at Stage 3 (seed 137).

All four configurations in Fig. 4.1 show a broadly similar qualitative pattern: a steady rise in NAV through late 2023 and 2024, a sharp drawdown around early 2025, and a subsequent recovery that finishes the test window at new highs. The equal-weight and minimum-variance portfolios differ mainly in the depth of the drawdown and the smoothness of the recovery, consistent with their respective Sharpe and maximum-drawdown statistics.

The price-only PPO agent (Fig. 4.1c) closely tracks the equal-weight benchmark for most of the horizon but ends with a higher terminal NAV, reflecting persistent risk-adjusted outperformance. The Stage 3 RQ3 curriculum agent in Fig. 4.1d follows a very similar path, including the common drawdown episode, but finishes with a slightly lower terminal NAV than the price-only policy. This is consistent with the generalisation results in Section 4.5, which show that while the curriculum can shape trading style and improve in-sample fit, it does not automatically

Table 4.11: Train and test performance for representative configurations (cost 10 bp, cash floor 0). Higher *test* NAV and test Sharpe are better; training metrics indicate how easily each objective is optimised. All rows except “short→long” aggregate over three seeds; the curriculum rows correspond to seed 137 at the end of each stage.

Config	train_nav	train_Sharpe	train_MDD	test_nav	test_Sharpe	test_MDD
Price-only (20k)	1.6280	0.4608	−0.4549	1.7120	1.6446	−0.1651
RQ1: +SEC signals (20k)	1.6345	0.4642	−0.4521	1.6747	1.5573	−0.1626
RQ2: +Tech risk (20k)	1.6652	0.4806	−0.4510	1.6813	1.5857	−0.1602
RQ3: all-in-one (60k)	1.9582	0.6438	−0.4570	1.6854	1.5630	−0.1584
short→long, stage 1	1.6544	0.4783	−0.4854	1.6964	1.5671	−0.1698
short→long, stage 2	1.9336	0.6415	−0.4458	1.6362	1.4264	−0.1762
short→long, stage 3	2.3484	0.8585	−0.4398	1.5237	1.2182	−0.1867

Table 4.12: Optimisation diagnostics for the same configurations as Table 4.11. Larger approx_kl and more negative policy_gradient_loss indicate more aggressive policy updates; high explained_variance suggests the value function fits the returns well.

Config	approx_kl	expl. var.	ep_rew_mean	val. loss	pg_loss
Price-only (20k)	0.0432	0.9942	−14.9590	0.0207	−0.0552
RQ1: +SEC signals (20k)	0.0597	0.9944	−14.9581	0.0204	−0.0729
RQ2: +Tech risk (20k)	0.0494	0.8732	−0.6311	0.0017	−0.0672
RQ3: all-in-one (60k)	0.1210	0.9957	−13.2085	0.0163	−0.0717
short→long, stage 1	0.0693	0.9943	−14.8236	0.0162	−0.0831
short→long, stage 2	0.1116	0.9969	−13.8574	0.0108	−0.0808
short→long, stage 3	0.1690	0.9966	−12.3381	0.0100	−0.0728

translate into substantially higher out-of-sample NAV on this finite test window.

4.5 Training stability and generalisation

To check whether the lack of clear LLM gains in Sections 4.4.1–4.4.3 is caused by numerical optimisation issues or by genuine generalisation limits, we inspect the training dynamics of four representative configurations: (i) price-only PPO (RQ0 baseline), (ii) RQ1 with SEC directional signals, (iii) RQ2 with technical-risk penalties, and (iv) RQ3 all-in-one multi-modal integration trained for 60k steps. For RQ3 we further follow the short→long curriculum for a representative seed (137) across its three stages, using its final model at each stage.

Table 4.11 reports train/test performance metrics, while Table 4.12 summarises the main optimisation diagnostics. Figures 4.2–4.3 show the evolution of the episode reward mean and approximate KL for the four key configurations for which we have full training curves: price-only, RQ2 tech risk, RQ3 all-in-one (60k), and RQ3 short→long (Stage 1, seed 137).

Price-only, RQ1, and RQ2 runs. For the price-only and LLM-augmented (RQ1, RQ2) configurations, the mean episode reward increases smoothly over 20k training steps for the runs with available curves (price-only and RQ2 tech; Figures 4.2a and 4.2b). The approximate KL trajectories remain in a moderate range (roughly 0.02–0.06), indicating that PPO stays within a stable trust region and does not suffer from catastrophic policy jumps. Table 4.12 confirms that the value function achieves high explained variance (≈ 0.99 for price-only and RQ1, slightly lower for RQ2 due to the heavier risk shaping), and the policy gradient losses remain well-behaved.

Adding SEC signals (RQ1) or technical risk penalties (RQ2) makes the training objective somewhat easier to optimise: `train_nav` increases from 1.63 (price-only) to 1.67 (RQ2), and `train_Sharpe` improves from 0.46 to 0.48. However, these gains are not matched by test performance: `test_nav` and `test_Sharpe` stay close to the price-only baseline and do not exhibit a consistent improvement, while `test MDD` remains virtually unchanged (Table 4.11). This suggests that the LLM-derived information does not fundamentally change the optimisation landscape; rather, it slightly reshapes the reward without yielding a more robust out-of-sample policy.

All-in-one 60k training. The RQ3 all-in-one configuration trained for 60k steps exhibits a stronger increase in training performance: `train_nav` reaches 1.96 and `train_Sharpe` exceeds 0.64, substantially higher than for any 20k run. The episode reward mean for this model rises almost linearly across the full 60k horizon (Figure 4.2c). At the same time, PPO becomes noticeably more aggressive: `approx_kl` grows to ≈ 0.12 , roughly double the 20k runs, and the policy gradient loss becomes more negative (Table 4.12). Despite this, the test metrics saturate: `test_nav` and `test_Sharpe` are only marginally above the RQ1 and RQ2 configurations and do not outperform the price-only baseline. This pattern is consistent with mild overfitting on a finite training window: the agent continues to fit its value function and reward proxy, but the additional optimisation does not translate into better generalisation.

Curriculum stages. The short→long curriculum (seed 137) makes the train–test gap particularly clear. As we progress from stage 1 to stage 3, `train_nav` increases from 1.65 to 2.35 and `train_Sharpe` from 0.48 to 0.86, while the approximate KL almost doubles from 0.07 to 0.17 (Table 4.12). The stage 1 training curves (Figure 4.2d) have the same smooth, monotone shape as the non-curriculum runs, with KL safely within the PPO target range. Yet the corresponding test performance deteriorates monotonically: `test_nav` drops from 1.70 to 1.52, `test Sharpe` from 1.57 to 1.22, and `test MDD` becomes more negative (Table 4.11). Later curriculum stages therefore continue to exploit idiosyncratic structure in the training period without improving, and in

fact harming, out-of-sample behaviour.

Takeaways. Across all four representative runs, the training traces tell a consistent and fairly concrete story. First, PPO is clearly able to optimise the specified objectives: mean episodic return increases smoothly over the 20k–60k updates, the approximate KL divergence remains in a moderate band (well below typical early-stopping thresholds), and the value function reaches high explained variance on the training data (around 0.8–0.9 for the price-only baseline and only slightly lower for the multi-modal variants). Second, adding LLM-based signals or extending training does not manifest as optimisation breakdown: the RQ2 tech-risk variant and the RQ3 all-in-one 60k run both achieve training Sharpe and NAV at least as strong as the price-only PPO, with no signs of divergence or collapse in the learning curves.

What changes is how much of this apparent progress carries over to the test period. The more complex configurations primarily improve in-sample metrics: they reduce training loss faster, attain higher training Sharpe, or further raise explained variance, but their test Sharpe and NAV remain comparable to, or weaker than, the price-only agent. In other words, within this finite weekly dataset, additional model complexity and longer curricula are used to fit noise rather than uncover robust extra signal. The main bottleneck is therefore statistical—limited sample size and modest signal-to-noise in the LLM features—rather than a failure of the PPO optimiser itself. We return to this diagnosis in Chapter 5 when discussing the broader implications for LLM-driven portfolio agents.

Generalisation across train–test splits

To quantify how well the learned policies generalise beyond the specific training window, we evaluate each configuration on an alternative train–test split and report both out-of-sample performance and simple train–test gap statistics. For each configuration we record: (i) the test-set NAV multiple, Sharpe ratio, and maximum drawdown (MDD); and (ii) the average change in NAV multiple, Sharpe ratio, and MDD between the original and alternative split, denoted ΔNAV , ΔSharpe and ΔMDD . Positive ΔNAV and ΔSharpe indicate that performance is at least as good on the alternative split, while smaller ΔMDD corresponds to more stable downside risk. Results are summarised in Table 4.13.

Across all configurations, the price-only PPO baseline provides a useful reference point. It attains strong test performance (Sharpe ≈ 1.64) and exhibits a positive train–test NAV gap and $\Delta\text{Sharpe} > 1$, indicating that its good performance is not purely an artefact of a single split.

The RQ1 single-source directional signals largely preserve this behaviour. All five channels

Table 4.13: Out-of-sample performance and train–test gaps on an alternative split. ΔNAV , ΔSharpe and ΔMDD are the mean differences between the original and alternative splits for NAV multiple, Sharpe ratio and maximum drawdown, respectively.

Configuration	Test NAV	Test Sharpe	Test MDD	ΔNAV	ΔSharpe	ΔMDD
Price-only PPO (avg. over feature extractors)	1.7120	1.6446	-0.1651	0.0840	1.184	0.175
RQ1: Earnings-call signal	1.6807	1.5792	-0.1633	0.0443	1.112	0.175
RQ1: News signal	1.6665	1.5484	-0.1623	0.0287	1.083	0.175
RQ1: SEC signal	1.6919	1.6035	-0.1637	0.0523	1.136	0.176
RQ1: Senior-summary signal	1.6747	1.5573	-0.1626	0.0402	1.093	0.176
RQ1: Technical-text signal	1.6515	1.5203	-0.1662	0.0131	1.053	0.173
RQ2: Earnings-call risk ($\gamma=0.005, \beta=0.03$)	1.6893	1.6153	-0.1613	0.0210	1.133	0.175
RQ2: News risk ($\gamma=0.005, \beta=0.03$)	1.6868	1.6045	-0.1611	0.0143	1.120	0.175
RQ2: SEC risk ($\gamma=0.005, \beta=0.02$)	1.6840	1.6028	-0.1605	0.0152	1.120	0.175
RQ2: Senior-summary risk ($\gamma=0.005, \beta=0.02$)	1.6779	1.5862	-0.1642	0.0103	1.105	0.174
RQ2: Technical-text risk ($\gamma=0.010, \beta=0.02$)	1.7013	1.6290	-0.1627	0.0360	1.149	0.175
RQ3: All-in-one, 20k steps (avg. over seeds)	1.6595	1.5381	-0.1627	0.0113	1.066	0.175
RQ3: All-in-one, 60k steps (avg. over seeds)	1.6854	1.5630	-0.1584	-0.2728	0.919	0.154
RQ3: Short→long curriculum, Stage 1 (seed 137)	1.6964	1.5671	-0.1698	0.0420	1.089	0.161
RQ3: Short→long curriculum, Stage 2 (seed 137)	1.6362	1.4264	-0.1762	-0.2974	0.785	0.129
RQ3: Short→long curriculum, Stage 3 (seed 137)	1.5237	1.2182	-0.1867	-0.8248	0.360	0.083

show positive ΔNAV and $\Delta\text{Sharpe} > 1$, with SEC filings delivering the strongest generalisation (largest ΔNAV and ΔSharpe), and technical-text features the weakest. Importantly, their ΔMDD values remain very close to the baseline, signalling that the downside risk profile is stable across splits.

Risk-aware RQ2 configurations behave similarly: each risk channel yields a modest but positive ΔNAV and ΔSharpe slightly above one. Technical risk scores stand out as the best generaliser in this group, but the overall picture is that adding risk-based penalties does not harm robustness. The near-identical ΔMDD values suggest that risk shaping primarily adjusts the reward landscape without introducing fragile dependence on a particular split.

The RQ3 experiments reveal the limits of this robustness. Training an all-in-one policy for

20k steps still produces a small positive ΔNAV and $\Delta\text{Sharpe} \approx 1.07$, with ΔMDD comparable to the baseline, indicating acceptable generalisation despite the richer observation space. Extending training to 60k steps, however, flips the picture: ΔNAV becomes strongly negative and $\Delta\text{Sharpe} < 1$, while ΔMDD drops from ≈ 0.175 to ≈ 0.154 , consistent with the policy overfitting idiosyncratic patterns in the original split and suffering a deterioration in both average performance and tail risk on the alternative split.

The short→long curriculum (seed 137) shows a similar progression. Stage 1, which sees only short-horizon technical information, exhibits healthy generalisation with positive ΔNAV , $\Delta\text{Sharpe} > 1$ and moderate ΔMDD . As longer-horizon and textual modalities are added in Stages 2 and 3, the train–test NAV gap turns sharply negative and ΔSharpe collapses to 0.79 and then 0.36, respectively, while ΔMDD shrinks to 0.13 and 0.08. In other words, the later curriculum stages fit the original split so aggressively that their apparent improvements in-sample do not translate out-of-sample.

Taken together, these results show that moderate LLM augmentation (price-only, RQ1, RQ2 and RQ3-20k) can deliver consistent improvements that generalise across splits, whereas heavily trained, fully multi-modal agents (RQ3-60k and late curriculum stages) are much more prone to overfitting, despite looking attractive in the original backtest. This complements the training-stability analysis and highlights the need to regularise both training dynamics and model capacity when deploying LLM-driven portfolio policies.

4.6 Trading behaviour analysis

This section studies how different ways of integrating LLM-derived information change the trading behaviour of the learned policies. We focus on three behavioural dimensions: (i) trading intensity (number of trades and average transaction cost per decision step); (ii) weekly turnover and portfolio concentration (HHI); and (iii) the cross-sectional pattern of portfolio weights over time on the 2023-10 to 2025-09 test window.

Tables 4.14 and 4.15 summarise these statistics for the configurations considered in this chapter. The price-only row corresponds to the best-performing feature extractor (flattened price windows with a convolutional backbone). For RQ1 we report each single-source directional signal separately (SEC, earnings calls, news, senior summaries, technical text), each averaged over three seeds. For RQ2 we aggregate over five risk-score channels (SEC, earnings calls, news, senior summaries, technical text) and over the (β, γ) settings, averaged over three seeds, because their effects are nuanced rather than dominated by a single configuration. For RQ3 we



(a) Price-only PPO (best seed)



(b) RQ2: Tech risk penalty (best seed)



(c) RQ3: All-in-one, 60k steps (best seed)

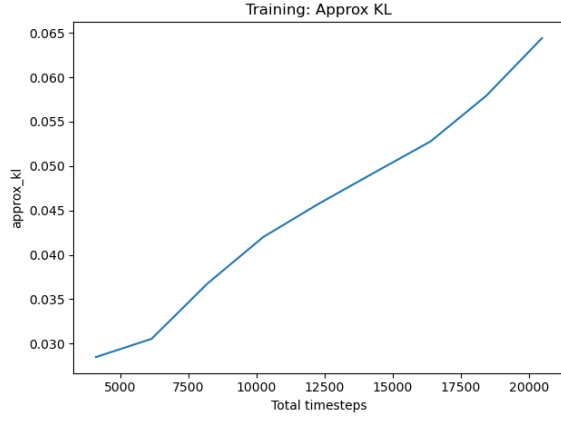


(d) RQ3: Short→Long Stage 3 (seed 137)

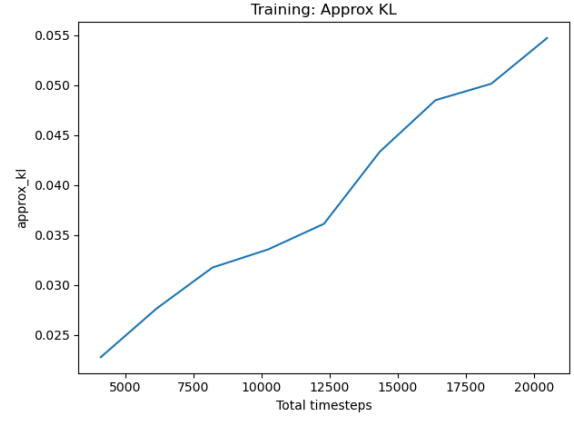
Figure 4.2: Evolution of mean episodic reward over training timesteps for the four representative configurations used in the stability analysis.

show both a single policy trained for 60k steps with all information sources available in one go (“all-in-one”) and the three stages of the short→long curriculum (seed 137), which reveals how trading style evolves as modalities are added.

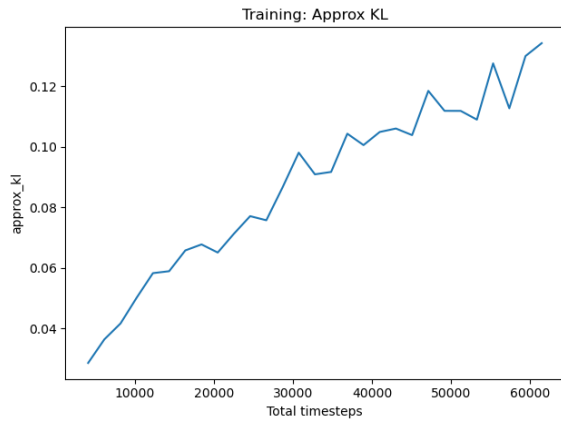
Aggregate trading patterns. The price-only PPO agent already trades fairly actively, with about 2,000 trades over the 100-week test set—roughly one trade per asset per week across the 20 equities—and median turnover around 1–2% per week, while keeping the portfolio close to equal-weighted (median HHI ≈ 0.0487). For reference, an exactly equal-weighted portfolio over 20 assets has $\text{HHI} = 1/20 = 0.05$, so values in the 0.048–0.050 range indicate a broadly diversified portfolio with only mild concentration. Directional LLM features in RQ1 separate into two regimes. Earnings-call and SEC signals induce behaviour very close to the baseline, with similar costs and turnover and essentially unchanged HHI. News, senior-summary and technical-text signals, in contrast, lead to much more aggressive trading: median turnover jumps to 5–7% and average cost per step increases by a factor of 2–3, but portfolio concentration remains anchored



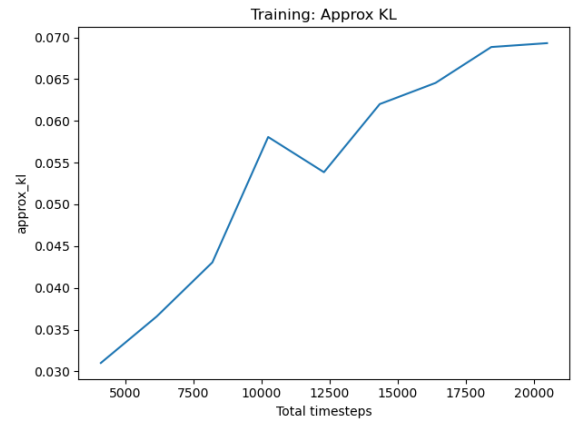
(a) Price-only PPO (best seed)



(b) RQ2: Tech risk penalty (best seed)



(c) RQ3: All-in-one, 60k steps (best seed)



(d) RQ3: Short→Long Stage 3 (seed 137)

Figure 4.3: Approximate KL divergence between consecutive policies for the same four configurations.

near the baseline. These signals appear to generate frequent small reallocations without driving the agent into persistent single-name bets.

The risk-aware RQ2 configuration stays very close to the price-only regime once we aggregate across risk channels and hyperparameters: trading intensity and HHI are almost unchanged (Tables 4.14 and 4.15). In this setting, risk penalties mainly reshape the reward without materially altering the mechanics of trading.

The RQ3 all-in-one agent operates in a different regime. With all information sources available for 60k training steps, it incurs around five times the transaction cost of the baseline and an order-of-magnitude higher median turnover ($\approx 13\%$ per week), together with a persistent upward shift in HHI. The curriculum run makes this trend explicit: as more modalities are added, median turnover rises from about 7.7% to 17.8%, and HHI moves from roughly 0.049 to 0.056, especially in the upper tail. Although the curriculum rows are from a single seed, they highlight a clear drift towards more aggressive and more concentrated trading as the information set

Table 4.14: Trading intensity on total trades and average cost per step(test set)

Configuration	Total trades	Avg. cost per step
Price-only PPO (baseline)	2,000	62.36
RQ1: Earnings-call signal only	2,000	59.50
RQ1: News signal only	1,999.7	201.35
RQ1: SEC signal only	2,000	67.37
RQ1: Senior-summary signal only	1,999.7	158.58
RQ1: Technical-text signal only	2,000	188.45
RQ2: Risk scores in reward (aggregated)	1,999.6	63.94
RQ3: All-in-one, 60k steps	2,000	350.44
RQ3: Curriculum, Stage 1 (short→long, seed 137)	2,000	203.40
RQ3: Curriculum, Stage 2 (seed 137)	2,000	323.35
RQ3: Curriculum, Stage 3 (seed 137)	2,000	476.35

Table 4.15: Trading intensity on turnover and HHI(test set)

Configuration	Med. turnover	90th perc. turnover	Med. HHI	90th perc. HHI
Price-only PPO (baseline)	0.0108	0.0204	0.0487	0.0495
RQ1: Earnings-call signal only	0.0111	0.0196	0.0486	0.0491
RQ1: News signal only	0.0716	0.0984	0.0491	0.0497
RQ1: SEC signal only	0.0120	0.0295	0.0486	0.0492
RQ1: Senior-summary signal only	0.0549	0.0770	0.0489	0.0498
RQ1: Technical-text signal only	0.0677	0.0922	0.0489	0.0497
RQ2: Risk scores in reward (aggregated)	0.0122	0.0223	0.0488	0.0494
RQ3: All-in-one, 60k steps	0.1291	0.1720	0.0525	0.0551
RQ3: Curriculum, Stage 1 (seed 137)	0.0771	0.0970	0.0492	0.0498
RQ3: Curriculum, Stage 2 (seed 137)	0.1205	0.1685	0.0520	0.0536
RQ3: Curriculum, Stage 3 (seed 137)	0.1776	0.2480	0.0564	0.0629

grows.

Temporal dynamics of concentration and turnover. Figure 4.4 plots the HHI trajectories over the test period for price-only, RQ1 with SEC features, RQ2 with technical risk scores, and RQ3 all-in-one. The price-only and RQ2 curves fluctuate in a tight band (≈ 0.048 – 0.050), with only mild spikes around early 2024 and mid-2025. The SEC-based RQ1 agent exhibits somewhat larger swings and temporary increases around filing-heavy periods but quickly mean-reverts. The RQ3 all-in-one agent, in contrast, shows an early spike to about 0.068 and then spends much of 2024–2025 at an elevated concentration level, consistent with Table 4.15.

Turnover trajectories in Figure 4.5 show a similar picture. Price-only and RQ2 mainly adjust positions by 1–3% per week, with occasional bursts around major market moves. The SEC-featured RQ1 agent fires more such bursts but remains in a moderate regime overall. The RQ3 all-in-one agent instead sustains high turnover between roughly 7% and 23% almost every week,

explaining the large cumulative transaction costs.

Cross-sectional allocation patterns. The portfolio-weight heatmaps in Fig. 4.6 reveal which individual stocks and sectors each configuration prefers, and thus provide evidence about what the policies have actually learned from their respective observation spaces.

For the price-only agent (Fig. 4.6a), the policy consumes only close-price based features over a rolling window (via convolutional or flattening feature extractors), and yet it still produces a distinctive allocation pattern. The policy consistently tilts towards a narrow subset of large growth and platform firms: AMZN (consumer discretionary), GOOGL and META (communication services), and NFLX. Other mega-cap names such as AAPL (information technology) and WMT (consumer staples) rarely receive comparable weights, indicating that the policy is not simply tracking market capitalisation or equal weights but has learned a state-dependent mapping from price dynamics to stock selection. Throughout much of the first half of the test window the agent also assigns above-average weight to XOM (energy), suggesting a persistent view on that sector given the realised return and covariance structure. Cash holdings are used intermittently as a buffer: the policy periodically increases its cash weight during periods where the learned mapping deems the cross-section less favourable, but generally remains substantially invested in equities. In practice, this behaviour aligns with realised price paths: META and NFLX experienced strong upward trends over the test period, whereas AMT, INTC and UNH saw drawdowns or pronounced volatility.

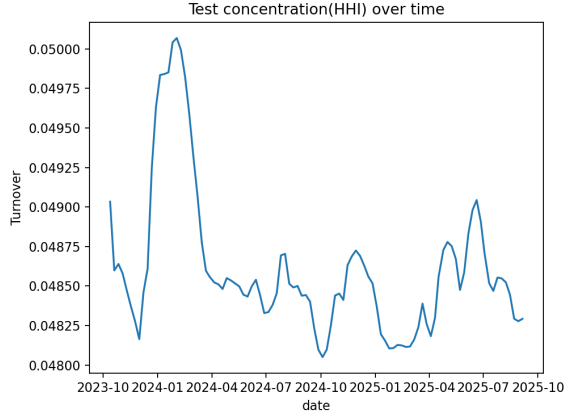
The RQ1 SEC configuration (Fig. 4.6b) preserves the same core cluster of AMZN, GOOGL, META and NFLX as in the price-only baseline, but exhibits a more distinctive rotation from around 2025-06 onwards. In this later period the portfolio becomes dominated by an “AI hardware and cloud” cluster comprising AMD, NVDA, MSFT and GOOGL, while weights assigned to cash and several more defensive names (UNH in healthcare, JPM and V in financials, and INTC) are visibly damped. This pattern is consistent with the agent using information in SEC filings to identify and concentrate on firms explicitly associated with the widely publicised AI investment boom in 2024–2025, producing an allocation trace that is not reproduced with comparable strength in the other configurations. In other words, the SEC-augmented agent appears to have learned a state-dependent mapping from filing-derived signals to an AI-intensive sector bet, rather than merely amplifying the price-only policy.

Under the RQ2 technical-risk configuration (Fig. 4.6c), the heatmap shows a different style. While AMZN remains important, the portfolio places persistent emphasis on XOM (energy), UNH (defensive healthcare) and MU (semiconductors), and generally spreads weight more evenly

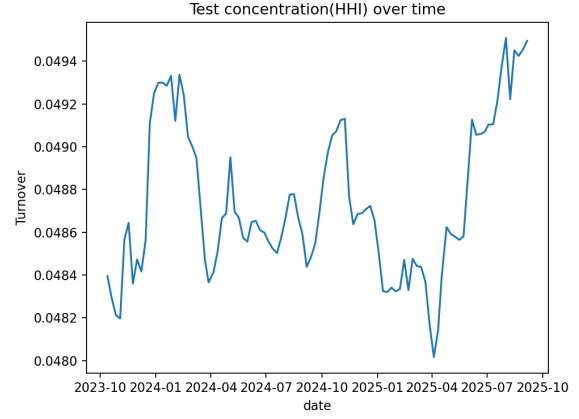
across sectors. These names are less purely “high-beta growth” and more defensive or diversifying. Combined with the lower median turnover and slightly higher concentration reported in Table 4.15, this suggests that the risk-penalised agent has learned to trade off return against second-order properties of the returns (volatility and co-movement), favouring stocks that offered more favourable risk–return trade-offs in this window. The fact that this more defensive cross-sectional pattern does not appear with the same intensity in the price-only or SEC-only agents supports the view that the risk-aware formulation induces a qualitatively different policy rather than a scaled version of the baseline.

The RQ3 all-in-one policy trained for 60k steps (Fig. 4.6d) uses all modalities jointly and produces yet another distinct allocation signature. Unlike the other configurations, cash is kept very close to zero across the entire test period: the agent is almost always fully invested and expresses its views entirely through cross-sectional rebalancing rather than market timing. The dominant holdings are concentrated in a small set of liquid, large-cap names such as AMZN, BAC, XOM and V, but these positions appear as sharp, short-lived spikes rather than the smoother, multi-week bands seen in the price-only and RQ1/RQ2 policies. The agent does not lock into the AI-related cluster identified by the SEC-only policy; instead, it rapidly rotates between sectors and names, with transient excursions into energy, financials and consumer-related stocks. Because such high-frequency cross-sectional rotations are largely absent from the other agents that see the same underlying market, they are best interpreted as a consequence of exposing the policy to all information sources simultaneously for a longer training horizon (60k timesteps versus 20k for the other configurations). This extended training increases the capacity for overfitting idiosyncratic patterns in the finite backtest window: the policy has learned a very aggressive, fully-invested mapping from the rich multimodal state space to portfolio weights, but that mapping is fragile and difficult to regularise, leading to highly non-stationary allocation patterns despite comparable average risk and turnover statistics.

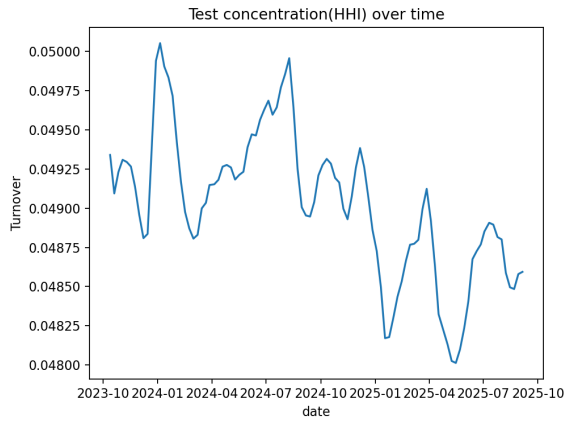
Summary. Overall, price-only PPO and the risk-aware RQ2 agent trade smoothly, remain well diversified, and make moderate use of cash: they implement relatively conservative, low-friction mappings from their respective observation spaces to actions. Single-source directional signals in RQ1 split into near-baseline regimes (earnings-call and SEC signals), which mainly refine the price-only allocation, and high-activity regimes (news, senior summaries, technical text) that substantially increase turnover and transaction costs while keeping concentration close to an equal-weight baseline. Among these, the SEC-based agent is the only one that clearly learns a persistent AI-intensive sector bet (AMD, NVDA, MSFT, GOOGL) during the 2024–2025 AI



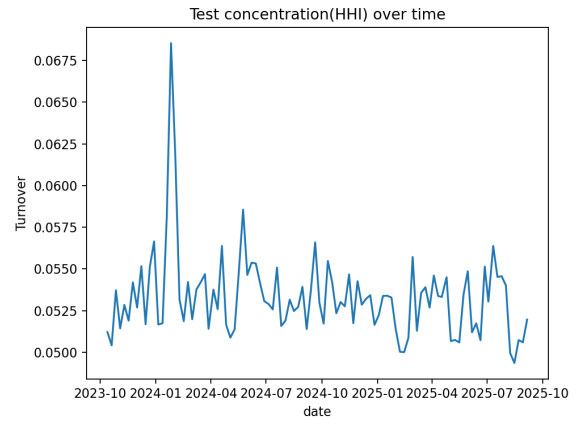
(a) Price-only PPO (baseline)



(b) RQ1: SEC directional features



(c) RQ2: Technical risk scores



(d) RQ3: All-in-one, 60k steps

Figure 4.4: Portfolio concentration (HHI) trajectories on the test set for four representative configurations.

boom, a pattern not replicated with comparable strength by the other agents.

Full all-in-one integration in RQ3 yields the most aggressive trading style: the 60k-step policy is almost always fully invested, rotates rapidly between a small set of large-cap names with short-lived spikes in weights, and exhibits structurally high turnover and elevated HHI. Given that this policy is trained three times longer than the other configurations (60k versus 20k steps), these behaviours are best interpreted as a high-capacity mapping that risks overfitting to idiosyncratic patterns in the finite backtest window rather than a stable, low-complexity strategy. Taken together, the results suggest that LLM-derived signals do induce qualitatively different trading behaviours across configurations, but there is a clear trade-off between risk-adjusted performance, interpretability and stability on one hand, and transaction cost and overfitting risk on the other. This trade-off is central for any practical deployment of LLM-driven portfolio agents.

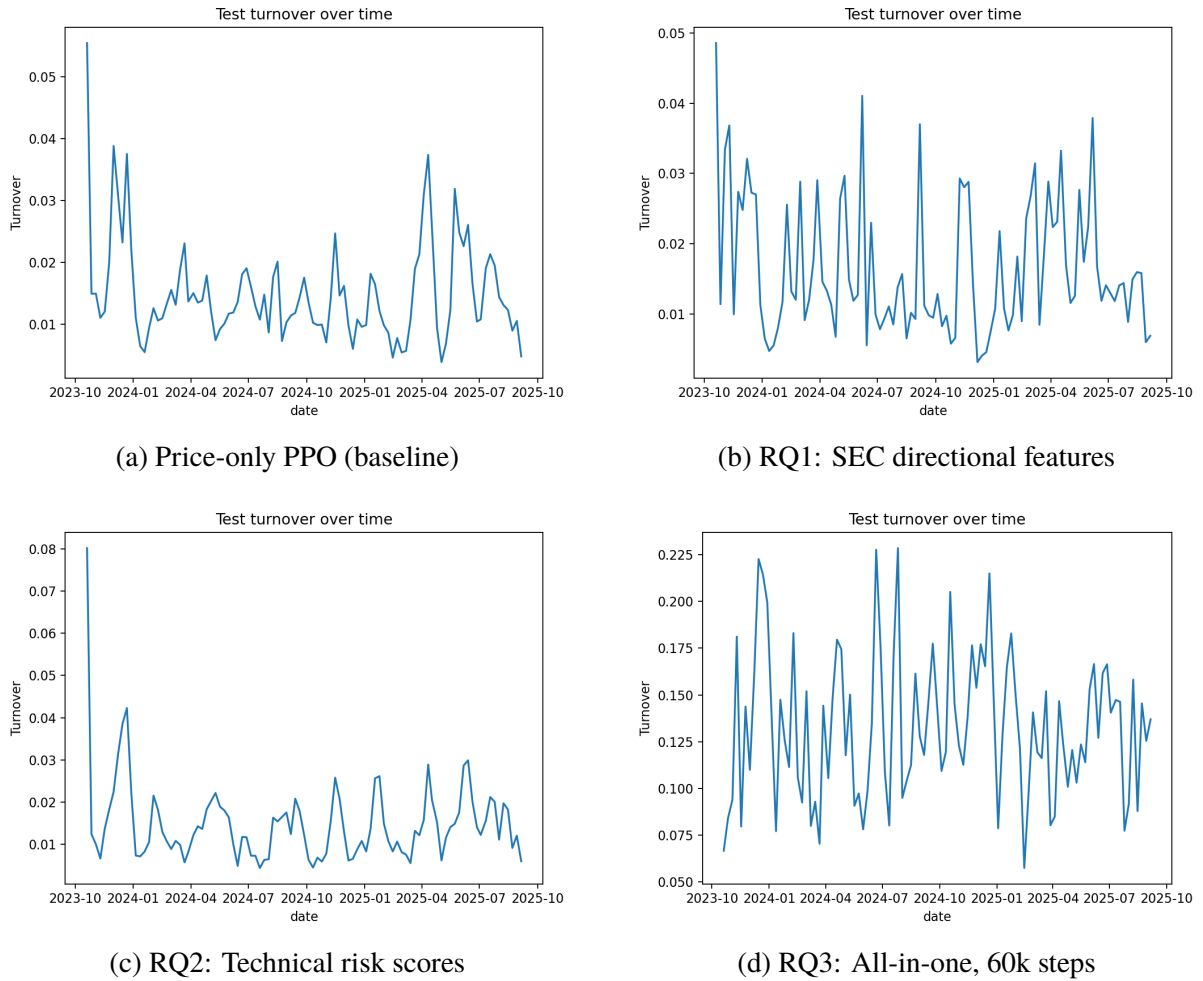
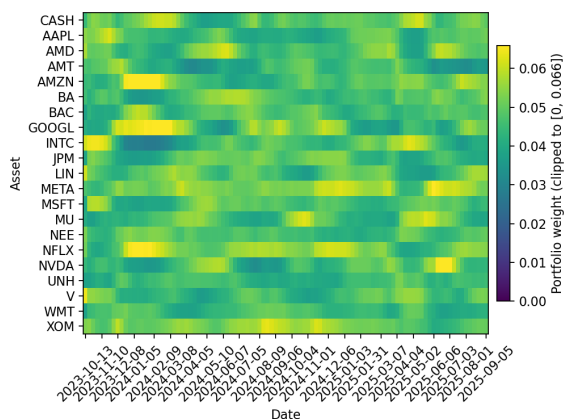


Figure 4.5: Weekly portfolio turnover on the test set for the same four configurations.

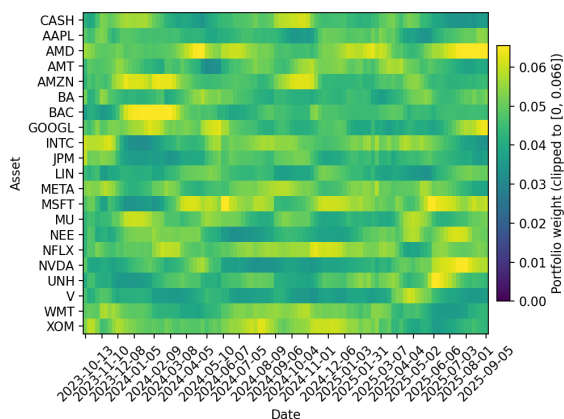
Market-efficiency perspective on LLM integration

The empirical results across RQ1–RQ3 can be summarised at a higher level. Table 4.16 aggregates test-window performance for all baselines and PPO variants. Three robust patterns emerge:

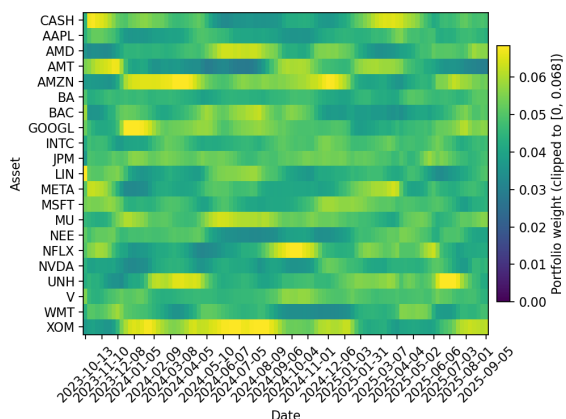
- Classical EW and MV baselines are already strong, delivering cumulative gains of roughly 65% and 45% over the test window despite several constituent stocks experiencing severe drawdowns. Any LLM-augmented agent is competing against these strong baselines, not against a weak benchmark.
- The price-only PPO agent reliably and slightly exceeds EW, with the best seed delivering about 8 percentage points more cumulative return than EW at similar Sharpe and MDD. This shows that a standard RL method can squeeze a modest edge out of price dynamics alone in this setting.
- Once LLM-derived signals are introduced, several configurations remain competitive with



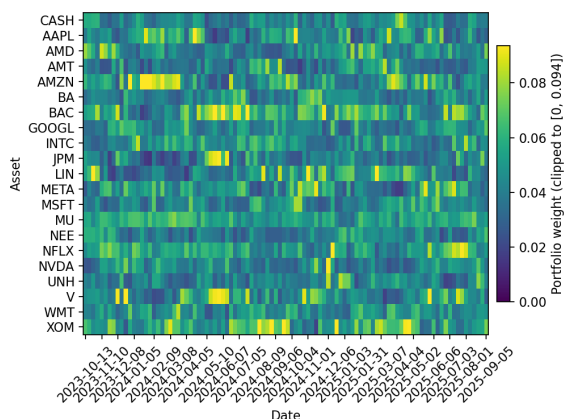
(a) Price-only PPO (baseline)



(b) RQ1: SEC directional features



(c) RQ2: Technical risk scores



(d) RQ3: All-in-one, 60k steps

Figure 4.6: Heatmaps of weekly portfolio weights (cash + 20 equities) on the test set. Colours are clipped to the 99th percentile of the distribution in each configuration.

or slightly stronger than EW in terms of NAV multiple, but none consistently beats the price-only PPO across seeds and research questions. In many cases, LLM integration amplifies trading intensity and concentration without producing a compensating improvement in risk-adjusted performance.

One natural interpretation is in terms of the semi-strong form of the efficient market hypothesis. The LLM analysts are deliberately restricted to public information channels—regulatory filings, earnings calls, news, and price-derived technical narratives—exactly the types of information that semi-strong efficiency posits are rapidly incorporated into prices in deep, liquid markets such as large-cap US equities. From this perspective, LLMs act primarily as compressive devices: they reorganise and summarise already-priced public information into structured scores. Any residual predictive edge in these summaries must therefore be weak, noisy, and highly sensitive to regime shifts and transaction costs. The results in Table 4.16 are consistent with this view: LLM signals occasionally help a specific configuration for a specific horizon,

but once transaction costs, finite sample size and hyperparameter sweeps are accounted for, the net effect is neutral to slightly negative relative to a strong, price-only PPO backbone.

At the same time, the behavioural analysis shows that LLM integration is not meaningless. The SEC-based RQ1 agent clearly learns to express an AI-intensive sector bet that is economically interpretable and aligned with the 2024–2025 narrative. The RQ2 technical-risk configuration leans into defensive, diversifying names and trades less aggressively. The all-in-one RQ3 agent behaves as a high-capacity, fully invested, high-turnover strategy. In each case, the LLM channels imprint a distinct trading style on top of the common PPO machinery. What they do *not* do, under the current design and data, is deliver statistically robust improvements over an already strong price-only agent. This gap between behavioural richness and limited headline performance is a key motivating factor for the broader discussion in Chapter 5 on how to re-design both LLM integration mechanisms and evaluation protocols for LLM-driven portfolio agents.

Table 4.16: Summary of out-of-sample trading performance across baselines, price-only PPO, and LLM integration regimes (cash floor = 0, cost = 10 bp).

Regime	Strategy	NAV mult.	NAV (ann.)	Weekly Sharpe	MDD
Classical baselines (test window)					
Baseline	EW	1.6539	1.2990	1.64	-0.16
Baseline	MV	1.4573	1.2160	1.27	-0.20
Price-only PPO (test window backbone)					
PPO (price-only)	seed 42	1.7267	1.3280	1.67	-0.16
PPO (price-only)	seed 137	1.7191	1.3250	1.65	-0.17
PPO (price-only)	seed 256	1.6902	1.3140	1.61	-0.16
PPO (price-only)	mean (3 seeds)	1.7120	1.3228	1.64	-0.17
RQ1: LLM features as observations (3-seed means, test window)					
RQ1	PPO (price-only backbone)	1.7120	1.3228	1.6446	-0.1651
RQ1	+ SEC signal	1.6919	1.3130	1.6035	-0.1637
RQ1	+ EC signal	1.6807	1.3080	1.5792	-0.1633
RQ1	+ Senior signal	1.6747	1.3052	1.5573	-0.1626
RQ1	+ News signal	1.6665	1.3016	1.5484	-0.1623
RQ1	+ Technical-text signal	1.6515	1.2951	1.5203	-0.1662
RQ2: LLM-based risk shaping (best-Sharpe configs, test window)					
RQ2	PPO (price-only backbone)	1.7120	1.3228	1.6446	-0.1651
RQ2	+ EC risk penalty	1.6893	1.3134	1.6153	-0.1613
RQ2	+ News risk penalty	1.6868	1.3124	1.6044	-0.1611
RQ2	+ SEC risk penalty	1.6840	1.3113	1.6028	-0.1605
RQ2	+ Senior risk penalty	1.6711	1.3061	1.5862	-0.1639
RQ2	+ Technical-text risk penalty	1.7013	1.3183	1.6290	-0.1627
RQ3: Curriculum vs. all-in-one feature integration (3-seed means, test window)					
RQ3	<i>Curriculum: Long → Short (SEC+EC → +News → +Tech)</i>				
RQ3	Stage 1 (SEC+EC)	1.6770	1.308	1.5622	-0.1658
RQ3	Stage 2 (SEC+EC+News)	1.6836	1.311	1.5694	-0.1669
RQ3	Stage 3 (SEC+EC+News+Tech)	1.6528	1.299	1.4905	-0.1748
RQ3	<i>Curriculum: Short → Long (+Tech → +News → +SEC+EC)</i>				
RQ3	Stage 1 (Tech)	1.6550	1.299	1.5206	-0.1662
RQ3	Stage 2 (Tech+News)	1.6048	1.279	1.3965	-0.1733
RQ3	Stage 3 (Tech+News+SEC+EC)	1.5462	1.254	1.2870	-0.1782
RQ3	<i>All-in-one: full feature set from start</i>				
RQ3	All-in-one (20k steps)	1.6595	1.301	1.5381	-0.1627
RQ3	All-in-one (60k steps)	1.6854	1.312	1.5630	-0.1584

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Overview

This thesis investigated whether and how large language models (LLMs) can be used as structured information providers for a reinforcement learning (RL) agent that allocates capital in a weekly portfolio management setting. The empirical study instantiated this question in a 20-asset U.S. equity universe over the period 2019–04–20 to 2025–09–14, using the custom portfolio management environment introduced in Chapter 3. This environment exposes a structured observation space, risk-aware reward shaping, and a realistic transaction-cost model. A multi-component LLM pipeline produced analyst-style summaries and risk assessments from multiple textual sources—regulatory filings, earnings-call transcripts, news, and technical commentary—which were then integrated into the RL system either as additional observation features or as exogenous risk signals in the reward.

Chapter 4 addressed three research questions:

- **RQ1 (LLM-derived features as observations).** How does a portfolio agent based on proximal policy optimisation (PPO), operating on a structured observation space that includes LLM-derived features, perform relative to a price-only PPO agent and classical, non-LLM baselines such as the equal-weight (EW) and minimum-variance (MV) portfolios? How do different textual channels (regulatory filings, earnings calls, news, and technical commentary) differ in their impact on trading behaviour and risk?
- **RQ2 (LLM-derived risk shaping).** When LLMs are used to produce risk scores that directly shape the reward (via penalties on risky exposures), do they materially alter the risk profile, tail behaviour, and robustness of the learned policies, particularly in stressed market regimes?
- **RQ3 (Curriculum versus all-in-one training).** Does introducing LLM-derived information via a curriculum (adding channels in stages) improve training stability and out-of-sample behaviour compared with training a single agent on the full multimodal feature set from the outset?

The experimental design in Chapter 4 systematically varied (i) the observation space (price-only; price plus single textual channel; price plus multiple channels), (ii) the reward (with and without LLM-based risk penalties), and (iii) the training schedule (20,000 versus 60,000 total gradient steps; curriculum versus all-in-one), with multiple random seeds per configuration. Performance was evaluated using both conventional risk–return metrics (net asset value (NAV), annualised return, Sharpe ratio, maximum drawdown (MDD)) and behavioural diagnostics (turnover, Herfindahl–Hirschman Index (HHI) of portfolio concentration, cash usage, and regime-specific behaviour during market stress events). Table 4.16 (Chapter 4, Table 4.16) provides a global summary of these metrics across baselines and all LLM integration regimes.

This chapter interprets these empirical findings, extracts design lessons for LLM–RL integration, and discusses limitations and future directions. Section 5.2 synthesises the empirical answers to RQ1–RQ3 and summarises the main robustness patterns and failure modes observed in Chapter 4. Section 5.3 draws broader design implications for LLM-driven RL systems beyond the specific portfolio setting. Section 5.4 discusses limitations and threats to validity. Section 5.5 outlines directions for further research, and Section 5.6 offers concluding remarks.

5.2 Summary of empirical findings

5.2.1 Empirical answers to RQ1–RQ3

RQ1: LLM-derived observations versus price-only PPO. The first research question asked whether augmenting the numerical observation space with LLM-derived signals from different textual channels materially improves trading performance and alters trading behaviour relative to a price-only PPO baseline.

(1) *Strength of classical baselines and price-only PPO.* On the 100-week test window, the classical EW and MV portfolios already provide strong non-LLM benchmarks: NAV multiples are in the range 1.45–1.65 with weekly Sharpe ratios above 1.2, despite several underlying names (e.g. INTC, UNH, AMT) experiencing pronounced drawdowns or collapses during the period. Against this backdrop, the tuned price-only PPO agent improves test performance modestly but consistently. The best seed attains a NAV multiple of approximately 1.73× (around 7–8% higher than the EW benchmark) with weekly Sharpe similar to or slightly above EW and comparable MDD (around –16%). The cross-seed average price-only PPO configuration in Table 4.16 therefore serves as a strong RL backbone: it is hard to beat, yet clearly not degenerate.

(2) *Trading performance under LLM-derived observations.* When single-channel LLM-

derived features are appended to the price-based observations (RQ1), no configuration surpasses the tuned price-only PPO in test Sharpe or NAV. SEC- and EC-augmented agents form the top tier among LLM observation variants: their test NAV multiples and Sharpe ratios are close to the price-only backbone and remain competitive with EW and MV, but they do not provide a robust performance gain beyond the pure-price PPO. Higher-frequency channels—news, technical commentary, and the senior aggregator—perform worse on average. Their Sharpe ratios often drop below the price-only agent and, in some seeds, even below EW/MV. Taken together, Table 4.16 shows that in this weekly, finite-horizon setting, LLM-derived observation features from public text do *not* deliver a clear, statistically robust improvement in risk-adjusted performance over a well-tuned price-only PPO.

(3) *Trading behaviour and risk profile.* The impact of LLM observations is more visible in trading style. For SEC and EC configurations, total trades, median turnover, and HHI stay close to the price-only PPO: these agents rebalance at similar frequency and maintain diversified portfolios with HHI around 0.049–0.050, not far from the equal-weight benchmark $1/20 = 0.05$. By contrast, news, technical, and senior-text configurations substantially increase median turnover (to 5–7% or more per week) and average transaction cost per step, and induce more frequent spikes in HHI. In other words, adding higher-frequency text channels amplifies the *risk profile* of the policy—more churning and more concentration—without commensurate gains in Sharpe or NAV. This is precisely the pattern highlighted in the defence: LLM integration can make the agent trade harder and take more concentrated bets, but does not reliably translate that extra activity into better risk-adjusted performance.

(4) *Cross-sectional allocation patterns.* The portfolio-weight heatmaps clarify what the price-only and SEC-augmented agents actually learn. The price-only PPO configuration already exhibits clear overweight and underweight patterns: names such as META and NFLX typically carry weights above 5% over extended periods, while AMT, INTC, and UNH are systematically underweight, often below 5%. Ex post, this aligns with realised price paths: META and NFLX experienced pronounced upward trends across the test window, whereas AMT, INTC, and UNH were either flat or volatile with significant drawdowns. This suggests that the price-only PPO has learned a sensible cross-sectional mapping from price dynamics to portfolio weights.

When SEC-derived features are added, the allocation heatmap changes in a distinctive way. After mid-2025, the SEC+price PPO agent shifts more capital towards an “AI-intensive” cluster (AMD, NVDA, MSFT, GOOGL) than the price-only PPO, while tilting away from several defensive or underperforming names. This behaviour matches the narrative in SEC filings during the AI investment boom: regulatory disclosures emphasise themes of AI infrastructure, cloud

build-out, and capex cycles for these firms. The SEC-augmented agent therefore appears to use filing-derived signals to strengthen its AI-sector tilt relative to the price-only policy. However, as Table 4.16 shows, this change in allocation does *not* produce a test Sharpe or NAV that clearly dominates the pure-price PPO: the SEC variant ends slightly below the price-only configuration on those metrics.

(5) *Efficient-market interpretation.* From a semi-strong efficient-market perspective, the RQ1 results are unsurprising. Regulatory filings, earnings calls, and mainstream news for deep, large-cap U.S. equities are public information that professional investors are expected to process rapidly. The LLM analysts in this thesis therefore compress information that is largely already priced in; any residual edge in their signals is small, noisy, and fragile relative to transaction costs, regime shifts, and the limited sample of weekly decisions. Within this interpretation, LLM-derived observation features act more like structured behavioural regularisers—modulating sector and name tilts and amplifying certain narratives—than like robust new sources of standalone alpha.

RQ2: LLM-based risk shaping and tail behaviour. RQ2 studied whether using LLM-derived signals as risk-shaping terms in the reward, rather than only as observations, can improve tail risk and trading robustness. Concretely, portfolio-level risk exposures z_t^{llm} from different channels were combined with an endogenous price-based tail-risk statistic E_t inside a linear reward:

$$r_t = r_t^{\text{return}} - \beta z_t^{\text{mkt}} E_t - \gamma z_t^{\text{llm}} \quad (\lambda = 1),$$

with (β, γ) swept over a small grid.

(1) *Trading performance under risk shaping.* The best Sharpe configurations for each risk channel (EC, SEC, news, technical, and senior) all remain close to, but slightly below, the price-only PPO in test NAV and Sharpe. For example, the technical-risk configuration achieves a test NAV multiple around 1.70 $\times$ and weekly Sharpe ≈ 1.63 , versus 1.71 $\times$ and 1.64 for the price-only PPO; SEC and EC risk penalties yield slightly smaller NAV and Sharpe again. No (β, γ) pair strictly dominates the price-only baseline on both NAV and Sharpe, and the improvements over the RQ1 observation-only variants are modest. Thus, in aggregate, the linear risk-shaping mechanism is performance-neutral to mildly harmful: it does not unlock higher risk-adjusted returns beyond the pure-return objective.

(2) *Tail behaviour and trading style.* Risk shaping produces more systematic and interpretable changes in behaviour than in headline performance. Maximum drawdowns are typi-

cally reduced by a few percentage points relative to the price-only PPO, and turnover and cost per step remain in a moderate regime. Allocation heatmaps show that the risk-shaped agents maintain diversified portfolios but tilt towards more defensive or diversifying names (e.g. energy and healthcare) and away from stocks that the LLM pipeline repeatedly flags as high risk. In this sense, LLM-derived risk signals function as soft guardrails: they discourage the policy from leaning too heavily into fragile exposures, even when those exposures look attractive on a pure-return basis.

(3) *Interpretation.* Taken together, the RQ2 experiments suggest that treating LLM risk scores as exogenous penalty terms is a sensible way to inject text-informed caution into the RL objective. However, the benefits are incremental and primarily visible as slightly improved tail metrics and more conservative sector tilts; they do not overturn the central finding that the strongest risk-adjusted performance comes from a well-calibrated price-only PPO backbone.

RQ3: Curriculum versus all-in-one training. RQ3 compared curriculum-style training—in which channels are added in three stages (for example, SEC+EC → +news → +technical) with 20,000 steps per stage—against an *all-in-one* regime that trains a single agent on the full feature set for 20,000 or 60,000 steps.

(1) *Final performance.* When total training budgets are matched, the 60,000-step all-in-one configuration generally matches or slightly outperforms the final curriculum stage in test NAV and Sharpe, while achieving comparable or slightly better MDD. In Table 4.16, the best 60,000-step all-in-one agent attains NAV and Sharpe close to the price-only PPO baseline but does not provide a decisive improvement; the final curriculum stages tend to sit below both all-in-one 60,000-step and price-only PPO configurations.

(2) *Training dynamics and behavioural instability.* Training curves and diagnostics reveal that curriculum stages are easy to optimise in-sample: mean episodic reward, training NAV, and training Sharpe all increase as modalities are added. However, approximate policy KL grows and the train–test gap widens at later stages: the final curriculum stage often has the highest training Sharpe but the lowest test Sharpe and most negative test MDD among curriculum stages. Behaviourally, turnover and HHI increase from stage to stage, and allocation heatmaps show increasingly aggressive, short-lived bets as more modalities are added. In contrast, the 60,000-step all-in-one configuration achieves smoother training curves and less abrupt behavioural shifts, even though it sees the full feature set from the outset.

(3) *Interpretation.* The RQ3 experiments indicate that, in this finite weekly dataset, staged feature introduction does not provide a robust advantage over a simple all-in-one training regime

with a sufficient step budget. Abruptly changing the observation space mid-training introduces a form of distribution shift that can lead to catastrophic interference and overfitting to idiosyncratic patterns in the training period. The 60,000-step all-in-one agent is itself more aggressive than the 20,000-step backbone—higher turnover and HHI— but, on balance, it is at least as stable and effective as the final curriculum stage. Curriculum design based solely on “turning on” more features over time is therefore not a guaranteed performance enhancement in data-limited, non-stationary RL settings.

5.2.2 Robustness and failure modes

Beyond headline metrics, Chapter 4 and the defence slides highlight several robustness patterns and failure modes.

Seed sensitivity and finite-sample variability. Across RQ1–RQ3, there is noticeable variation across random seeds. Some configurations that appear attractive under one seed are much less compelling under another. This is typical for PPO in small-data, non-stationary environments and reinforces the need to report cross-seed averages, not single-run anecdotes. In most tables, the price-only PPO remains strong and stable across seeds, whereas some LLM-heavy configurations fluctuate more.

Training stability versus generalisation. The training diagnostics in Chapter 4 show that the main configurations (price-only, RQ1 SEC, RQ2 technical risk, RQ3 all-in-one) are numerically stable: mean episodic reward increases smoothly; approximate KL remains within acceptable bounds; value-function explained variance stays high. There is no evidence of catastrophic optimisation failure or runaway policy divergence. The main issue is not instability of the PPO optimiser, but rather limited generalisation: more complex, LLM-heavy or curriculum configurations continue to improve in-sample metrics without translating those gains into better test NAV or Sharpe. The generalisation tables confirm that train–test gaps widen as model complexity and training budget increase.

Behavioural failure modes under LLM integration. The most salient failure modes arise at the behavioural level. Configurations that rely heavily on higher-frequency LLM channels (news, technical) or on long all-in-one/curriculum training horizons tend to:

- exhibit elevated and persistent turnover, leading to large cumulative transaction costs;
- develop spikes in HHI, indicating more concentrated bets on a small subset of names; and

- react sharply to the addition of new channels (in curricula), with sudden changes in trading style and poorer out-of-sample behaviour.

These behaviours correspond to “risk amplification” in the sense used in the defence: the agent trades harder, concentrates more, and becomes more fragile to regime changes, without delivering a robust gain in Sharpe or NAV.

In summary, the empirical evidence depicts a landscape where (i) classical baselines and a tuned price-only PPO provide strong anchors; (ii) LLM integration is feasible and behaviourally meaningful, but does not yet improve risk-adjusted performance beyond the price-only backbone; and (iii) the main dangers of LLM integration lie in hidden behavioural pathologies—excessive churn and concentration—rather than in obvious training collapse.

5.3 Design lessons and implications

5.3.1 Baseline-first, guarantee-seeking use of LLM signals

A central lesson is the importance of strong, transparent baselines. The EW and MV portfolios set a high bar for non-LLM methods; the price-only PPO agent, with its modest but consistent improvements over EW/MV under realistic transaction costs, sets the bar for RL. Most LLM-based configurations fail to clear this bar on risk-adjusted performance and, in some cases, fall below it while incurring higher trading costs and more concentrated positions.

This motivates a baseline-first, guarantee-seeking approach to LLM integration:

- **Start from a robust price-only PPO backbone.** Before adding any LLM-derived signals, the RL agent should be tuned and stress-tested on price and basic technical features alone. Behavioural diagnostics (turnover, HHI, cash usage) and test metrics (NAV, Sharpe, MDD) should indicate a stable and interpretable policy that is competitive with EW/MV.
- **Require “no-regret” behaviour from LLM additions.** New LLM channels should not substantially degrade the risk profile relative to the price-only backbone. In the experiments, SEC and EC observation channels and LLM-based risk shaping generally satisfied this criterion; news and technical-text observation channels often did not.
- **Evaluate behaviour, not just averages.** As seen in Chapter 4, agents with similar Sharpe ratios can exhibit dramatically different trading patterns and tail risks. Any LLM integration should therefore be evaluated on turnover, HHI, regime behaviour, and sensitivity to seed and split, not just on summary performance metrics.

This conservative stance is consistent with a semi-strong efficient-market view of large-cap equities: when public information is quickly reflected in prices, LLM-extracted fundamentals and sentiment are more likely to act as structured regularisers of behaviour and tail risk than as reliable new sources of excess return. In practice, LLM channels should be treated as *optional* modules that must earn their place relative to a strong price-only PPO, rather than as default components.

5.3.2 LLM risk as exogenous signals, not direct controllers

The architecture in this thesis deliberately separates LLMs and RL in their roles. LLMs act as structured information providers: they transform raw text into numeric signals (directional scores, risk scores, confidences, and rationales) under carefully designed schemas and leakage-aware prompts. PPO remains the sole decision-maker, learning a policy that maps the combined numeric observation space and reward to actions under explicit transaction costs and portfolio constraints.

The experiments support this division of labour:

- **Separation of concerns.** LLMs are not asked to output actions or trade lists, which mitigates the risk that hallucinations or idiosyncratic LLM behaviour directly drive execution. Instead, they provide structured summaries that influence, but do not deterministically dictate, the policy.
- **Guardrail interpretation of risk shaping.** When LLM risk scores are injected into the reward, they function as soft constraints rather than as hard rules. The policy can still take risky positions if the expected return justifies the risk penalty. This yields more conservative behaviour in stressed regimes and modest MDD reductions, at the cost of some upside—a trade-off that can be tuned via (β, γ) .
- **Auditability.** Numeric LLM outputs are easier to calibrate and audit than free-form action proposals. Misaligned or noisy risk scores can be detected by comparing them against realised losses and alternative risk metrics, and the reward-shaping coefficients can be adjusted accordingly.

These observations suggest that, at least in risk-sensitive domains like portfolio management, LLMs are best used initially as exogenous information sources whose influence is mediated by a separate RL controller. More direct LLM-based action generation could be explored in future work, but only after the structured interface is well understood.

5.3.3 Broader implications for LLM–RL systems

Although the empirical study is specific to weekly equity portfolios, several lessons appear to generalise to other LLM–RL applications:

- **Curriculum design in finite-data regimes.** Changing the observation space mid-training can be as disruptive as it is helpful. In finite datasets, this change acts as a distribution shift that may induce catastrophic interference and overfitting. Curriculum designs must therefore be evaluated not only on training ease but also on out-of-sample robustness.
- **Behavioural evaluation as a first-class objective.** For safety-critical or regulated domains, it is not enough for an LLM–RL system to achieve good mean performance. Its behaviour under stress (tail events, regime shifts, rare configurations) must also be scrutinised. The HHI, turnover, and regime-specific analyses in this thesis are examples of such behavioural evaluation.
- **Structured schemas as interfaces.** The pattern LLM $\rightarrow$ structured numeric features $\rightarrow$ RL provides a flexible way to incorporate rich context into RL systems without relinquishing control over the decision rule. This pattern can be transplanted to domains such as operations, logistics, or resource allocation, where unstructured information is abundant but decisions must respect tight constraints.

Overall, the evidence favours LLM–RL architectures that use LLMs to *summarise and constrain* rather than to end-to-end replace domain models and controllers.

5.4 Limitations and threats to validity

Despite the breadth of configurations explored, this work has several limitations that constrain the scope of its conclusions.

Data and environment limitations. The experiments use a 20-asset U.S. equity universe over a fixed historical window and a weekly rebalancing schedule. This omits many real-world complexities: larger universes, non-U.S. markets, intraday dynamics, corporate actions, liquidity constraints, and transaction cost heterogeneity. Survivorship bias and selection bias may also be present. Consequently, the observed patterns may not directly translate to other markets, asset classes, or trading frequencies.

Model and training limitations. The RL component is restricted to PPO with a particular network architecture and a limited hyperparameter grid (learning rates, rollout lengths, and training budgets of 20,000 and 60,000 steps). Other algorithms (e.g. soft actor–critic, deterministic policy-gradient methods, distributional RL, or explicitly risk-constrained RL) could interact differently with LLM-derived signals and risk shaping. Moreover, the finite budgets and small number of seeds mean that some configurations may not have fully converged and that statistical uncertainty remains.

LLM pipeline limitations. The LLM pipeline relies on a specific family of frontier models, prompts, and schemas. Although leakage was controlled and outputs were manually spot-checked, the quality of the extracted signals and risk scores is imperfect. No explicit calibration of LLM confidences or uncertainty estimates is performed. Different models, fine-tuning strategies, or prompting techniques could yield different signal quality and hence different downstream behaviour.

Evaluation limitations. The evaluation uses a particular transaction cost model, risk metrics (NAV, Sharpe, MDD, and related ratios), and backtest methodology. While these are standard, they are not exhaustive. Other risk measures (e.g. conditional value-at-risk, Omega ratio, or drawdown-duration metrics) and alternative cost structures might change the relative attractiveness of some configurations. Furthermore, all results are based on historical backtests; no live or paper trading was conducted, so implementation risks (slippage, market impact, latency, monitoring errors) are not captured.

Internal and external validity. Internally, multiple design choices (feature sets, reward shaping, curriculum schedules) are entangled with a single RL algorithm and data window, which limits causal attribution. Externally, the main threat is over-generalisation: while the thesis provides concrete evidence in one realistic but narrow setting, its conclusions about LLMs “helping RL” should be treated as hypotheses for further testing in other domains, markets, and regimes, not as universal claims.

5.5 Future research directions

Several directions emerge naturally from this work.

RL-side methodological extensions. Future work could test alternative RL algorithms (e.g. soft actor-critic, trust-region methods, distributional RL) and explicit risk-constrained formulations (e.g. constrained RL with CVaR or drawdown constraints) to assess whether the patterns observed here persist. Representation-learning and regularisation techniques (e.g. stronger KL control, auxiliary tasks, or contrastive objectives) could be used to stabilise training with richer feature sets or longer horizons.

Improving LLM signal quality and calibration. On the LLM side, refining schemas, prompts, and model choice could improve signal quality. Domain-specific fine-tuning on financial filings and earnings-call data may yield more calibrated risk scores and directional signals. Techniques for uncertainty estimation (ensembles, Monte Carlo sampling, or explicit confidence training) could make risk shaping more principled by tying penalties to calibrated tail probabilities rather than heuristic scores.

Richer environment design and evaluation. Extending the environment to larger universes, multiple asset classes, and multi-frequency decision-making would test the scalability of the framework. Integrating more realistic execution models (slippage, impact) and stress testing agents with synthetic or scenario-based regimes would provide stronger evidence of robustness. Cross-period and cross-market evaluations would help disentangle genuine generalisation from overfitting to a particular historical window.

Transplanting the pattern beyond finance. The architectural pattern explored here—LLMs as structured information providers, RL as controller—is applicable beyond finance. Sequential decision problems in logistics, supply chains, and operations often involve rich unstructured context (reports, emails, documentation) alongside strict, quantifiable objectives and constraints. Replicating the core design in those domains would clarify which aspects of the framework are domain-specific and which are generic.

Towards agentic systems with memory and hierarchy. Finally, this thesis studied a relatively simple interaction pattern: LLMs produce per-period signals, and PPO maps them to actions within a Markovian environment. An important extension is to incorporate persistent memory and hierarchy. LLMs could maintain and update stateful narratives or latent factors over time, or coordinate specialised sub-agents that operate at different time scales. Hierarchical RL and options frameworks may provide a natural way to integrate such agentic LLM components while preserving risk-aware control.

5.6 Concluding remarks

This thesis presented a case study of LLM-driven reinforcement learning for weekly portfolio allocation. It introduced a custom RL framework with risk-aware reward shaping and a multi-stage LLM pipeline that extracts structured signals and risk scores from multi-genre textual sources. The empirical analysis across RQ1–RQ3 led to three main conclusions:

- On this weekly, large-cap U.S. benchmark, LLM-derived observation features did not deliver a clear, robust improvement in risk-adjusted performance over the tuned price-only PPO baseline. Slow-moving channels such as regulatory filings and earnings calls produced policies whose test NAV and Sharpe remained close to the price-only backbone and competitive with EW/MV, while shaping allocations in interpretable ways (for example, stronger AI-related tilts under SEC integration). Higher-frequency news and technical-commentary features, by contrast, tended to amplify turnover and concentration without commensurate performance gains.
- LLM-derived risk shaping can tilt policies towards more conservative behaviour in stressed regimes and reduce worst-case drawdowns, at the cost of some upside in calm periods. In this setup, LLM-based risk signals function most effectively as guardrails—soft constraints that nudge the policy away from fragile exposures—rather than as primary sources of excess return.
- Curriculum-style staged introduction of LLM channels did not outperform a straightforward 60,000-step all-in-one training regime. In finite-data environments, changing the observation space mid-training can induce behavioural instability and overfitting, so curricula based purely on staged feature introduction should be designed and evaluated with care.

More broadly, the work advocates for a baseline-first, evaluation-driven approach to LLM–RL integration, with a clear separation between LLMs as structured information providers and RL algorithms as decision-makers. While the results do not show that LLMs dramatically improve RL performance in equity portfolio management, they demonstrate that carefully designed LLM signals can be incorporated into a risk-aware RL framework without catastrophic failures, and that they can sometimes yield meaningful behavioural and robustness benefits.

In this sense, the thesis contributes both a concrete, reproducible framework for LLM-driven portfolio allocation and a set of design lessons that may inform future LLM–RL systems in finance and beyond. The broader question of how to reliably harness large language models

in safety-critical, sequential decision-making remains open, but the evidence here suggests that structured interfaces, conservative integration strategies, and rigorous behavioural evaluation are promising ingredients of a viable answer.

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APPENDIX A

TICKER UNIVERSE

Table A.1 lists the 20 large-cap U.S. equities that constitute the fixed trading universe used throughout this thesis. All results in Chapter 4 are based on this universe.

Table A.1: Universe of 20 U.S. equities used in this study.

Ticker
AAPL
AMD
AMT
AMZN
BA
BAC
GOOGL
INTC
JPM
LIN
META
MSFT
MU
NEE
NFLX
NVDA
UNH
V
WMT
XOM

APPENDIX B

ADDITIONAL BASELINE ANALYSES

Table B.1: Cost sensitivity of classical baselines on the 100-week test window. “Cost” denotes the round-trip proportional trading cost. Annualised NAV is obtained by compounding the realised NAV multiple over a 52-week year.

Strategy	Cost	NAV multiple	Annualised NAV	Weekly Sharpe	Max drawdown
Equal-weight	10 bp	1.6539	1.299	1.6404	-0.1631
Equal-weight	5 bp	1.6563	1.300	1.6448	-0.1631
Min-variance	10 bp	1.4573	1.216	1.2692	-0.2009
Min-variance	5 bp	1.5986	1.276	1.5936	-0.1819

APPENDIX C

FORWARD-ROLLING VALIDATION EXPERIMENTS

This appendix reports the results of preliminary forward-rolling train–validation experiments used to assess whether a multi-fold scheme was viable for hyperparameter selection. The overall data set consists of 334 weekly decision dates. As in Section 4.1.1, we first fix a 70% prefix comprising the earliest 234 weeks; the remaining 100 weeks form the final test window used in the main experiments.

Within the 234-week prefix we construct three chronological folds by keeping the training start date fixed at the beginning of the sample and rolling a 30-week validation window forward in time. In Fold 1 the model is trained on weeks 1–144 and validated on weeks 145–174; in Fold 2 it is trained on weeks 1–174 and validated on weeks 175–204; and in Fold 3 it is trained on weeks 1–204 and validated on weeks 205–234. Thus, later folds have longer effective training prefixes but all use a 30-week validation segment.

For a pure price-based PPO baseline we consider three training budgets (20k, 60k, and 80k environment steps) in each fold. Within every (fold, budget) configuration we train three independent policies with different random seeds and evaluate them on the corresponding validation slice using the same metrics as in Section 4.2. The table below reports averages across these three seeds.

Table C.1 shows, for each configuration, the validation NAV multiple, validation Sharpe ratio, validation maximum drawdown, and the differences between validation and train means for these quantities. Folds 1 and 2, which have the shortest training prefixes, exhibit NAV multiples below one and negative Sharpe ratios across all training budgets, together with large negative gaps between validation and train NAV and Sharpe. In contrast, Fold 3, whose training prefix is substantially longer, achieves NAV multiples above one and positive Sharpe ratios around 2.2 on the validation segment, with much smaller performance gaps. These patterns support the interpretation that the early folds suffer from underfitting due to limited data and motivate the single-split protocol adopted in the main experiments.

Table C.1: Forward-rolling validation results for a price-only PPO baseline. Each row corresponds to one (fold, training-budget) configuration. Metrics are averaged over three random seeds. Δ values are val minus train means.

Fold	Steps	NAV _{val}	Sharpe _{val}	MDD _{val}	Score	Δ NAV	Δ Sharpe
1	20 000	0.908	-0.797	-0.199	-0.187	-0.666	-1.698
1	60 000	0.913	-0.746	-0.197	-0.107	-0.696	-1.715
1	80 000	0.915	-0.722	-0.196	-0.070	-0.712	-1.729
2	20 000	0.956	-0.420	-0.148	0.399	-0.444	-0.950
2	60 000	0.951	-0.491	-0.144	0.288	-0.491	-1.078
2	80 000	0.952	-0.488	-0.143	0.292	-0.503	-1.098
3	20 000	1.147	2.297	-0.049	4.618	-0.150	1.947
3	60 000	1.145	2.246	-0.050	4.539	-0.170	1.869
3	80 000	1.135	2.155	-0.051	4.394	-0.205	1.750

APPENDIX D

RUBRICS FOR LLM-BASED ANALYST EVALUATION

This appendix documents the qualitative rubrics used to evaluate the quality of the LLM-generated RL signals and rationales for earnings calls (EC) and SEC filings.

All rubric dimensions are scored on a 1–5 scale:

- 1 = very poor / misaligned,
- 3 = acceptable but mixed,
- 5 = clearly good / strongly aligned with the intended behaviour.

D.1 Rubric for Earnings-Call (EC) RL Outputs

This rubric is applied to the EC RL outputs for a given ticker-week: `signal`, `risk_score`, `confidence`, and `rationale`, evaluated together with the corresponding condensed EC summary.

D.1.1 Scoring Dimensions

Horizon alignment (1–5). Does the rationale and overall assessment clearly focus on the intended horizon of the next 1–4 quarters (about 3–12 months)?

- 1: Mostly vague or long-term commentary; the 1–4 quarter horizon is unclear or absent.
- 3: Mixed horizon; some near-term statements, but blended with long-term or vague views.
- 5: Consistently framed around the 1–4 quarter horizon, with no ambiguity.

Business anchoring (1–5). To what extent is the assessment grounded in business fundamentals and concrete call content?

- 1: Dominated by generic sentiment or stock-price talk; weak link to revenues, margins, products, etc.
- 3: Mix of business facts and generic language; some relevant details but not consistently used.

- 5: Strongly grounded in fundamentals and call details (e.g., revenue/margin trends, guidance, segments, products, customers, demand).

Risk specificity (1–5). How concrete and mechanism-based are the identified risks underlying `risk_score`?

- 1: Generic statements (e.g., “competition”, “macro uncertainty”, “regulation”) with no clear mechanisms.
- 3: Some concrete risks, but mixed with vague or boilerplate phrasing.
- 5: Clearly articulated downside mechanisms tied to the call (e.g., execution risk in a ramp-up, customer concentration, liquidity constraints, credible demand shock).

Calibration quality (1–5). Are the magnitudes of `signal` and `risk_score` reasonable given the EC summary?

- 1: Clearly miscalibrated (e.g., extreme scores with weak evidence, or almost all scores near zero).
- 3: Roughly plausible, but with limited cross-sectional differentiation or inconsistent use of the scale.
- 5: Strong evidence corresponds to strong scores; mild or ambiguous calls receive mild scores; the full range (low/medium/high) is used sensibly.

Rationale quality (1–5). How well does the textual rationale support and explain the numeric outputs?

- 1: Mainly generic or templated language; weakly connected to the EC summary.
- 3: Partially grounded in the EC summary; some references to specific numbers or risks.
- 5: Clearly references key items (numbers, risks, opportunities) from the condensed EC analysis and explains the chosen `signal` and `risk_score` in 2–5 coherent sentences.

D.2 Rubric for SEC RL Outputs

This rubric is applied to the SEC RL outputs for a given ticker-week: `signal`, `risk_score`, `confidence`, and `rationale`, evaluated together with the condensed SEC filing (`SECReduceFinalJSON`), including `summary_text`, `top_risks`, `exposures`, `opportunities`, and `flags`.

D.2.1 Scoring Dimensions

Horizon alignment (1–5). Does the assessment focus on the intended structural horizon of roughly 1–3 years?

- 1: Mainly short-term or very vague commentary; little sense of a multi-year outlook.
- 3: Mixed horizon; partially structural, partially short-term or generic.
- 5: Clearly framed as a 1–3 year structural view on the business and equity risk.

Structural focus (1–5). To what extent does the rationale emphasise structural opportunities and risks (business model, regulatory dependence, leverage, concentration, etc.) rather than transient news flow?

- 1: Largely short-term or event-style comments, weak link to structural features.
- 3: Some reference to structural aspects, but mixed with transient or superficial remarks.
- 5: Strong emphasis on persistent exposures, long-lived strengths/weaknesses, and structural risk channels.

Risk specificity (1–5). How concrete and mechanism-based are the structural downside risks supporting `risk_score`?

- 1: Mostly boilerplate risk language (e.g., “may be affected by economic conditions”) with no clear mechanisms.
- 3: Some specific risks, but mixed with unspecific or generic text.
- 5: Well-articulated structural threats (e.g., high leverage, single-region exposure, concentrated customers, regulatory investigations) that are explicitly linked to the business and time horizon.

Calibration quality (1–5). Are the magnitudes of `signal` and `risk_score` appropriate for a structural SEC-based view?

- 1: Very weak evidence driving very strong scores, or almost no variation in the scores.
- 3: Broadly plausible values but limited or inconsistent differentiation across filings.
- 5: Strong, persistent structural advantages/threats map to higher magnitudes; mild or mixed structural situations map to moderate scores; the available range is used in a discriminative way.

Rationale quality (1–5). How well does the rationale connect to the SEC summary fields?

- 1: Generic or template-like text with little traceable link to `top_risks`, `exposures`, or `opportunities`.
- 3: Partially grounded; some explicit references, but also vague or unsupported claims.
- 5: Clearly references specific items from `top_risks`, `exposures`, and `opportunities`, and explains in 2–5 sentences why the chosen signal and `risk_score` are appropriate.

D.3 Rubric for News RL Outputs

This rubric is applied to the weekly news RL outputs for a given ticker-week: the textual `summary_text`, the structured `key_events` list, and the RL head (`signal`, `risk_score`, `confidence`, `rationale`). Evaluation is done in the context of the intended 1–4 quarter (3–12 month) horizon and the role of news as an incremental update on top of the slower EC and SEC views.

All dimensions are scored on a 1–5 scale:

- 1 = very poor / misaligned,
- 3 = acceptable but mixed,
- 5 = clearly good / strongly aligned with the intended behaviour.

D.3.1 Scoring Dimensions

Horizon alignment and incremental focus (1–5). Does the news analysis correctly treat weekly news as an incremental update to the 1–4 quarter fundamental outlook (rather than short-term price noise or very long-term speculation)?

- 1: Dominated by intraday/weekly price chatter or vague long-term commentary; the 1–4 quarter horizon is unclear or absent.
- 3: Mixed horizon; some statements clearly relate to the next 1–4 quarters, but blended with micro-noise or very long-term speculation.
- 5: Consistently framed as “what this week’s news changes for the next 1–4 quarters,” with no confusion between short-term noise and structural SEC-style information.

Noise filtering and relevance (1–5). How well does the analysis filter out non-material, non-ticker-specific headlines and focus on genuinely relevant news?

- 1: Key events are dominated by generic macro headlines, trivial announcements, or weakly related stories; clear ticker-level fundamentals are often ignored.

- 3: Some irrelevant or low-materiality items are included, but the main ticker-relevant stories are still captured.
- 5: Strong emphasis on a small number of high-quality, company-specific events; generic sector or macro news appears only when it is clearly linked to this company's fundamentals.

Event structuring and coverage (1–5). Are the `key_events` well-chosen, non-duplicative, and informative as a structured summary of the week?

- 1: Key events are poorly titled, overlapping, or missing; major stories are either omitted or fragmented across many redundant entries.
- 3: Main stories are present but with some duplication, inconsistent categories (e.g., earnings vs product), or underspecified rationales.
- 5: 3–8 clear, distinct events that together cover the dominant themes of the week, with sensible use of categories and concise rationales that explain why each event matters.

Risk and opportunity specificity (1–5). How concrete and mechanism-based are the upside and downside implications implied by the news (even if not explicitly encoded as separate lists)?

- 1: Mostly generic language (e.g., “may be positive”, “could be a risk”) with little explanation of how fundamentals or risk channels are affected.
- 3: Some concrete mechanisms (e.g., demand shifts, regulatory actions) are described, but mixed with boilerplate phrases or ambiguous impacts.
- 5: Clear, company-specific mechanisms are articulated, linking events to earnings power, margins, balance sheet, or tail risks over the next 1–4 quarters.

Calibration quality of `signal` and `risk_score` (1–5). Are the magnitudes and signs of `signal` and `risk_score` reasonable given the week's news and the cross-sectional calibration logic?

- 1: Clearly miscalibrated: extreme signals or risk scores for modest events, almost all weeks clustered near zero, or frequent sign errors (e.g., negative signal for clearly positive news).
- 3: Roughly plausible but with limited differentiation across weeks or categories; moderate overuse of large magnitudes relative to the actual strength of evidence.
- 5: Strong, concentrated positive (negative) news flows map to clearly positive (negative) signals within the prescribed ranges; elevated tail risks (e.g., serious investi-

gations, accidents, liquidity concerns) are reflected in higher `risk_score`, while noisy or routine weeks stay in the mild-to-moderate range.

Rationale quality and cross-modal consistency (1–5). How well does the textual `rl.rationale` connect the main `key_events` to the numeric outputs and to the intended role of news relative to EC/SEC?

- 1: Generic or template-like rationale that barely references the specific events; little explanation of why the chosen signal, `risk_score`, or confidence values are appropriate.
- 3: Partially grounded rationale that cites some key events and distinguishes positives vs negatives, but with limited detail on mechanisms or how confidence is determined; occasional tension with the implied EC/SEC horizon.
- 5: 3–6 coherent sentences that explicitly reference the most important events, explain how they update the 1–4 quarter outlook, justify both signal and `risk_score` magnitudes, and briefly motivate the confidence level; the overall story is consistent with the idea that EC/SEC provide slower baselines and news provides incremental updates.

D.4 Rubric for Technical RL Outputs

This rubric is applied to the weekly technical analysis outputs for a given ticker-week: the textual `summary_text`, the regime label, the list of `key_signals`, and the RL head (`signal, risk_score, confidence, rationale`).

All dimensions are scored on a 1–5 scale:

- 1 = very poor / misaligned,
- 3 = acceptable but mixed,
- 5 = clearly good / strongly aligned with the intended behaviour.

The technical head is interpreted as a short- to medium-horizon view on price action and volatility over roughly the next 1–8 weeks, used as a faster overlay on top of the slower EC (1–4 quarters) and SEC (1–3 years) views.

D.4.1 Scoring Dimensions

Horizon and price-action alignment (1–5). Does the analysis focus on short- to medium-term price and volatility dynamics (approximately 1–8 weeks), using the provided weekly his-

tory and window statistics, rather than very long-term fundamentals or microstructure noise?

- 1: Dominated by vague long-term commentary or ultra-short-term noise (e.g., intraday ticks) with no clear 1–8 week view.
- 3: Mixed horizon; some coherent statements about near-term trend or mean-reversion, but blended with long-horizon or purely backward-looking remarks.
- 5: Consistently framed as a view on the next several weeks of trend, momentum, mean-reversion pressure, and volatility regime, clearly grounded in the recent-history window.

Technical purity and input respect (1–5). Does the analysis respect the constraint of using ONLY the provided technical indicators (no fundamentals, earnings, news, or macro that are not implied by price/volume behaviour)?

- 1: Frequently references earnings, products, macro news, or company fundamentals not derivable from OHLCV-derived indicators.
- 3: Mostly technical, but occasionally drifts into vague fundamental language (e.g., “business outlook”) without clear ties to price/volume features.
- 5: Fully technical; all statements are grounded in indicators such as trend, momentum, volatility, and volume, or in their recent changes and window statistics.

Regime classification quality (1–5). Is the chosen `regime` label consistent with the described price behaviour and the underlying indicators (trend, volatility, ranges)?

- 1: Regime label appears arbitrary or contradictory to the description (e.g., “trending” when price has been range-bound and choppy).
- 3: Roughly plausible regime choice but with some tension (e.g., “volatile” vs a description closer to range-bound).
- 5: Regime clearly matches the narrative and key signals (e.g., persistent uptrend with higher highs/lows and positive `dist_from_ema/macd_hist` → “trending”; choppy mean-reversion around moving averages → “mean_reverting” or “range_bound”; large, irregular swings with elevated `realized_vol_20`, turbulence, and `bb_width` → “volatile”; conflicting indicators → “unclear”).

Key signal specificity and coverage (1–5). How informative and non-duplicative are the `key_signals` as a structured summary of the technical picture?

- 1: Signals are generic (e.g., “the stock is volatile”), repetitive, or inconsistent with

the data; titles are vague and rationales add little beyond the label.

- 3: Main ideas (trend direction, momentum, overbought/oversold, support/resistance, volatility) appear, but with some redundancy or weak rationales. References to indicators are occasional but not systematic.
- 5: 3–7 clear, distinct signals that capture the central technical story (trend, momentum, mean-reversion setup, key levels, volatility), each with a sensible direction/strength and a concise rationale that explicitly references fields from `current_week`, `recent_history`, or `window_stats` (e.g., changes in `macd_hist`, `rsi_14`, `log_ret_1w`, or volatility deltas).

Calibration of `signal` and `risk_score` (1–5). Are the magnitudes and signs of the RL-facing `signal` and `risk_score` reasonable given the technical picture and a cross-sectional calibration across weeks and tickers?

- 1: Clearly miscalibrated: extreme signals or risk scores for modest moves; almost all weeks near zero; or frequent sign errors (e.g., strongly negative signal for a clean uptrend).
- 3: Broadly plausible but with limited differentiation across weeks; some tendency to overuse large magnitudes or cluster around a single value (e.g., `signal` $\approx$ 5 for any up-move, `risk_score` $\approx$ 5 regardless of regime).
- 5: Strong, sustained uptrends (downtrends) with confirmation from multiple indicators map to clearly positive (negative) signals within a sensible band (e.g., 4–8 or –4 to –8); sharply elevated volatility or clear downside tail risk (e.g., breakdown from a long range, large gaps, regime shifts) is reflected in higher `risk_score`, while quiet, stable regimes with low `realized_vol_20`, `bb_width`, and `turbulence` remain in the lower range (e.g., 1–4).

Confidence and rationale quality (1–5). Does `rl.rationale` clearly connect the observed technical features (regime and key signals) to the chosen `signal`, `risk_score`, and `confidence`?

- 1: Rationale is generic, templated, or disconnected from the described signals; confidence appears arbitrary or almost always the same across weeks.
- 3: Rationale references some key signals and mentions why the view is bullish/bearish/neutral, but gives limited detail on why risk is low/normal/elevated or why the confidence level is appropriate.
- 5: 3–6 coherent sentences that explicitly link the main technical features (trend,

momentum, volatility, levels) to the sign and magnitude of signal, explain the `risk_score` in terms of realized and recent volatility (e.g., `realized_vol_20_delta`, `turbulence_delta`, `bb_width_max`), and justify the confidence based on indicator agreement, data sufficiency, and regime clarity.

Discard behaviour and guardrails (1–5). Is the `discarded/discard_reason` mechanism used appropriately when input data are insufficient or inconsistent?

- 1: The model almost never discards even when inputs are clearly unusable (e.g., fewer than four valid history rows, many NaNs in critical fields), or discards valid inputs without a clear reason.
- 3: Occasional appropriate use of discarding, but some borderline cases where either a weak-but-valid view is returned or a usable case is discarded.
- 5: Discarding is used sparingly and only when there is genuine data insufficiency or inconsistency, with a clear `discard_reason`; otherwise, the model produces a cautious but valid technical view with appropriately low confidence when data are thin.

D.5 Rubric for Tier-2 Investment-Committee Outputs

This rubric evaluates the quality of Tier-2 LLM agents that aggregate Tier-1 source-specific analyses into a weekly senior verdict: the Business Analyst, Risk Analyst, Skeptic, and Judge.

All dimensions are scored on a 1–5 scale:

- 1 = very poor / misaligned,
- 3 = acceptable but mixed,
- 5 = clearly good / strongly aligned with the intended behaviour.

D.5.1 Business Analyst Rubric

The Business Analyst produces `signal_proposal`, `thesis`, and `key_drivers`.

Horizon alignment (1–5). Does the thesis clearly frame the business trajectory as a short- to medium-horizon tilt for the next 1–4 weeks, *anchored* in the 1–4 quarter fundamental outlook?

- 1: Focuses mainly on long-term hopes or day-to-day price noise; little connection to the 1–4 quarter view.

- 3: Mixed horizon; some quarterly themes, but blended with speculative near-term commentary.
- 5: Explicitly links this week's tilt to fundamental information from EC/SEC/news over the next 1–4 quarters, avoiding microstructure-style prediction.

Business anchoring and evidence use (1–5). To what extent is the thesis grounded in Tier-1 summaries and concrete fundamentals (revenues, margins, products, segments, catalysts)?

- 1: Dominated by generic sentiment; weak link to Tier-1 evidence.
- 3: Mix of specific drivers and vague statements; some use of Tier-1 content but not systematic.
- 5: Clearly references key EC/SEC/news points (e.g. guidance changes, structural strengths, major catalysts), with minimal generic filler.

Signal calibration (1–5). Is `signal_proposal` a reasonable use of the [-10, 10] scale?

- 1: Frequent extreme values (near ± 10) for routine cases, or almost no variation across names.
- 3: Roughly plausible magnitudes but with occasional overreaction or underuse of the scale.
- 5: Typical large-cap weeks cluster in a moderate band (e.g. [-3, 3]); extreme values are reserved for unusually strong multi-source evidence.

Thesis quality (1–5). How well does the 2–6 sentence `thesis` describe the business trajectory?

- 1: Vague, repetitive, or internally inconsistent; unclear view.
- 3: Partially coherent; main ideas present but missing some important context or trade-offs.
- 5: Clear, coherent summary of the near- to medium-term business trajectory, explicitly balancing strengths, weaknesses, and catalysts.

Key-drivers quality (1–5). Are the 3–7 `key_drivers` concrete and non-redundant?

- 1: Generic bullet points; heavy duplication or boilerplate.
- 3: Mixed; some specific drivers but also vague or redundant items.
- 5: Each driver is specific (e.g., products, segments, geographies, catalysts), non-overlapping, and directly supports the thesis and signal.

D.5.2 Risk Analyst Rubric

The Risk Analyst produces `risk_score_proposal`, `risk_factors`, and `tail_events`.

Near-term risk focus (1–5). Does the analysis concentrate on incremental downside risk over the next 1–4 weeks, using structural information as background?

- 1: Mostly long-dated structural risks; little sense of the 1–4 week horizon.
- 3: Mixed; some near-term risks, but blurred with generic, multi-year statements.
- 5: Clearly distinguishes baseline structural risk from incremental near-term risk; focuses on triggers that could matter over the next few weeks.

Risk specificity and mechanisms (1–5). How concrete and mechanism-based are the listed risk factors?

- 1: Generic phrases (“competition”, “macro uncertainty”) with no clear channels.
- 3: Some specific risks, but mixed with vague or boilerplate text.
- 5: Each risk factor explains how and why downside could occur (e.g., a refinancing cliff, customer loss, regulatory decision) with a plausible near-term timeline.

Tail-event identification (1–5). Are genuine binary or tail events properly separated into `tail_events`?

- 1: Tail events are missing when evident, or everything is incorrectly labelled as a tail event.
- 3: Some tail events are correctly identified, but coverage or separation is incomplete.
- 5: Clear identification of true binary events (e.g., trial outcomes, approvals) when present; empty list when none exist.

Risk-score calibration (1–5). Is `risk_score_proposal` a reasonable use of the [0, 10] range?

- 1: Frequent use of very high scores without clear triggers, or scores clustered in a narrow band with little differentiation.
- 3: Broadly reasonable but with some overuse or underuse of high scores.
- 5: Typical diversified large-caps lie in a moderate range (e.g. 2–5); elevated scores (≥ 7) appear only when specific, near-term risks are present.

Coverage and non-duplication (1–5). Do `risk_factors` and `tail_events` jointly cover the main foreseeable risks without double-counting?

- 1: Important risks are omitted or repeated in multiple bullets.
- 3: Mixed; most key risks appear, but with some redundancy.

- 5: Concise set of bullets covering all important near-term risks without unnecessary duplication.

D.5.3 Skeptic Rubric

The Skeptic produces `disagreement_points` and `disagreement_score`.

Issue coverage (1–5). Does the Skeptic surface the main weaknesses in the Business and Risk analyses when present?

- 1: Misses obvious omissions, contradictions, or overstatements.
- 3: Identifies some issues but overlooks others of similar importance.
- 5: Highlights all materially important issues; stays quiet (low disagreement) only when the case is genuinely well supported.

Specificity and grounding (1–5). How specific and evidence-based are `disagreement_points`?

- 1: Very generic (e.g., “too optimistic”) with no reference to the underlying summaries or metrics.
- 3: Mixed; some points reference evidence, others are vague.
- 5: Each point references concrete claims, horizons, or recency (e.g., “relies on a 90-day-old filing as a fresh catalyst”).

Disagreement-score calibration (1–5). Is `disagreement_score` in $[0, 1]$ calibrated to the materiality of the issues?

- 1: Often extreme (near 0 or 1) regardless of context.
- 3: Roughly aligned but with weak differentiation across cases.
- 5: Low scores for minor nitpicks, moderate for real but non-fatal issues, and high only when the view is poorly grounded or strongly contradicted.

D.5.4 Judge Rubric

The Judge produces the final weekly verdict: `senior_signal`, `senior_risk_score`, `senior_confidence`, and `senior_rationale`.

Integration quality (1–5). How well does the Judge integrate Tier-1 sources (EC/SEC/news/technical) with the Business and Risk proposals and the Skeptic critique?

- 1: Largely repeats one analyst (typically Business) and ignores others.
- 3: Uses multiple inputs but without clear trade-offs.

- 5: Explicitly balances structural, fundamental, news, and technical signals, adjusting for disagreement and recency.

Horizon consistency (1–5). Is the final verdict framed as a weekly allocation tilt that remains consistent with 1–4 quarter fundamentals and 1–3 year structural views?

- 1: Treats the decision as a pure short-term price bet or a purely long-term view.
- 3: Mixed; acknowledges multiple horizons but blurs them.
- 5: Clearly interprets the weekly decision in light of the underlying 1–4 quarter and structural outlooks.

Signal and risk-score calibration (1–5). Are `senior_signal` and `senior_risk_score` reasonable given the combined evidence and disagreement?

- 1: Frequent extreme values or almost no variation; risk and signal appear entangled.
- 3: Mostly plausible, but sometimes double-counts risk or overreacts to single sources.
- 5: Uses moderate scores for typical large-cap weeks, reserves extremes for cases with strong multi-source alignment, and maintains clear separation between signal (directional view) and `risk_score` (downside risk).

Rationale quality (1–5). How well does `senior_rationale` justify the final scores?

- 1: Vague or generic; does not mention specific strengths, risks, or time scales.
- 3: Partially grounded; mentions some key points but misses important contradictions or uncertainties.
- 5: In 3–7 sentences, clearly summarises the main positive and negative drivers, refers to recency and horizon, and explains how they justify the chosen signal, `risk_score`, and confidence.